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Cartographie et Navigation Autonome sur Champs Physiques
par Filtrage Particulaire à Processus Gaussiens

Autonomous Navigation and Mapping on Physical Fields
by Gaussian Process-based Particle Filtering

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NOMENCLATURE

Abbreviations

APF	Auxiliary Particle Filter
ARMSE	Averaged Root Mean Squared Error
a.s.	almost surely
BLUE	Best Linear Unbiased Estimator
CDF	Cumulative Distribution Function
EKF	Extended Kalman Filter
flop	Floating-Point Operations
GP	Gaussian Process
IEKF	Iterative Extended Kalman Filter
i.i.d.	independent and identically distributed
KL	Karhunen-Loève
MC	Monte Carlo
MAP	Maximum <i>A Posteriori</i>
MISE	Mean Integrated Squared Error
ML	Maximum Likelihood
MMAE	Minimum Mean Absolute Error
MMSE	Minimum Mean Squared Error
OK	Ordinary Kriging
PDF	Posterior Probability Density Function
PF	Particle Filter
RBPF	Rao-Blackwellised Particle Filter
RMSE	Root Mean Squared Error
RPF	Regularised Particle Filter
SE	Squared Exponential
SIR	Sampling Importance Resampling
SIS	Sequential Importance Sampling
SK	Simple Kriging
SPD	Symmetric Positive Definite
SPSD	Symmetric/Self-adjoint Positive Semi-Definite
UK	Universal Kriging

Symbols

$\triangleq$	Equality by definition
$\propto$	Proportionality
$\sim$	"is distributed according to"
$\overset{i.i.d.}{\sim}$	Independently and identically distributed according to
$\overset{a.s.}{\rightarrow}$	Almost sure convergence
$\xrightarrow{\mathcal{L}}$	Convergence in law
$\cdot \succeq 0$	Symmetric positive semi-definite
$\cdot \succ 0$	Symmetric positive definite
$\langle \cdot \cdot \rangle, \langle \cdot \cdot \rangle_{\mathcal{E}}$	Inner product, subscript: of the space $\mathcal{E}$
$\langle \cdot, \cdot \rangle$	Measure/function duality pairing, $\langle \mu, \varphi \rangle = \int \varphi d\mu$
$\ \cdot \ , \ \cdot \ _2, \ \cdot \ _F$	Generic, Euclidean and Frobenius norms
$\cdot^{\top}$	Transpose of a matrix
$\cdot^{-1}$	Inverse of a matrix
$*$	Convolution operator
$\circ$	Function composition operator
$\otimes$	Kronecker product
$\mathbb{1}$	Indicator function
$[\cdot, \cdot]$	Discrete interval
$\cdot_{m n}$	"at time m conditioned on the observations from 1 to n "
Γ	Gamma function
δ_x	Dirac measure at x
$\varepsilon, \varepsilon_i$	Query and training noises
θ	Kernel hyperparameters
$\lambda_k^{(i)}$	Auxiliary weights
$\lambda_{\min}, \lambda_i$	Smallest and i -th eigenvalues of a matrix
μ	Target probability measure, of density p
$\mu_k^{(i)}$	Auxiliary particle
ν	Proposal (importance) measure, of density q
π_k	Projection of a trajectory onto its last component
ρ	Distance on the estimator space $\mathcal{F}$
φ	Mapping of the state
$\hat{\psi}_{\text{MMSE}}, \hat{\psi}_{\text{MMAE}}, \hat{\psi}_{\text{MAP}}, \hat{\psi}_{\text{ML}}$	State estimators
ψ_i, Ψ, Λ	Mercer eigenfunctions and eigenvalue matrix
$\operatorname{argmin}, \operatorname{argmax}$	Argument of the minimum, of the maximum
$b_k^{(e)}, b_k^{(o)}$	Evolution and observation bias vectors
C_{μ}, C_{ν}	Normalising constants of μ and ν
cdf	Cumulative distribution function
d_x, d_z, d_{φ}	Dimensions of the state, observation, and mapping
d_w, d_n	Dimensions of the evolution and observation noises
$\det$	Determinant of a matrix
$\mathbb{E}[\cdot], \mathbb{E}_p[\cdot]$	Expectation, subscript: under the law p
$\exp$	Exponential function
$f_k, f_k^l, f_k^n, \bar{f}_k^l$	Evolution functions
$F_k, F_k^l, F_k^n, \bar{F}_k^l$	Evolution matrices
$\mathcal{F}$	Space of admissible estimators
g, g_l	Trend functions

$\mathcal{GP}(m, k)$	Gaussian process
G, G_θ	Gram matrix
h_k	Observation function
H_k	Observation matrix
I, I_n	Identity matrix, subscript: of size n
$\mathbf{1}_n$	Vector of ones of size n
J_f	Jacobian matrix of a function f
$k(\cdot, \cdot), k_{\text{SE}}, k_\nu, k_{\text{sep}}, k_{\text{MS}}, k_B, k_{r_{\text{KL}}}$	Covariance kernels
$K, K_h, K_{\text{Epa}}, K_{\text{Gauss}}$	Regularisation kernels
K_ν	Modified Bessel function of the second kind
K_k	Kalman gain
$\mathcal{K}, \mathcal{K}_{r_{\text{KL}}}$	Mercer operator and its reduced-rank truncation
$\mathcal{L}^0, \mathcal{L}^p$	Sets of measurable functions and p -integrable random variables
m	Mean function of a Gaussian process
$\hat{m}_{\text{OK}}, \hat{m}_{\text{UK}}$	Estimated ordinary and universal-kriging means
$\mathcal{M}_{m,n}(\mathbb{R})$	Set of real $m \times n$ matrices
n_k	Observation noise
$N, \tilde{N}$	Number of particles, of training points
$\hat{N}_{\text{eff}}, N_{\text{eff}}$	Estimated and real effective sample size
N_{th}	Resampling threshold
$\mathbb{N}, \mathbb{N}^*$	Sets of natural and strictly positive natural integers
$\mathcal{N}(m, P)$	Gaussian distribution
$o(\cdot)$	Landau symbol for asymptotic negligibility
$\mathcal{O}(\cdot)$	Landau symbol for asymptotic boundedness
p	Target probability density
p_0	Probability density of the initial state
$p_k^{(e)}, p_k^{(o)}$	Probability densities of the evolution and observation noises
$p_{A B}(a b)$	Conditional density of A given B
$P_0, P_{k k-1}, P_{k k}$	State covariance matrices
$\mathbb{P}$	Probability of an event
q	Proposal (importance) density
$Q_k, Q_k^n, Q_k^l, Q_k^{nl}, \bar{Q}_k^l$	Evolution noise covariances
$r_{\text{UK}}, r_{\text{KL}}$	Number of trend functions and KL rank
rank	Rank of a kernel or operator
$R, R_k, \tilde{R}$	Observation and training noise covariances
$R_{\text{SK}}, R_{\text{OK}}, R_{\text{UK}}, R_K, R_{\hat{m}}$	Kriging error covariances
$\mathbb{R}, \mathbb{R}_+, \mathbb{R}_+^*$	Sets of real, positive real, and strictly positive real numbers
$S^N(\mu), S^N(\mu, \nu), \tilde{S}^N, S_{K_h}^N$	Monte Carlo measures
S_k	Innovation covariance
$\mathcal{S}_d^+, \mathcal{S}_d^{++}$	Sets of SPSP and SPD matrices
supp	Support of a measure or function
tr	Trace of a matrix
$\mathcal{U}$	Input set of a stochastic process
$\mathcal{U}([a, b])$	Uniform distribution
$\mathbb{V}[\cdot], \mathbb{V}_\mu[\cdot]$	Variance, subscript: under the measure μ
$w_k, w_k^l, w_k^n, \bar{w}_k^l$	Evolution noises
$w_k^{(i)}, \tilde{w}_k^{(i)}$	Normalised and unnormalised particle weights
$x_k, x_k^n, x_k^l, \mathbf{x}_{0:k}$	System states and trajectory

$\mathbf{x}, \mathbf{x}', \tilde{\mathbf{x}}$	Input tuples and training inputs
$x_k^{(i)}$	Particle
$x_{k k-1}, x_{k k}, \hat{x}_k$	Predicted, corrected and estimated state
y_k	Innovation vector
$z_k, \mathbf{z}_{1:k}$	Observation vector and sequence
$z, \tilde{\mathbf{z}}$	Query and training observations
$\hat{z}, \hat{z}_{\text{SK}}, \hat{z}_{\text{OK}}, \hat{z}_{\text{UK}}, \hat{z}_{\text{K}}$	Kriging estimators

CHAPTER 1 PLAN GLOBAL

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1.1 Introduction

- contexte : navigation autonome, pas de GNSS mais des champs physiques
- motivation : limites des méthodes classiques (Kalman, carte embarquée connue)
- objectif : reconstruction de la fonction d'observation par GP, intégration dans le cadre de la navigation bayésienne
- principales contributions
- annonce du plan

Ne pas oublier la progression des problèmes:

1. drift of inertial navigation
2. GNSS data blurred or unavailable
3. high initial uncertainty
4. non-injective, non-linear, multimodal environments
5. undersampled, poorly-known environment model
6. weight degeneracy, particle impoverishment
7. non-stationarity of the field renders distant samples locally irrelevant
8. high computational cost of matrix inversion and multiplications
9. no preexisting data about the environment

1.2 3) Processus gaussiens

- méthodes de réduction de rang
- lien avec les RKHS + Kalman sur RKHS (ailleurs ?)

1.3 4) CAK-PF

- fenêtrage, clustering par Mean-Shift
- apprentissage des hyperparamètres
- étude d'ablation
- précision en fonction du nombre de données disponibles
- résultats numériques (RMSE, temps de calcul, comparaisons avec la BCR)

sous-échantillonnage:

- erreurs théoriques
- heuristiques de sous-échantillonnage (fenêtrage, clusterisation, sous-échantillonnage aléatoire)
- parler de l'intérêt pour la navigation ? permet de reconstruire la fonction d'observation ?
- résultats numériques sans navigation

1.4 SLAM

- GP stationnaires ou stationnaires par morceaux
- multi capteurs en mailles impossibles car dimensions trop proches
- RBPF/méthode de Manon
- RKHS/méthode de Karim

1.5 Ouverture sur la fusion de champs ?

- à voir mais information de Fisher théorique
- tests sur les 2 cas précédents

1.6 Conclusion et perspectives

- résumé des contributions : GP + FP
- adaptation du modèle en fonction de l'état (carte dégradée)
- reconstruction à partir d'un dictionnaire de fonctions de base (sans carte)
- limites des méthodes
- perspectives futures (modèles, méthodologie, ...)

Part I

Mathematical Foundations

CHAPTER 2

BAYESIAN ESTIMATION

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2.1 Introduction

The goal of this chapter is to formally introduce the tools necessary to solve the general filtering problem. More precisely, given a dynamical system whose state evolves according to a known stochastic model and is partially observed through noisy measurements, we seek to sequentially estimate the system state at each time step in an optimal sense. We first define what optimality means in a rigorous setting, then derive exact and approximate solutions to this problem. We also discuss the limitations of such methods.

Formally, for $d_x, d_z, d_\varphi \in \mathbb{N}^*$, we consider a hidden Markov chain $(x_k)_{k \geq 0}$ taking values in $\mathbb{R}^{d_x}$ and a series of observations $(z_k)_{k \geq 1}$ taking values in $\mathbb{R}^{d_z}$, each associated with the unknown state x_k . The quantity of interest is a mapping of the state under a known function $\varphi : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_\varphi}$, which allows us to target any derived quantity of the state (*e.g.* position, velocity, or a subset of coordinates). When solving a navigation problem, the state and observation processes are linked by the state-space model [1]:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \quad (2.1)$$

where the processes $w_k \in \mathbb{R}^{d_w}$ and $n_k \in \mathbb{R}^{d_n}$ denote the evolution and observation noises respectively, and are assumed to be white noises (*i.e.* temporally independent, centered noises of finite variance), mutually independent and independent of x_0 .

The known data of this problem are:

$$\begin{cases} p_0 & \text{the probability density of the initial state } x_0 \\ f_k & \text{the evolution function} \\ p_k^{(e)} & \text{the probability density of the evolution noise } w_k \\ z_k & \text{the current noisy observation} \\ h_k & \text{the observation function} \\ p_k^{(o)} & \text{the probability density of the observation noise } n_k \end{cases} \quad (2.2)$$

Remark 2.1 (Control input). *In many navigation applications, a control input u_k is also available, and the evolution function takes the form $f_k(x_{k-1}, u_k, w_k)$. For the sake of clarity, we omit u_k from the general formulation as it can always be absorbed into f_k or treated as a known deterministic input.*

Depending on how observations are used relative to the time index of the state to estimate, three problems arise [2]:

- **Filtering:** estimate $\varphi(x_k)$ given $\mathbf{z}_{1:k} \triangleq (z_1, \dots, z_k)$. It is the most common case in real-time navigation.
- **Smoothing:** estimate $\varphi(x_l)$ for $l < k$ given $\mathbf{z}_{1:k}$. It corresponds to refining past estimates using future observations.
- **Prediction:** estimate $\varphi(x_l)$ for $l > k$ given $\mathbf{z}_{1:k}$. This amounts to anticipating future states from current observations.

This thesis focuses on the filtering problem. The rest of this chapter is organised as follows:

- Section 2.2 rigorously defines what it means to optimally estimate $\varphi(x_k)$ from the observations $\mathbf{z}_{1:k}$, and shows that all classical optimality criteria reduce to computing the posterior probability density function (PDF) $p(x_k|\mathbf{z}_{1:k})$.
- Section 2.3 derives analytical filtering algorithms to propagate this density over time, and discusses their limitations.
- Section 2.4 then presents asymptotically optimal numerical methods based on Monte Carlo (MC) sampling that alleviate such flaws.
- Finally, section 2.5 presents advanced particle filtering methods that improve upon standard MC techniques.

2.2 State Estimation

The first issue to tackle is to define precisely the meaning of "best estimator" of $\varphi(x_k)$ given $\mathbf{z}_{1:k}$ [3]. Formally, given a functional metric space $(\mathcal{F}, \rho)$, it amounts to finding a measurable function $\psi \in \mathcal{F}$ that minimises the expected loss:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\rho(\psi(\mathbf{z}_{1:k}), \varphi(x_k))) \quad (2.3)$$

Depending on the considered space and distance, different optimal estimators arise. We present three classical ones below. For the sake of simplicity, we write Z for a generic observation vector and X for the underlying state, as opposed to the time-indexed processes x_k and z_k . At a fixed time step k , (2.3) thus becomes:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\rho(\psi(Z), \varphi(X))) \quad (2.4)$$

For the remainder of this document, we let, for $d, d', p \in \mathbb{N}^*$:

$$\begin{aligned} \mathcal{L}^0(\mathbb{R}^{d'}, \mathbb{R}^d) &\triangleq \left\{ f : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d \mid f \text{ measurable} \right\} \\ \mathcal{L}^p(\mathbb{R}^d) &\triangleq \left\{ A \text{ an } \mathbb{R}^d\text{-valued random variable} \mid \mathbb{E}[\|A\|^p] < \infty \right\} \end{aligned} \quad (2.5)$$

where almost surely equal random variables are identified.

2.2.1 Posterior Expectation Estimator

Let:

$$\mathcal{F} \triangleq \left\{ \psi \in \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi}) \mid \mathbb{E}[\|\psi(Z)\|^2] < \infty \right\} \quad (2.6)$$

To simplify notations, we identify each estimator $\psi \in \mathcal{F}$ with the random variable $\psi(Z)$ it produces. Under this identification, $\mathcal{F}$ is a closed subspace of $\mathcal{L}^2(\mathbb{R}^{d_\varphi})$, and can be equipped with the inner product:

$$(A, B) \in \mathcal{F}^2 \mapsto \langle A|B \rangle \triangleq \mathbb{E}[A(Z)^\top B(Z)] \quad (2.7)$$

where $\mathbb{E}[X]$ denotes the expectation of the random variable X .

Let $\|\cdot\|$ be the norm associated with $\langle \cdot | \cdot \rangle$. When $\varphi(X) \in \mathcal{L}^2(\mathbb{R}^{d_\varphi})$, the minimisation problem (2.4) is equivalent to:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\mathbb{E}[\|\psi(Z) - \varphi(X)\|^2]) \quad (2.8)$$

An estimator solving (2.8) is called a minimum mean squared error (MMSE) estimator.

Proposition 2.1 (MMSE estimator). *The unique solution to (2.8) is the posterior expectation:*

$$\hat{\psi}_{\text{MMSE}}(z) \triangleq \mathbb{E}[\varphi(X)|Z = z] \quad (2.9)$$

Proof. Let $\hat{\psi}(z) \triangleq \mathbb{E}[\varphi(X)|Z = z]$ and $\psi \in \mathcal{F}$. It comes that:

$$\begin{aligned} \rho(\psi(Z), \varphi(X))^2 &= \langle \psi(Z) - \varphi(X) | \psi(Z) - \varphi(X) \rangle \\ &= \langle \psi(Z) - \hat{\psi}(Z) + \hat{\psi}(Z) - \varphi(X) | \psi(Z) - \hat{\psi}(Z) + \hat{\psi}(Z) - \varphi(X) \rangle \\ &= \langle \psi(Z) - \hat{\psi}(Z) | \psi(Z) - \hat{\psi}(Z) \rangle + \langle \hat{\psi}(Z) - \varphi(X) | \hat{\psi}(Z) - \varphi(X) \rangle \\ &\quad + 2\langle \psi(Z) - \hat{\psi}(Z) | \hat{\psi}(Z) - \varphi(X) \rangle \end{aligned} \quad (2.10)$$

The cross term vanishes by the tower property of conditional expectation:

$$\begin{aligned} \langle \psi(Z) - \hat{\psi}(Z) | \hat{\psi}(Z) - \varphi(X) \rangle &= \mathbb{E} \left[(\psi(Z) - \hat{\psi}(Z))^\top (\hat{\psi}(Z) - \varphi(X)) \right] \\ &= \iint (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \varphi(x)) p(dx, dz) \\ &= \iint (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \varphi(x)) p(dx|z) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \left(\int (\hat{\psi}(z) - \varphi(x)) p(dx|z) \right) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \left(\hat{\psi}(z) - \mathbb{E}[\varphi(X)|Z = z] \right) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \times 0 p(dz) \\ &= 0 \end{aligned} \quad (2.11)$$

by definition of $\hat{\psi}(z)$.

Therefore:

$$\begin{aligned} \rho(\psi(Z), \varphi(X))^2 &= \rho(\psi(Z), \hat{\psi}(Z))^2 + \rho(\hat{\psi}(Z), \varphi(X))^2 \\ &\geq \rho(\hat{\psi}(Z), \varphi(X))^2 \end{aligned} \quad (2.12)$$

with equality if and only if $\|\psi(Z) - \hat{\psi}(Z)\|^2 = 0$, i.e. $\psi = \hat{\psi}$ almost surely (a.s.). $\square$

Remark 2.2 (Geometric interpretation). *The MMSE estimator is the orthogonal projection of $\varphi(X)$ onto $\mathcal{F}$, viewed as a closed subspace of $\mathcal{L}^2(\mathbb{R}^{d_\varphi})$, i.e. the set of all square-integrable random variables that are measurable with respect to Z . The disappearance of the cross term in the proof above is precisely the orthogonality condition characterising this projection.*

Remark 2.3 (MMSE assumptions). *Square-integrability of $\varphi(X)$ is only required for the minimisation (2.8) to be well-posed, not the estimator itself. The posterior expectation $\mathbb{E}[\varphi(X)|Z = z]$ remains well-defined for any $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$, but it only minimises the mean squared error inside of $\mathcal{L}^2(\mathbb{R}^{d_\varphi})$.*

2.2.2 Posterior Median Estimator

Because the MMSE estimator minimises a quadratic loss, it is highly sensitive to extreme values and may perform poorly for heavy-tailed distributions. To mitigate this sensitivity to outliers, one can use an absolute loss instead. In dimension $d_\varphi = 1$, let:

$$\mathcal{F} \triangleq \{\psi \in \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}) \mid \mathbb{E}[|\psi(Z)|] < \infty\} \quad (2.13)$$

As previously, $\mathcal{F}$ is identified as a subspace of $\mathcal{L}^1(\mathbb{R})$, and is equipped with the following non-Euclidean distance:

$$(A, B) \in \mathcal{F}^2 \mapsto \rho(A, B) \triangleq \mathbb{E}[|A(Z) - B(Z)|] \quad (2.14)$$

The minimisation problem (2.4) thus becomes:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\mathbb{E}[|\psi(Z) - \varphi(X)|]) \quad (2.15)$$

While the terminology is less standard than MMSE, an estimator solving (2.15) is often called a minimum mean absolute error (MMAE) estimator.

Proposition 2.2 (MMAE estimator, simplified assumptions). *Assume that, for every z , the conditional distribution of $\varphi(X)$ given $Z = z$ is absolutely continuous with a strictly positive density. Then, the unique solution to (2.15) is the posterior median of $\varphi(X)$ given $Z = z$, i.e. the value $\hat{\psi}_{\text{MMAE}}(z) \in \mathbb{R}$ that satisfies:*

$$\mathbb{P}(\varphi(X) \geq \hat{\psi}_{\text{MMAE}}(z) | Z = z) = \mathbb{P}(\varphi(X) < \hat{\psi}_{\text{MMAE}}(z) | Z = z) = \frac{1}{2} \quad (2.16)$$

Proof. By the law of total expectation, for any $\psi \in \mathcal{F}$, we have:

$$\mathbb{E}[|\psi(Z) - \varphi(X)|] = \mathbb{E} \left[\int |\psi(z) - \varphi(x)| p(dx | Z = z) \right] \quad (2.17)$$

Since the outer expectation is over non-negative quantities, this is minimised by minimising the inner integral pointwise for each z , i.e. by solving:

$$\operatorname{argmin}_{u \in \mathbb{R}} \int |\varphi(x) - u| p(dx | Z = z) \quad (2.18)$$

As $u \mapsto |\varphi(x) - u|$ is convex, so is the integral, hence any stationary point is a global minimum. Since the conditional law is absolutely continuous, the integral is differentiable in u . By differentiating this expression with respect to $u \in \mathbb{R}$, it follows that:

$$\begin{aligned} & \frac{d}{du} \left(\int |\varphi(x) - u| p(dx | Z = z) \right) = 0 \\ \iff & \frac{d}{du} \left(\int ((\varphi(x) - u) \mathbb{1}_{\varphi(x) \geq u}(x) - (\varphi(x) - u) \mathbb{1}_{\varphi(x) < u}(x)) p(dx | Z = z) \right) = 0 \\ \iff & \int (-\mathbb{1}_{\varphi(x) \geq u}(x) + \mathbb{1}_{\varphi(x) < u}(x)) p(dx | Z = z) = 0 \\ \iff & \mathbb{P}(\varphi(X) \geq u | Z = z) = \mathbb{P}(\varphi(X) < u | Z = z) \end{aligned} \quad (2.19)$$

Since the total probability is one, this is equivalent to $u = \hat{\psi}_{\text{MMAE}}(z)$ as defined in (2.16), with the strict positivity of the density ensuring uniqueness. $\square$

Remark 2.4 (Median assumptions). *The absolute continuity and positivity assumptions are only required for the clean statement above, not for the existence of a solution. Since the inner integral is convex, continuous and coercive in u , a minimum always exists for any integrable conditional distribution. In full generality, the set of solutions is the closed interval of medians:*

$$\left\{ u \in \mathbb{R} \left| \mathbb{P}(\varphi(X) \leq u | Z = z) \geq \frac{1}{2} \text{ and } \mathbb{P}(\varphi(X) \geq u | Z = z) \geq \frac{1}{2} \right. \right\} \quad (2.20)$$

The absence of atoms then guarantees that this interval is characterised by the equality (2.16), while the strict positivity of the density ensures that it reduces to a single point.

Remark 2.5 (Generalised median). *The extension to higher dimensions $d_\varphi > 1$ is non-trivial. The $\mathcal{L}^1(\mathbb{R}^{d_\varphi})$ median, also called geometric median, is the most common generalisation. In the remainder of this thesis, we focus on the MMSE and MAP estimators (presented below), which extend naturally to any dimension.*

2.2.3 Maximum *a Posteriori* Estimator

As they produce a single point estimate that may lie in a low-probability region, both the MMSE and the MMAE estimators may not be suited for multimodal posteriors. In such cases, a natural alternative is to select a maximum of the posterior density.

Let $\mathcal{F} \triangleq \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi})$, with no integrability constraint. The maximum *a posteriori* (MAP) estimator is defined as the global mode of the posterior density of $\varphi(X)$:

$$\hat{\psi}_{\text{MAP}} \triangleq \operatorname{argmax}_{\psi \in \mathcal{F}} (\mathbb{E} [p_{\varphi(X)|Z}(\psi(Z)|Z)]) \quad (2.21)$$

As the objective is separable over z , the maximisation can be carried out pointwise, yielding for each z :

$$\hat{\psi}_{\text{MAP}}(z) = \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{\varphi(X)|Z}(u|z)) \quad (2.22)$$

Using Bayes' theorem, this is equivalent to:

$$\hat{\psi}_{\text{MAP}}(z) = \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{Z|\varphi(X)}(z|u) p_{\varphi(X)}(u)) \quad (2.23)$$

since the normalising constant $p_Z(z)$ is independent of u .

Although the MAP estimator avoids low-density regions, it still reduces the posterior to a single point. In particular, in multimodal situations, only one mode is retained, while the remaining ones are discarded.

Remark 2.6 (MAP optimisation problem). *The MAP estimator can be seen as the solution of an optimisation problem of the form (2.4), associated with the threshold loss:*

$$\forall \varepsilon > 0, (A, B) \in \mathcal{F}^2 \mapsto \rho_\varepsilon(A, B) \triangleq \mathbb{1}_{\|A(Z) - B(Z)\| > \varepsilon} \quad (2.24)$$

in the limit $\varepsilon \rightarrow 0^+$:

$$\hat{\psi}_{\text{MAP}} = \lim_{\varepsilon \rightarrow 0^+} \operatorname{argmin}_{\psi \in \mathcal{F}} (\rho_\varepsilon(\psi(Z), \varphi(X))) \quad (2.25)$$

Maximum Likelihood Estimator

In the special case where no prior distribution on $\varphi(X)$ is assumed, (2.23) reduces to the maximum likelihood (ML) estimator:

$$\hat{\psi}_{\text{ML}}(z) \triangleq \underset{u \in \mathbb{R}^{d_\varphi}}{\operatorname{argmax}} (p_{Z|\varphi(X)}(z|u)) \quad (2.26)$$

As it does not take any prior information on $\varphi(X)$ into account, this estimator is not a Bayesian estimator and may be poorly suited when the available data is scarce. While it coincides with the MAP estimator under a uniform prior, it will not be considered in the rest of this document.

2.3 Analytical Density Estimation

As established in section 2.2, all three estimators ultimately rely on the PDF $p(x_k|\mathbf{z}_{1:k})$, from which an estimate of $\varphi(x_k)$ can be derived. Point estimators provide a concise and directly interpretable representation of the state, which is often desirable for our applications. However, reducing the posterior density to a single point estimate may discard valuable information, particularly when it is heavy-tailed or multimodal.

The goal of the following section is thus to present algorithms that propagate the PDF itself at each discrete time step k in order to solve the inference problem presented by (2.1).

2.3.1 Optimal Bayesian Filter

In the Bayesian filtering framework, the PDF can be propagated recursively through two steps. However, each step relies on a specific conditional independence hypotheses as presented by Assumptions 2.1 and 2.2.

Assumption 2.1 (Markov property). *The state sequence $(x_k)_{k \geq 0}$ is a Markov chain. As such, it verifies:*

$$p(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) = p(x_k|x_{k-1}) \quad (2.27)$$

with $\mathbf{x}_{0:k} \triangleq (x_0, \dots, x_k)$.

Assumption 2.2 (Conditional observation independence). *The current observation z_k only depends on the current state x_k :*

$$p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) = p(z_k|x_k) \quad (2.28)$$

Both assumptions follow from the state-space model (2.1) and, in particular, from the whiteness of w_k and n_k and their independence from x_0 . Building on them, the optimal estimation algorithm computes the PDF in two steps: prediction and correction.

Prediction. The prior density at time k can be obtained by marginalising the previous PDF:

$$\begin{aligned} p(x_k|\mathbf{z}_{1:k-1}) &= \int p(x_k, x_{k-1}|\mathbf{z}_{1:k-1}) dx_{k-1} \\ &= \int p(x_k|x_{k-1}, \mathbf{z}_{1:k-1}) p(x_{k-1}|\mathbf{z}_{1:k-1}) dx_{k-1} \end{aligned} \quad (2.29)$$

Assumption 2.1 simplifies (2.29) into the Chapman-Kolmogorov equation [4]:

$$p(x_k | \mathbf{z}_{1:k-1}) = \int p(x_k | x_{k-1}) p(x_{k-1} | \mathbf{z}_{1:k-1}) dx_{k-1} \quad (2.30)$$

where $p(x_k | x_{k-1})$ is the evolution, or transition, density induced by the state equation $x_k = f_k(x_{k-1}, w_k)$ and the density $p_k^{(e)}$ of the evolution noise w_k .

Correction. Upon receiving the new observation z_k , Bayes' theorem gives:

$$p(x_k | \mathbf{z}_{1:k}) = \frac{p(z_k | x_k, \mathbf{z}_{1:k-1}) p(x_k | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \quad (2.31)$$

Applying Assumption 2.2 to (2.31) yields:

$$p(x_k | \mathbf{z}_{1:k}) = \frac{p(z_k | x_k) p(x_k | \mathbf{z}_{1:k-1})}{\int p(z_k | x_k) p(x_k | \mathbf{z}_{1:k-1}) dx_k} \quad (2.32)$$

where the likelihood $p(z_k | x_k)$ depends on the observation model $z_k = h_k(x_k, n_k)$ and the density $p_k^{(o)}$ of the observation noise n_k .

These two equations allow for a recursive algorithm to compute the PDF of the system at each time step. This algorithm, usually referred to as the optimal Bayesian filter [5], is given by Algorithm 2.1.

Algorithm 2.1 Optimal Bayesian Filter

Compute $p(x_k \mathbf{z}_{1:k-1})$ using (2.30)	▷ Prediction
Compute $p(x_k \mathbf{z}_{1:k})$ using (2.32)	▷ Correction

While theoretically optimal, this algorithm has no closed-form solution in general. The integrands in both (2.30) and (2.32) are analytically intractable for arbitrary f_k , h_k .

2.3.2 Kalman Filter

While the optimal estimation algorithm cannot be used in the form given by Algorithm 2.1, a closed-form solution can be obtained by introducing additional hypotheses on (2.1), known as the Kalman assumptions [6]. In this section, we assume that:

$$\left\{ \begin{array}{l} x_0 \sim \mathcal{N}(\bar{x}_0, P_0) \\ f_k : x_{k-1} \mapsto F_k x_{k-1} + b_k^{(e)} \\ w_k \sim \mathcal{N}(0, Q_k), \text{ independent of } x_0 \\ z_k \text{ is known} \\ h_k : x_k \mapsto H_k x_k + b_k^{(o)} \\ n_k \sim \mathcal{N}(0, R_k), \text{ independent of } x_0 \text{ and } w_k \end{array} \right. \quad (2.33)$$

where $F_k \in \mathcal{M}_{d_x, d_x}(\mathbb{R})$ and $H_k \in \mathcal{M}_{d_z, d_x}(\mathbb{R})$ are known matrices, and $b_k^{(e)} \in \mathbb{R}^{d_x}$ and $b_k^{(o)} \in \mathbb{R}^{d_z}$ are known bias vectors. $P_0 \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$ is the initial state covariance, and $Q_k \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$ and $R_k \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ are the evolution and observation noise covariance matrices. Here, $\mathcal{S}^{++}$ denotes the set of symmetric positive definite (SPD) matrices [7], as introduced by Definition 2.1.

Definition 2.1 (symmetric positive definite matrix). Let $n \in \mathbb{N}^*$ and $M \in \mathcal{M}_{n,n}(\mathbb{R})$. M is symmetric positive semi-definite (SPSD), denoted $M \succeq 0$, if it is:

$$\begin{cases} M^\top = M & \text{symmetric} \\ \forall a \in \mathbb{R}^n, a^\top M a \geq 0 & \text{positive semi-definite} \end{cases} \quad (2.34)$$

If the inequality is strict for every non-zero vector, then M is SPD. We write $\mathcal{S}_n^+(\mathbb{R})$ and $\mathcal{S}_n^{++}(\mathbb{R})$ respectively for the sets of PSD and SPD matrices of size n .

In particular, the state-space model (2.1) becomes:

$$\begin{cases} x_k = F_k x_{k-1} + b_k^{(e)} + w_k \\ z_k = H_k x_k + b_k^{(o)} + n_k \end{cases} \quad (2.35)$$

Finally, x_0 , $(w_k)_{k \geq 0}$ and $(n_k)_{k \geq 0}$ are assumed to be mutually independent. Under these assumptions, the true state x_k follows a Gaussian distribution at each time step as an affine transformation of Gaussian random variables.

Proposition 2.3 (Gaussian closure under affine assumptions). For $m, n \in \mathbb{N}$, let the index $m|n$ denote "at time m conditioned on the observations from 1 to n ". Under the hypotheses of (2.33), if $p(x_{k-1}|\mathbf{z}_{1:k-1}) = \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$, then:

$$\begin{cases} p(x_k|\mathbf{z}_{1:k-1}) = \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1}) \\ p(x_k|\mathbf{z}_{1:k}) = \mathcal{N}(\hat{x}_{k|k}, P_{k|k}) \end{cases} \quad (2.36)$$

where the predicted and corrected states $\hat{x}_{k|k-1}$ and $\hat{x}_{k|k}$, and covariances $P_{k|k-1} \succ 0$ and $P_{k|k} \succ 0$ are given by Algorithm 2.2.

Proof. See section B.1. □

The resulting algorithm, known as the Kalman filter [8], is depicted by Figure 2.1 and presented in Algorithm 2.2.

Algorithm 2.2 Kalman Filter

$\hat{x}_{k k-1} = F_k \hat{x}_{k-1 k-1} + b_k^{(e)}$	▷ State prediction
$P_{k k-1} = F_k P_{k-1 k-1} F_k^\top + Q_k$	▷ Covariance prediction
$y_k = z_k - (H_k \hat{x}_{k k-1} + b_k^{(o)})$	▷ Innovation
$S_k = H_k P_{k k-1} H_k^\top + R_k$	▷ Innovation covariance
$K_k = P_{k k-1} H_k^\top S_k^{-1}$	▷ Kalman gain
$\hat{x}_{k k} = \hat{x}_{k k-1} + K_k y_k$	▷ State correction
$P_{k k} = (I - K_k H_k) P_{k k-1}$	▷ Covariance correction

Remark 2.7 (Kalman state estimation). Under the Kalman assumptions, the PDF is Gaussian, and so every Bayesian estimator presented in section 2.2 coincides with $\hat{x}_{k|k}$:

$$\hat{\psi}_{\text{MMSE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\text{MMAE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k}) = \hat{x}_{k|k} \quad (2.37)$$

However, since the above equations use a predictive prior, the ML estimator does not generally correspond to the corrected state.

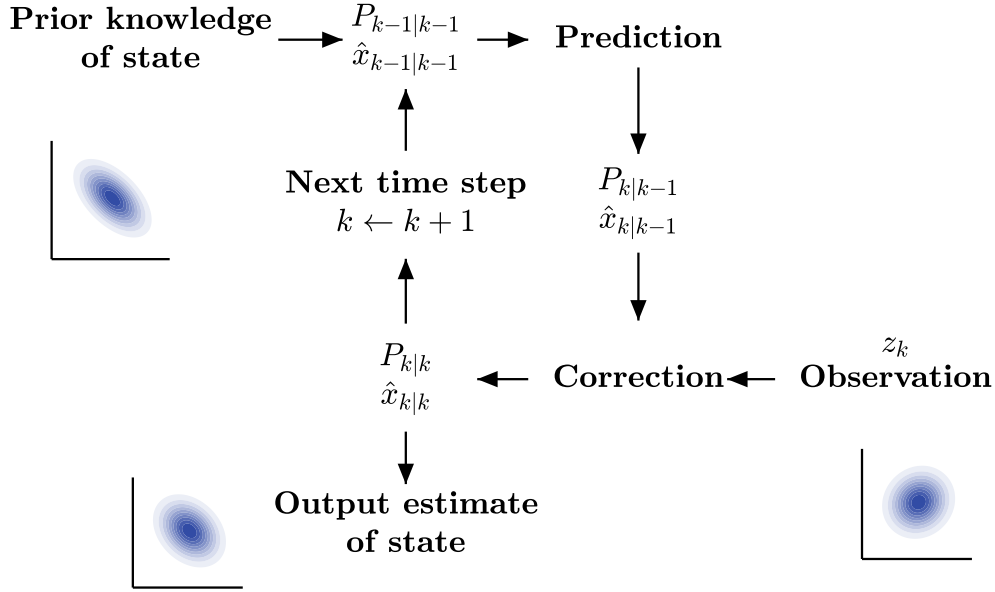


Figure 2.1: Schematic representation of the Kalman filter prediction/correction cycle. Each ellipsoid represents a confidence region associated with its corresponding Gaussian density.

Remark 2.8 (Precomputation of gains). *Under the assumptions of (2.33), the matrices $P_{k|k-1}$, K_k and $P_{k|k}$ depend only on the system matrices F_k , H_k , Q_k , R_k and the initial covariance P_0 , but not on the observations z_k . They can therefore be computed before any measurement is received.*

Joseph's Formula

In finite-precision arithmetic, the standard update $P_{k|k} = (I - K_k H_k) P_{k|k-1}$ may lose symmetry or positive definiteness due to rounding errors. Joseph's formula [9, 10] gives an alternative way to compute $P_{k|k}$ while ensuring that this property remains numerically satisfied:

$$\begin{aligned}
 P_{k|k} &= (I - K_k H_k) P_{k|k-1} \\
 &= (I - K_k H_k) P_{k|k-1} - K_k S_k K_k^\top + K_k S_k K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} - (P_{k|k-1} H_k^\top S_k^{-1}) S_k K_k^\top + K_k (H_k P_{k|k-1} H_k^\top + R_k) K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} - P_{k|k-1} H_k^\top K_k^\top + K_k H_k P_{k|k-1} H_k^\top K_k^\top + K_k R_k K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} - (I - K_k H_k) P_{k|k-1} H_k^\top K_k^\top + K_k R_k K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^\top + K_k R_k K_k^\top
 \end{aligned} \tag{2.38}$$

Under this more robust form, $P_{k|k}$ is SPSD as the sum of two SPSD matrices.

2.3.3 Extended Kalman Filter

The hypothesis that the evolution and observation functions are affine in x_k is often too restrictive in practice. The extended Kalman filter (EKF) [11] relaxes this assumption by locally linearising f_k and h_k around the current state estimate via a first-order Taylor expansion.

Let $m, n \in \mathbb{N}^*$, let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at some $a \in \mathbb{R}^n$. f admits the following first-order expansion:

$$\forall x \in \mathbb{R}^n, f(x) = f(a) + J_f(a) (x - a) + o(\|x - a\|_2) \quad (2.39)$$

where $J_f(a) \in \mathcal{M}_{m,n}(\mathbb{R})$ is the Jacobian matrix of f evaluated at a , and defined as:

$$J_f \triangleq \frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f^{(1)}}{\partial x^{(1)}} & \cdots & \cdots & \frac{\partial f^{(1)}}{\partial x^{(n)}} \\ \vdots & \ddots & \ddots & \vdots \\ \frac{\partial f^{(m)}}{\partial x^{(1)}} & \cdots & \cdots & \frac{\partial f^{(m)}}{\partial x^{(n)}} \end{bmatrix} \quad (2.40)$$

and $f^{(i)}$ and $x^{(j)}$ respectively denote the i -th and j -th component of f and x .

Assuming that f_k and h_k are differentiable along the state trajectory, we can apply the first-order expansion (2.39) to the evolution and observation model. Successive linearisations at each time step can then be used to apply the Kalman filter equations locally. The resulting algorithm is given by Algorithm 2.3.

Algorithm 2.3 Extended Kalman Filter

$\tilde{F}_k = J_{f_k}(\hat{x}_{k-1 k-1})$	▷ Evolution Jacobian
$\hat{x}_{k k-1} = f_k(\hat{x}_{k-1 k-1}) + b_k^{(e)}$	▷ State prediction
$P_{k k-1} = \tilde{F}_k P_{k-1 k-1} \tilde{F}_k^\top + Q_k$	▷ Covariance prediction
$\tilde{H}_k = J_{h_k}(\hat{x}_{k k-1})$	▷ Observation Jacobian
$y_k = z_k - \left(h_k(\hat{x}_{k k-1}) + b_k^{(o)} \right)$	▷ Innovation
$S_k = \tilde{H}_k P_{k k-1} \tilde{H}_k^\top + R_k$	▷ Innovation covariance
$K_k = P_{k k-1} \tilde{H}_k^\top S_k^{-1}$	▷ Kalman gain
$\hat{x}_{k k} = \hat{x}_{k k-1} + K_k y_k$	▷ State correction
$P_{k k} = (I - K_k \tilde{H}_k) P_{k k-1}$	▷ Covariance correction

While less restrictive than the standard Kalman filter, the EKF is suboptimal for three reasons:

- First, f_k and h_k must be differentiable along the state trajectory, which excludes discontinuous or piecewise-defined models.
- Second, the linearisation in (2.39) neglects higher-order terms of the Taylor expansion, which introduces an approximation error that grows with the degree of nonlinearity of f_k and h_k . This error increases over time as it propagates through the equations of the Kalman filter due to the sequential formulation.
- Finally, it relies on Gaussian assumptions and thus fails to handle multimodal or heavy-tailed distributions, as does its affine counterpart.

2.3.4 Other Suboptimal Kalman Filters

Iterative and Higher-Order EKF

The iterative EKF (IEKF) [12] improves upon the EKF correction step by iteratively re-linearising h_k around successive state estimates within the same measurement update, while keeping the prediction step unchanged. Starting from the initialisation $\hat{x}_{k|k}^{(0)} = \hat{x}_{k|k-1}$, each iteration produces a new $\hat{x}_{k|k}^{(i)}$ by repeating the correction step from the predicted state, with the Jacobian evaluated at the latest estimate. This is done until:

$$\left\| \hat{x}_{k|k}^{(i)} - \hat{x}_{k|k}^{(i-1)} \right\|_2 \leq \varepsilon \quad (2.41)$$

for some $\varepsilon > 0$, or a maximum number of iterations is reached. The corrected state and covariance matrix are then defined as the result of the last loop.

This procedure is equivalent to performing a Gauss-Newton descent [13] on the posterior density maximisation problem, and converges to the MAP estimator under Gaussian assumptions. It reduces the linearisation error of the correction step when h_k is strongly non-linear, at the cost of several Jacobian evaluations per time step.

The second-order EKF (SO-EKF) [14] can also be used to extend the linearisation (2.39) to second-order terms via the Hessians of f_k and h_k , which reduces the approximation error for highly nonlinear models. However, it requires computing and storing the Hessian matrices of f_k and h_k at each time step, making it computationally prohibitive in high dimensions.

Unscented Kalman Filter

Rather than linearising the model, the unscented Kalman filter (UKF) [15] propagates a small deterministic set of points through the non-linear functions themselves. Let $d \in \mathbb{N}^*$, and let X be a random vector of $\mathbb{R}^d$ with mean μ and covariance $P \succ 0$. The $2d + 1$ sigma points associated with X , and their weights, are:

$$\begin{cases} \mathcal{X}^{(0)} \triangleq \mu, & W^{(0)} \triangleq \frac{\lambda}{d + \lambda} \\ \mathcal{X}^{(\pm i)} \triangleq \mu \pm \sqrt{d + \lambda} (A)_i, & W^{(\pm i)} \triangleq \frac{1}{2(d + \lambda)}, i \in \llbracket 1, d \rrbracket \end{cases} \quad (2.42)$$

where A is a Cholesky factor [16] of P , $(A)_i$ is its i -th column, and $\lambda > -d$ is a scaling parameter. In particular, the weights are independent of X and add up to 1.

For a function φ , the first and second moments of $\varphi(X)$ can then be approximated by the weighted moments of the images of the sigma points:

$$\begin{cases} \hat{\varphi} \triangleq \sum_{i=-d}^d W^{(i)} \varphi(\mathcal{X}^{(i)}) \\ \hat{P} \triangleq \sum_{i=-d}^d W^{(i)} (\varphi(\mathcal{X}^{(i)}) - \hat{\varphi}) (\varphi(\mathcal{X}^{(i)}) - \hat{\varphi})^\top \end{cases} \quad (2.43)$$

This procedure is known as the unscented transform [17], and is applied twice per time step by the UKF. It is first used with $\mu = \hat{x}_{k-1|k-1}$, $P = P_{k-1|k-1}$ and $\varphi = f_k$

to compute the predicted state, then with $\mu = \hat{x}_{k|k-1}$, $P = P_{k|k-1}$ and $\varphi = h_k$ to obtain the predicted observation and thus the innovation. In both cases, Q_k and R_k are added to the transformed covariance $\hat{P}$, as in Algorithm 2.3, to account for evolution and observation uncertainty.

This removes the linearisation shortcomings of the EKF, at the cost of $2d_x + 1$ evaluations of f_k and h_k per time step. A square-root formulation [18] propagates a factor of the covariance rather than the covariance itself, which preserves its symmetry and positive definiteness throughout the recursion. However, the UKF still relies on the Gaussian assumption, the posterior remaining summarised by a mean and a covariance matrix.

Ensemble Kalman Filter

The ensemble Kalman filter (EnKF) [19] avoids both the local linearisation of the EKF and the deterministic sigma-point approximation of the UKF. Instead, the state distribution is represented by a set of $N \in \mathbb{N}^*$ samples $(x_{k|k}^{(1)}, \dots, x_{k|k}^{(N)})$, which are propagated through the true evolution and observation functions.

For two sets $\mathbf{X} \triangleq (X^{(1)}, \dots, X^{(N)})$ and $\mathbf{Z} \triangleq (Z^{(1)}, \dots, Z^{(N)})$, we let:

$$\left\{ \begin{array}{l} \hat{X} \triangleq \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \hat{Z} \triangleq \frac{1}{N} \sum_{i=1}^N Z^{(i)} \\ \hat{P}_{\mathbf{XZ}} \triangleq \frac{1}{N-1} \sum_{i=1}^N \left(X^{(i)} - \hat{X} \right) \left(Z^{(i)} - \hat{Z} \right)^\top \end{array} \right. \quad (2.44)$$

Each predicted sample $x_{k|k-1}^{(i)}$ is obtained by propagating the corresponding sample $x_{k-1|k-1}^{(i)}$ through f_k . The predicted observation $z_k^{(i)}$ is then computed using h_k . The approximate Kalman gain is defined as:

$$\hat{K}_k \triangleq \hat{P}_{\mathbf{X}_{k|k-1} \mathbf{Z}_k} \left(\hat{P}_{\mathbf{Z}_k \mathbf{Z}_k} + R_k \right)^{-1} \quad (2.45)$$

where $\mathbf{X}_{k|k-1}$ is the predicted sample set, and $\mathbf{Z}_k$ is the predicted observation set. Finally, each corrected sample $x_{k|k}^{(i)}$ is computed from its associated predicted sample through a standard Kalman update, using $\hat{K}_k$ in place of the exact gain.

The computational cost of the EnKF is primarily determined by the value of N instead of the dimensions d_x and d_z . It is therefore particularly attractive for high-dimensional systems, such as those encountered in meteorological and oceanographic data assimilation [20].

It should be noted that, while the EnKF is more flexible than its counterparts, the correction step remains a Kalman update based on the empirical mean and covariance matrix of the sample set. As a result, it may provide a poor approximation when the PDF is strongly non-Gaussian.

2.4 Monte Carlo Methods

While Kalman-based extensions can alleviate some of the limitations of the original Kalman filter, they all enforce the PDF to be Gaussian at each time step. This

assumption is often violated in practice, as navigation problems frequently involve highly non-linear and non-injective functions. In turn, these functions induce multimodal or heavy-tailed distributions, which cannot be adequately characterised by Gaussian hypotheses.

Monte Carlo (MC) methods [21, 22] offer an alternative approach that remains consistent under very mild assumptions. Their accuracy improves asymptotically with the number of samples, at the expense of an increased computational cost. This section introduces the main concepts for MC-based algorithms to solve the navigation problem (2.1).

2.4.1 Monte Carlo Integration

Even though MC methods originally stem from the computation of expectations, they approximate the entire distribution [23], from which every estimator of section 2.2 can be derived. Being itself an expectation, the MMSE estimator is simply the most immediate, and is where we begin.

Let X be a random variable taking values in $\mathbb{R}^{d_x}$ with probability density p , and let $\varphi \in \mathcal{L}^0(\mathbb{R}^{d_x}, \mathbb{R}^{d_\varphi})$ such that $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$. We wish to compute the expectation of $\varphi(X)$:

$$\mathbb{E}_p[\varphi(X)] = \int \varphi(x) p(x) dx \quad (2.46)$$

Let $\mu(dx) \triangleq p(x) dx$ be the probability measure with density p with respect to the Lebesgue measure. (2.46) can be rewritten as:

$$\langle \mu, \varphi \rangle \triangleq \int \varphi(x) \mu(dx) = \mathbb{E}_p[\varphi(X)] \quad (2.47)$$

Since μ is a continuous measure, evaluating (2.47) exactly is intractable in most practical situations. The MC method aims to approximate μ by an empirical measure induced by N independent samples, as presented by Definition 2.2 and Proposition 2.4.

Definition 2.2 (Empirical measure). *Let $N \in \mathbb{N}^*$ and $X^{(1)}, \dots, X^{(N)}$ be independent and identically distributed (i.i.d.) random variables drawn from μ . The empirical measure associated with $(X^{(i)})_{i \in [1, N]}$ is defined as:*

$$S^N(\mu) \triangleq \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}} \quad (2.48)$$

where δ_x denotes the Dirac measure at x .

Proposition 2.4 (Monte Carlo estimator). *The MC estimator:*

$$\langle S^N(\mu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \langle \delta_{X^{(i)}}, \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \varphi(X^{(i)}) \quad (2.49)$$

is unbiased and consistent:

$$\mathbb{E}_p[\langle S^N(\mu), \varphi \rangle] = \mathbb{E}_p[\varphi(X)] = \langle \mu, \varphi \rangle \quad (2.50)$$

$$\langle S^N(\mu), \varphi \rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \quad (2.51)$$

If moreover $\varphi(X) \in \mathcal{L}^2(\mathbb{R}^{d_\varphi})$, its mean squared error is:

$$\mathbb{E}_p \left[\left| \langle S^N(\mu) - \mu, \varphi \rangle \right|^2 \right] = \frac{1}{N} \text{tr } \mathbb{V}_\mu[\varphi] \quad (2.52)$$

where $\mathbb{V}_\mu[\varphi] \triangleq \langle \mu, (\varphi - \langle \mu, \varphi \rangle)(\varphi - \langle \mu, \varphi \rangle)^\top \rangle$ is the covariance matrix of φ under μ , and its rescaled error converges in distribution to a centred Gaussian:

$$\sqrt{N} \langle S^N(\mu) - \mu, \varphi \rangle \xrightarrow{\mathcal{L}} \mathcal{N}(0, \mathbb{V}_\mu[\varphi]) \quad (2.53)$$

Proof. By linearity of the expectation and the transfer theorem, it comes that:

$$\begin{aligned} \mathbb{E}_p [\langle S^N(\mu), \varphi \rangle] &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_p [\varphi(X^{(i)})] \\ &= \frac{1}{N} \sum_{i=1}^N \langle \mu, \varphi \rangle \\ &= \langle \mu, \varphi \rangle \end{aligned} \quad (2.54)$$

Furthermore, by independence of the $X^{(i)}$:

$$\begin{aligned} \mathbb{E}_p \left[\left| \langle S^N(\mu) - \mu, \varphi \rangle \right|^2 \right] &= \mathbb{E}_p \left[\left| \frac{1}{N} \sum_{i=1}^N (\varphi(X^{(i)}) - \langle \mu, \varphi \rangle) \right|^2 \right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}_p \left[\left| \varphi(X^{(i)}) - \langle \mu, \varphi \rangle \right|^2 \right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E} \left[\left| \varphi(X) - \langle \mu, \varphi \rangle \right|^2 \right] \\ &= \frac{1}{N} \text{tr } \mathbb{V}_\mu[\varphi] \end{aligned} \quad (2.55)$$

Finally, the a.s. convergence and the convergence in distribution follow respectively from the strong law of large numbers and the central limit theorem, applied to the i.i.d. variables $\varphi(X^{(i)})$. $\square$

Informally, this means that:

$$S^N(\mu) = \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}} \approx \mu \quad (2.56)$$

in the sense that, for any measurable function φ such that $\varphi(X)$ is integrable:

$$\langle S^N(\mu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \varphi(X^{(i)}) \approx \langle \mu, \varphi \rangle \quad (2.57)$$

2.4.2 Importance Sampling

The standard MC method requires sampling directly from μ , which can be complex in practice. Importance sampling [24] circumvents this issue by drawing from a chosen importance, or proposal, distribution ν with density q .

Assuming that $\text{supp}(q) \subseteq \text{supp}(p)$, and for $x \in \text{supp}(q)$, it follows from the identity $p(x) = \frac{p(x)}{q(x)} q(x)$ and the transfer theorem that:

$$\mathbb{E}_p[\varphi(X)] = \int \left(\varphi(x) \frac{p(x)}{q(x)} \right) q(x) dx = \mathbb{E}_q \left[\varphi(X) \frac{p(X)}{q(X)} \right] \quad (2.58)$$

Using measure theory notations, this is equivalent to writing:

$$\langle \mu, \varphi \rangle = \left\langle \nu, \frac{d\mu}{d\nu} \varphi \right\rangle \quad (2.59)$$

The density $\frac{d\mu}{d\nu}$ is known as the Radon-Nikodym derivative. This motivates the introduction of the weighted empirical measure, given by Definition 2.3.

Definition 2.3 (Weighted empirical measure). *Let $\nu(dx) = q(x) dx$ with $\text{supp}(\mu) \subseteq \text{supp}(\nu)$, and let $X^{(1)}, \dots, X^{(N)} \stackrel{i.i.d.}{\sim} \nu$. The associated weighted empirical measure is:*

$$S^N(\mu, \nu) \triangleq \frac{d\mu}{d\nu} S^N(\nu) = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \delta_{X^{(i)}} \quad (2.60)$$

Pairing this measure with a test function φ yields the weighted MC estimator of Proposition 2.5.

Proposition 2.5 (Weighted MC estimator). *The weighted MC estimator:*

$$\langle S^N(\mu, \nu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \varphi(X^{(i)}) \quad (2.61)$$

is unbiased and consistent:

$$\mathbb{E}_p [\langle S^N(\mu, \nu), \varphi \rangle] = \mathbb{E}_p[\varphi(X)] \quad (2.62)$$

$$\langle S^N(\mu, \nu), \varphi \rangle \xrightarrow{a.s.} \mathbb{E}_p[\varphi(X)] \quad (2.63)$$

Furthermore, when $\mathbb{V}_\nu \left[\frac{d\mu}{d\nu} \varphi \right] < \infty$, its mean squared error and rescaled error satisfy:

$$\mathbb{E}_q \left[|\langle S^N(\mu, \nu) - \mu, \varphi \rangle|^2 \right] = \frac{1}{N} \text{tr} \mathbb{V}_\nu \left[\frac{d\mu}{d\nu} \varphi \right] \quad (2.64)$$

$$\sqrt{N} \langle S^N(\mu, \nu) - \mu, \varphi \rangle \xrightarrow{\mathcal{L}} \mathcal{N} \left(0, \mathbb{V}_\nu \left[\frac{d\mu}{d\nu} \varphi \right] \right) \quad (2.65)$$

Proof. This is direct using the results of Proposition 2.4. □

For $\varphi \equiv 1$, the total mass:

$$\langle S^N(\mu, \nu), 1 \rangle = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \quad (2.66)$$

converges a.s. to 1, but each finite sample is not necessarily normalised. Furthermore, p and q are often known only up to normalising constants $\tilde{p} = C_\mu p$ and $\tilde{q} = C_\nu q$. To alleviate these issues, Definition 2.4 introduces the self-normalised measure [25], whose estimator remains consistent despite these unknown constants, as proven by Proposition 2.6.

Definition 2.4 (Self-normalised measure). *Let $X^{(1)}, \dots, X^{(N)} \stackrel{i.i.d.}{\sim} \nu$, with $\tilde{\mu} = C_\mu \mu$ and $\tilde{\nu} = C_\nu \nu$. The self-normalised measure is:*

$$\tilde{S}^N(\mu, \nu) \triangleq \sum_{i=1}^N w^{(i)} \delta_{X^{(i)}} \quad (2.67)$$

$$w^{(i)} \triangleq \tilde{w}^{(i)} \left/ \sum_{j=1}^N \tilde{w}^{(j)} \right. \quad (2.68)$$

$$\tilde{w}^{(i)} \triangleq \frac{\tilde{p}(X^{(i)})}{\tilde{q}(X^{(i)})} = \frac{d\tilde{\mu}}{d\tilde{\nu}}(X^{(i)}) \quad (2.69)$$

Proposition 2.6 (Self-normalised estimator). *The self-normalised estimator:*

$$\langle \tilde{S}^N(\mu, \nu), \varphi \rangle = \sum_{i=1}^N w^{(i)} \varphi(X^{(i)}) = \frac{\langle S^N(\tilde{\mu}, \tilde{\nu}), \varphi \rangle}{\langle S^N(\tilde{\mu}, \tilde{\nu}), 1 \rangle} \quad (2.70)$$

is biased at finite N , but consistent as soon as $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$ and $\text{supp}(\mu) \subseteq \text{supp}(\nu)$:

$$\langle \tilde{S}^N(\mu, \nu), \varphi \rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \quad (2.71)$$

Proof. Although the bias of $\tilde{S}^N(\mu, \nu)$ admits no simple closed-form expression in general, it can be computed for $N = 1$. In this situation, the self-normalisation forces the single weight to 1, such that:

$$\langle \tilde{S}^1(\mu, \nu), \varphi \rangle = \varphi(X^{(1)}) \quad (2.72)$$

and thus:

$$\mathbb{E} \left[\langle \tilde{S}^1(\mu, \nu), \varphi \rangle \right] = \langle \nu, \varphi \rangle \neq \langle \mu, \varphi \rangle \quad (2.73)$$

in general.

Furthermore, applying the strong law of large numbers separately to both the numerator and denominator of the self-normalised estimator yields:

$$\langle S^N(\tilde{\mu}, \tilde{\nu}), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \tilde{w}^{(i)} \varphi(X^{(i)}) \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}} \varphi \right\rangle = \frac{C_\mu}{C_\nu} \langle \mu, \varphi \rangle \quad (2.74)$$

and:

$$\langle S^N(\tilde{\mu}, \tilde{\nu}), 1 \rangle = \frac{1}{N} \sum_{j=1}^N \tilde{w}^{(j)} \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}} \right\rangle = \frac{C_\mu}{C_\nu} \langle \mu, 1 \rangle = \frac{C_\mu}{C_\nu} \neq 0 \quad (2.75)$$

Combining both expressions finally gives the desired result. $\square$

2.4.3 Sequential Importance Sampling

In the filtering context, the target distribution is the joint posterior $p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ over the entire state trajectory. As new observations arrive, this distribution grows in dimension, making a static importance sampling approach impractical. Sequential importance sampling (SIS) [26, 27] addresses this by updating the weights recursively rather than re-approximating the full trajectory at every time step.

Assumption 2.3 (SIS hypotheses). *In addition to Assumptions 2.1 and 2.2, we assume that the proposal density of the previous state does not depend on the current observation:*

$$q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k}) = q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \quad (2.76)$$

This assumption enables the following factorisation:

$$\begin{aligned} q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) &= q(x_k, \mathbf{x}_{0:k-1}|\mathbf{z}_{1:k}) \\ &= q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k}) \\ &= q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \end{aligned} \quad (2.77)$$

While Assumption 2.3 is sufficient to derive the SIS algorithm, an additional assumption is often considered:

$$q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) = q(x_k|x_{k-1}, z_k) q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \quad (2.78)$$

so that a new particle $x_k^{(i)}$ can be propagated from $x_{k-1}^{(i)}$ and z_k alone.

Using Assumptions 2.1 and 2.2 at time k , the PDF factorises as:

$$\begin{aligned} p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) &= p(\mathbf{x}_{0:k}|z_k, \mathbf{z}_{1:k-1}) \\ &= \frac{p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(x_k, \mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k|x_k) p(x_k|x_{k-1}) p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &\propto p(z_k|x_k) p(x_k|x_{k-1}) p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \end{aligned} \quad (2.79)$$

Let $\mathbf{x}_{0:k}^{(i)}$ denote the sequence $(x_0^{(i)}, \dots, x_k^{(i)})$. The recursive weight update is obtained by combining (2.79) with Assumption 2.3:

$$\begin{aligned} w_k^{(i)} &\propto \frac{\tilde{p}(\mathbf{x}_{0:k}^{(i)}|\mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k}^{(i)}|\mathbf{z}_{1:k})} \\ &\propto \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)}) \tilde{p}(\mathbf{x}_{0:k-1}^{(i)}|\mathbf{z}_{1:k-1})}{\tilde{q}(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}) \tilde{q}(\mathbf{x}_{0:k-1}^{(i)}|\mathbf{z}_{1:k-1})} \\ &\propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})} \end{aligned} \quad (2.80)$$

with the further simplification under the hypothesis of (2.78):

$$w_k^{(i)} \propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} \quad (2.81)$$

While the current formulation allows us to approximate the density of the full trajectory $p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})$ through the self-normalised measure of Definition 2.4, we often only want to estimate the current state x_k . This can be done naturally by letting π_k denote the projection of a $(k+1)$ -tuple onto its last element:

$$\pi_k : \begin{cases} (\mathbb{R}^{d_x})^{k+1} \rightarrow \mathbb{R}^{d_x} \\ (x_0, \dots, x_k) \mapsto x_k \end{cases} \quad (2.82)$$

and computing the self-normalised estimator for $\varphi = \pi_k$, which approximates the MMSE estimator of section 2.2:

$$\hat{x}_k \triangleq \left\langle \tilde{S}^N(\mu_k, \nu_k), \pi_k \right\rangle = \sum_{i=1}^N w_k^{(i)} x_k^{(i)} \approx \mathbb{E}_{p(\cdot | \mathbf{z}_{1:k})}[x_k] \quad (2.83)$$

where $\mu_k(dx) \triangleq \tilde{p}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) dx$ and $\nu_k(dx) \triangleq \tilde{q}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) dx$. The corresponding empirical covariance matrix is:

$$\hat{P}_k \triangleq \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top \quad (2.84)$$

The other estimators of section 2.2 remain accessible, though they are not as immediate:

- For a scalar state, the MMAE estimator only requires the cumulative distribution function, which $\tilde{S}^N(\mu_k, \nu_k)$ does define, and is obtained by accumulating the weights of the sorted particles until the total exceeds $\frac{1}{2}$. Its extension to a vector state is however not straightforward.
- The MAP estimator, on the other hand, calls for a density, which the particle approximation does not possess. In particular, $\hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k})$ cannot be computed using the weights, since they are importance ratios rather than density values. The heaviest particle maximises the Radon-Nikodym derivative rather than the PDF itself, and maximising $\tilde{p}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})$ over the particles only locates a mode among the sampled locations.

As for the optimal and Kalman filters, the SIS algorithm follows the prediction/correction loop. The prediction step corresponds to the sampling of the new particle set through the proposal density, and is often called particle propagation. Conversely, the correction step is the weight update of (2.80). The SIS algorithm is given by Algorithm 2.4.

Remark 2.9 (Feynman-Kac distribution). *The recurrence (2.79) can be expanded:*

$$p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) \propto p(x_0) \prod_{j=1}^k p(x_j | x_{j-1}) \times \prod_{j=1}^k p(z_j | x_j) \quad (2.85)$$

Algorithm 2.4 Sequential Importance Sampling

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot x_{k-1}^{(i)}, z_k)$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k x_k^{(i)}) \tilde{p}(x_k^{(i)} x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})}$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation

to show that the PDF is a Feynman-Kac distribution [28, 29], with the transitions $p(x_j | x_{j-1})$ acting as the mutation and the likelihoods $p(z_j | x_j)$ as the selection function. Rather than being an artifice used to cancel the unknown constants, the self-normalisation introduced in Definition 2.4 is thus, more fundamentally, the normalisation of this distribution.

Numerical correction of the covariance matrix

For numerical reasons, the estimated covariance matrix $\hat{P}_k$ may not satisfy SPD properties in practice. While an analytic reformulation existed for the Kalman filter, the numerical nature of MC methods prevents a similar solution. Formally, we want to find the closest matrix $M_+^* \succ 0$ to $\hat{P}_k$ in the sense of the Frobenius norm:

$$A \in \mathcal{M}_{d_x, d_x}(\mathbb{R}) \mapsto \|A\|_F \triangleq \sqrt{\sum_{i=1}^{d_x} \sum_{j=1}^{d_x} A_{i,j}^2} \quad (2.86)$$

If $d_x = 1$, the positive real $m_+ \in \mathbb{R}_+$ closest to some real $m \in \mathbb{R}$ is:

$$m_+ = \frac{m + |m|}{2} \quad (2.87)$$

While this only applies to scalars, a similar operation can be performed in higher dimensions by using the SPSPD term of the polar decomposition. The singular values decomposition is used to simplify the computation of the polar decomposition.

Formally, the correction algorithm [30] is as follows:

- Symmetrisation of $\hat{P}_k$: $M \triangleq \frac{\hat{P}_k + \hat{P}_k^\top}{2}$
- Singular values decomposition: $M = U D V^\top$
- Polar decomposition: $M = \Omega \Sigma$ with $\Omega \triangleq U V^\top$ and $\Sigma \triangleq V D V^\top$
- Computation of M_+ : $M_+ \triangleq \frac{M + \Sigma}{2}$

With this algorithm, M_+ is ensured to be SPSPD. However, we wish to achieve strict positive definiteness. To solve this issue, we recall that a symmetric matrix is SPD if and only if all its eigenvalues are strictly positive, and we introduce a small correction $\varepsilon > 0$. Let $\lambda_{\min}$ denote the smallest eigenvalue of M_+ . Let:

$$\delta \triangleq \max(0, -\lambda_{\min}) + \varepsilon \quad (2.88)$$

where the first term is introduced to compensate any numerical errors introduced during the symmetrisation and SVD steps.

Since M_+ and δI commute, they are simultaneously diagonalisable, and the eigenvalues of:

$$M_+^* \triangleq M_+ + \delta I \quad (2.89)$$

are thus:

$$\lambda_i(M_+^*) = \lambda_i(M_+) + \delta \geq \lambda_{\min} + \delta > 0 \quad (2.90)$$

For small enough values of ε , M_+^* is the desired matrix.

2.4.4 Weight Degeneracy

A fundamental limitation of SIS is that the variance of the importance weights accumulates along the trajectory [31] as shown by Proposition 2.7.

Proposition 2.7 (Weight degeneracy). *Let $\mathbf{x}_{0:k} \sim \nu_k$, and let $\tilde{w}_k$ denote the non-normalised weight associated with the entire trajectory:*

$$\tilde{w}_k \triangleq \frac{\tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})} \quad (2.91)$$

Its variance decomposes as:

$$\mathbb{V}_{\nu_k}[\tilde{w}_k] = \mathbb{E}_{\nu_k}[\mathbb{V}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}]] + \mathbb{V}_{\nu_k}[\mathbb{E}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}]] \quad (2.92)$$

where the first term is non-negative, and vanishes if and only if $\tilde{w}_k$ is a.s. determined by $\mathbf{x}_{0:k-1}$.

Proof. (2.92) is the law of total variance. Being the expectation of a non-negative quantity, its first term vanishes if and only if $\mathbb{V}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}]$ is a.s. zero, that is if and only if $\tilde{w}_k$ is a.s. determined by $\mathbf{x}_{0:k-1}$. $\square$

Given $\mathbf{x}_{0:k-1}$, the only randomness remaining in (2.92) is the draw of x_k . The inner variance measures its effect, while the inner expectation integrates it out and leaves a function of $\mathbf{x}_{0:k-1}$. The first term therefore corresponds to the variance injected by x_k and the second is the variance inherited from the past.

As each step adds to the previous ones, the weight distribution typically concentrates on a small number of particles, and so (2.67) becomes a poor approximation of the PDF. While degeneracy occurs in most cases, Proposition 2.8 exhibits a proposal density that eliminates the additional variance introduced at the current step.

Proposition 2.8 (Optimal proposal density). *When the proposal density is the posterior transition:*

$$q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k|x_{k-1}, z_k) \quad (2.93)$$

the trajectory weight is determined by $\mathbf{x}_{0:k-1}$, so that the variance introduced at step k vanishes:

$$\mathbb{V}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}] = 0 \quad (2.94)$$

Proof. Plugging (2.93) into (2.80) yields:

$$\begin{aligned}
\tilde{w}_k &\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k|x_k) \tilde{p}(x_k|x_{k-1})}{\tilde{q}(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k})} \\
&\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k|x_k) \tilde{p}(x_k|x_{k-1})}{\tilde{p}(x_k|x_{k-1}, z_k)} \\
&\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k|x_k) \tilde{p}(x_k|x_{k-1})}{\tilde{p}(z_k|x_k, x_{k-1}) \tilde{p}(x_k|x_{k-1})} \\
&\quad \tilde{p}(z_k|x_{k-1}) \\
&\propto \tilde{w}_{k-1} \tilde{p}(z_k|x_{k-1}) \frac{\tilde{p}(z_k|x_k)}{\tilde{p}(z_k|x_k, x_{k-1})} \\
&\propto \tilde{w}_{k-1} \tilde{p}(z_k|x_{k-1})
\end{aligned} \tag{2.95}$$

Both factors being functions of $\mathbf{x}_{0:k-1}$ alone, $\tilde{w}_k$ is determined by the conditioning, and its conditional variance is therefore zero. $\square$

Using Proposition 2.8, the draw of x_k thus leaves $\tilde{w}_k$ untouched, but reaches $\tilde{w}_{k+1}$ through $\tilde{p}(z_{k+1}|x_k)$. Its contribution is deferred by one step rather than suppressed, and so this density delays the degeneracy rather than completely removing it. Furthermore, sampling from the posterior transition requires computing the marginal likelihood:

$$p(z_k|x_{k-1}^{(i)}) = \int p(z_k|x_k^{(i)}) p(x_k^{(i)}|x_{k-1}^{(i)}) dx_k^{(i)} \tag{2.96}$$

which is generally intractable.

A common alternative is to use the prior, or transition, density:

$$q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k|x_{k-1}) \tag{2.97}$$

which is easier to compute, but ignores the current observation during propagation.

The prior density also simplifies the weight update (2.80) to:

$$w_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k|x_k^{(i)}) \tag{2.98}$$

However, because the particles are propagated without regard to z_k , their incremental weight is simply the likelihood $\tilde{p}(z_k|x_k^{(i)})$. When this likelihood is peaked, most particles land where it is negligible. Only a few particles then carry most of the weight, which causes the degeneracy to accelerate. The prior is thus a less effective proposal in highly informative observation regimes.

2.4.5 Sampling Importance Resampling

Weight degeneracy is intrinsic to SIS and worsens with time unless the proposal distribution is chosen extremely close to the optimal proposal density. The sampling importance resampling (SIR) filter [32], also known as the bootstrap filter [33], mitigates this by introducing a resampling step after each weight update. Resampling replaces the weighted particle set $\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N$ by an unweighted set $\{x_k^{*(j)}, 1/N\}_{j=1}^N$ drawn according to a chosen resampling algorithm:

$$\{x_k^{*(j)}\}_{j=1}^N = \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right) \tag{2.99}$$

It also uses the transition density as proposal (2.97), which further reduces (2.98) to:

$$w_k^{(i)} \propto \tilde{p}(z_k | x_k^{(i)}) \quad (2.100)$$

since all weights are reset to $1/N$ after resampling. This makes the SIR applicable under very weak assumptions, only requiring the ability to sample from $p(\cdot | x_{k-1})$ and compute the likelihood $p(z_k | x_k)$. Its pseudo-code is given by Algorithm 2.5.

Algorithm 2.5 Sampling Importance Resampling

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot x_{k-1}^{(i)})$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = \tilde{p}(z_k x_k^{(i)})$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\{x_k^{(j)}, w_k^{(j)}, -\}_{j=1}^N \leftarrow \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right)$	▷ Resampling

Resampling Algorithms

Several resampling algorithms [34, 35] can be used in (2.99). We present three algorithms, which all select a new particle set from the particle approximation of the PDF:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} \delta_{x_k^{(i)}} \quad (2.101)$$

Drawing a particle from this discrete measure amounts to drawing its index $i \in \llbracket 1, N \rrbracket$ from the categorical distribution of the weights $(w_k^{(i)})_{i \in \llbracket 1, N \rrbracket}$ through its cumulative distribution function (CDF):

$$\begin{cases} \text{cdf}_i \triangleq \sum_{j=1}^i w_k^{(j)}, & i \in \llbracket 1, N \rrbracket \\ \text{cdf}_0 = 0 \end{cases} \quad (2.102)$$

As illustrated by Figure 2.2, a draw of $u_j \in [0, 1]$ uniquely determines the new particle $x_k^{*(j)}$ as the parent particle $x_k^{(i)}$ of index i satisfying:

$$\text{cdf}_{i-1} \leq u_j < \text{cdf}_i \quad (2.103)$$

The main difference between the three algorithms lies in how the N samples $\{u_j\}_{j \in \llbracket 1, N \rrbracket}$ are generated:

- **Multinomial resampling.** All the draws are independent:

$$u_j \stackrel{i.i.d.}{\sim} \mathcal{U}([0, 1]) \quad (2.104)$$

This is the most natural method, but has the highest resampling variance.

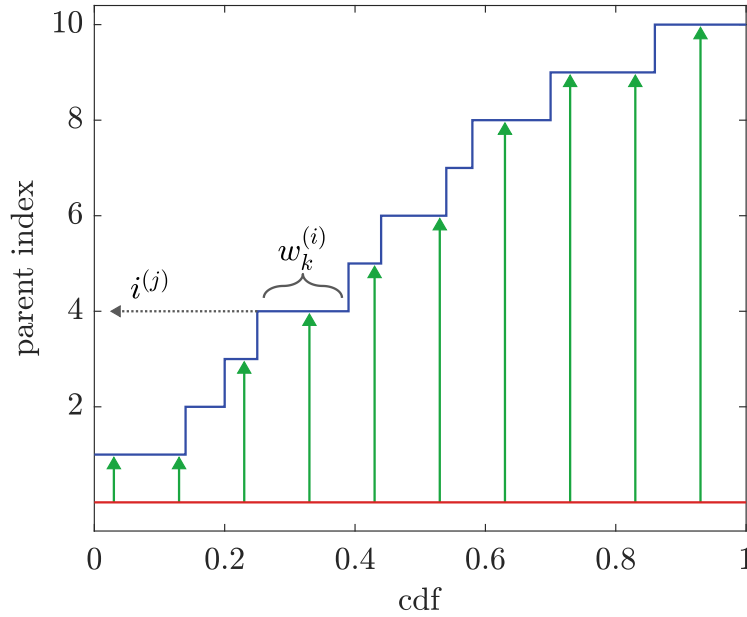


Figure 2.2: An example CDF of the particle weights. The width of each step is equal to the corresponding weight, so heavier particles are more likely to be selected.

- **Systematic or Kitagawa resampling [36].** A single draw generates all samples via:

$$\begin{cases} u_1 \sim \mathcal{U}([0, 1/N]) \\ u_j = u_1 + (j - 1)/N, j \in \llbracket 2, N \rrbracket \end{cases} \quad (2.105)$$

This significantly reduces the variance of the resampling step and is generally preferred in practice.

- **Deterministic resampling.** All samples are deterministic such that, for some $m \in [0, 1]$:

$$u_j = (j - m)/N, j \in \llbracket 1, N \rrbracket \quad (2.106)$$

This eliminates resampling variance entirely at the cost of a deterministic bias.

Some advanced filters, such as the auxiliary particle filter, presented in section 2.5, require keeping track of the parent index $i^{(j)}$ of each resampled particle. This value is thus recorded at the creation of each new particle. Algorithm 2.6 presents a linear implementation of the Kitagawa resampling algorithm, using the fact that both the CDF values and the Kitagawa samples are increasing to compute the parent indices sequentially.

Remark 2.10 (Loss of independence). *Because resampling duplicates particles, the resulting set is no longer independent, and the i.i.d. results of Propositions 2.4 and 2.5 no longer apply. However, the resampling is unbiased and the propagation step draws the particles independently conditionally on the past. The Feynman-Kac formalism proves [37, 38] that consistency and asymptotic Gaussianity still hold in this scenario, at the price of a variance larger than in the independent case and which accumulates over time.*

Algorithm 2.6 Kitagawa Resampling

```

cdf0 = 0                                ▷ Computation of the CDF
∀i ∈ [1, N], cdfi = cdfi-1 + wk(i)

u1 ~ U([0, 1/N])                        ▷ Kitagawa samples
∀j ∈ [2, N], uj = u1 + (j - 1)/N

i = 1
for j = 1 : N do
    while uj ≥ cdfi do                  ▷ Computation of the parent index
        i ← i + 1
    end while
    xk*(j) = xk(i)                        ▷ Generation of the new particle
    wk*(j) = 1/N
    i(j) = i
end for

```

2.5 Advanced Particle Filters

Even though the SIR applies under very mild assumptions, it has several caveats. Indeed, while resampling at every step eliminates the weight degeneracy of the SIS, it wastes computational load, and erodes particle diversity. Furthermore, the number of particles required grows quickly with the state dimension, a phenomenon known as the curse of dimensionality [39].

The SIR admits several refinements [40], each targeting a specific issue. We first present the generic particle filter, which reduces the SIR's computational cost by only resampling when necessary. We then introduce the regularised and auxiliary particle filters to restore particle diversity and improve on the prior proposal. Finally, we derive the equations of the Rao-Blackwellised particle filter, which reduces the dimension of the explored space under linear-Gaussian substructure assumptions.

2.5.1 Generic Particle Filter

The SIR resamples at every time step, which eliminates weight degeneracy but comes with two drawbacks:

- It introduces unnecessary computational cost when weights are already well-distributed, and increases the variance of the estimated state, as mentioned by Remark 2.10.
- It also replicates the heaviest particles and discards the lightest, so the set progressively concentrates on a few distinct locations. This loss of diversity, known as particle impoverishment [41], is especially severe in low evolution noise or high-dimensional settings, where little diversity is re-introduced at the propagation step. It is illustrated by Figure 2.3.

To limit both issues, the generic particle filter (PF) [42] resamples only when the weight distribution has degraded significantly. This degradation is measured by the

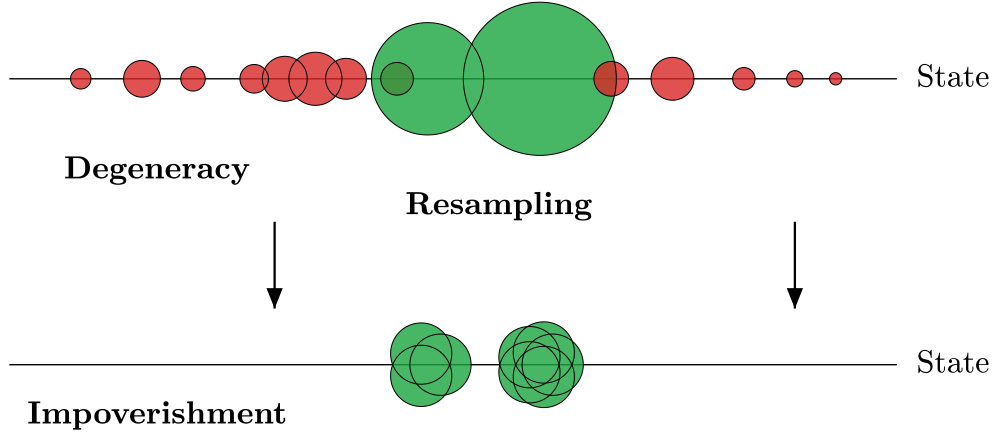


Figure 2.3: Particle impoverishment after resampling. Discarding non-resampled particles progressively shrinks the support of the particle approximation, degrading the filter’s ability to explore the state-space.

effective sample size:

$$N_{\text{eff}} \triangleq \frac{N}{1 + \mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})} \left[\frac{p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})} \right]} \quad (2.107)$$

Its denominator increases with the variance of the importance weights, whose growth over time corresponds to the weight degeneracy characterised by Proposition 2.7. When all the weights are uniform, the variance is zero, and thus $N_{\text{eff}} = N$. Conversely, the more unevenly they are distributed, the smaller N_{eff} , with $N_{\text{eff}} \rightarrow 1$ when the mass concentrates on a single particle.

Since the true importance weights are unknown, N_{eff} is estimated by replacing $\mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})}[\cdot]$ with the empirical spread of the normalised weights [43], which gives:

$$\hat{N}_{\text{eff}} \triangleq \frac{1}{\frac{1}{N} \sum_{i=1}^N \left(w_k^{(i)} \right)^2} \leq N \quad (2.108)$$

Like N_{eff} , this quantity ranges from N for uniform weights, down to 1 in the fully degenerate case. The PF then performs the resampling step only when this estimate falls below a threshold:

$$\hat{N}_{\text{eff}} < N_{\text{th}} \quad (2.109)$$

where N_{th} is a predetermined value, typically chosen proportional to N .

Unlike the SIR, which traditionally uses the transition density as its proposal distribution, the generic PF is not typically restricted to any particular choice of proposal distribution. It is presented by Algorithm 2.7.

2.5.2 Regularised Particle Filter

Since each resampling step still collapses diversity, the PF only mitigates impoverishment, but does not prevent it. Indeed, the self-normalised measure of Defini-

Algorithm 2.7 Generic Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot x_{k-1}^{(i)}, z_k)$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k x_k^{(i)}) \tilde{p}(x_k^{(i)} x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k})}$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N (w_k^{(i)})^2$	▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ then $\left\{ x_k^{(j)}, w_k^{(j)}, - \right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right)$ end if	▷ Resampling

tion 2.4 approximates the PDF by a mixture of Diracs:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} \delta_{x_k^{(i)}} \quad (2.110)$$

Since this measure is singular, resampling from it can only reproduce the existing particles. The regularised particle filter (RPF) [44] addresses this issue through kernel density estimation [45, 46, 47], by replacing each Dirac with a smooth kernel to obtain the absolutely continuous measure:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} K_h(x_k - x_k^{(i)}) dx_k \quad (2.111)$$

New particles are then drawn from this continuous density rather than from the discrete set, which restores diversity. This is illustrated by Figure 2.4 and formalised by Definition 2.5 and proposition 2.9. Being absolutely continuous, this approximation also supplies the density that (2.110) lacks, and thus makes the MAP estimator of section 2.2 well-defined at the cost of a numerical maximisation.

Definition 2.5 (Regularisation kernel). *Let $K : \mathbb{R}^{d_x} \rightarrow \mathbb{R}$. K is a regularisation kernel if:*

$$\left\{ \begin{array}{ll} \forall x \in \mathbb{R}^{d_x}, K(x) \geq 0 & \text{positive} \\ \forall x \in \mathbb{R}^{d_x}, K(-x) = K(x) & \text{symmetric} \\ \int K(x) dx = 1 & \text{normalised} \\ \int \|x\|^2 K(x) dx < \infty & \text{of finite variance} \end{array} \right. \quad (2.112)$$

Furthermore, for K a regularisation kernel and $h > 0$, we define the rescaled kernel K_h of bandwidth h as:

$$K_h : \left\{ \begin{array}{l} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ x \mapsto \frac{1}{h^{d_x}} K\left(\frac{x}{h}\right) \end{array} \right. \quad (2.113)$$

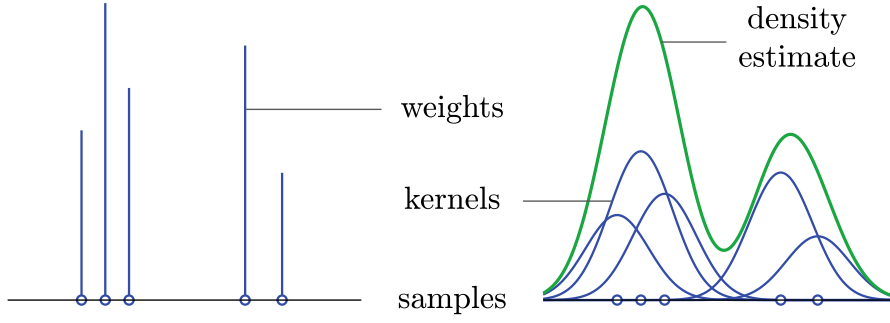


Figure 2.4: An example Monte Carlo approximation of the PDF (2.110) on the left and its kernel regularisation (2.111) on the right, for a given particle set. The latter approximation bridges the gaps between particles and spans a larger region of the state space.

Definition 2.6 (Regularised measure). *Let $*$ denote the convolution operator. The regularised measure $S_{K_h}^N$ of some empirical measure S^N with respect to a regularisation kernel K_h is:*

$$S_{K_h}^N(\mu) \triangleq K_h * S^N(\mu) \quad (2.114)$$

In particular, the regularised self-normalised measure verifies:

$$\tilde{S}_{K_h}^N(\mu, \nu) = \sum_{i=1}^N w^{(i)} (K_h * \delta_{X^{(i)}}) \quad (2.115)$$

Proposition 2.9 (Regularised estimator). *The regularised self-normalised estimator is:*

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle = \sum_{i=1}^N w^{(i)} (K_h * \varphi)(X^{(i)}) = \left\langle \tilde{S}^N(\mu, \nu), K_h * \varphi \right\rangle \quad (2.116)$$

It is a biased and inconsistent estimator in general:

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle \xrightarrow{a.s.} \langle \mu, K_h * \varphi \rangle \quad (2.117)$$

However, it becomes consistent when φ is linear. In particular:

$$\left\langle \tilde{S}_{K_h}^N(\mu_k, \nu_k), \pi_k \right\rangle \xrightarrow{a.s.} \langle \mu_k, \pi_k \rangle = \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k] \quad (2.118)$$

Proof. Let $i \in \llbracket 1, N \rrbracket$. By definition of $*$, we have:

$$(K_h * \delta_{X^{(i)}})(dx) = K_h(x - X^{(i)}) dx \quad (2.119)$$

and thus, by symmetry of K_h :

$$\begin{aligned} \langle K_h * \delta_{X^{(i)}}, \varphi \rangle &= \int \varphi(x) K_h(x - X^{(i)}) dx \\ &= \int \varphi(x) K_h(X^{(i)} - x) dx \\ &= (K_h * \varphi)(X^{(i)}) \end{aligned} \quad (2.120)$$

In turn, this result gives:

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle = \sum_{i=1}^N w^{(i)} (K_h * \varphi) (X^{(i)}) = \left\langle \tilde{S}^N(\mu, \nu), K_h * \varphi \right\rangle \quad (2.121)$$

and Proposition 2.6, applied to $K_h * \varphi$, yields the a.s. limit $\langle \mu, K_h * \varphi \rangle$.

Assume that φ is linear, that is $\varphi : x \mapsto A x$ for some $A \in \mathcal{M}_{d_\varphi, d_x}(\mathbb{R})$. Then, by normalisation and symmetry of K_h , for all $x \in \mathbb{R}^{d_x}$:

$$\begin{aligned} (K_h * \varphi)(x) &= \int K_h(y) \varphi(x - y) dy \\ &= \varphi(x) \int K_h(y) dy - A \int y K_h(y) dy \\ &= \varphi(x) \end{aligned} \quad (2.122)$$

Hence $K_h * \varphi = \varphi$, and the limit reduces to $\langle \mu, \varphi \rangle$. In particular, for the linear projection π_k , it becomes $\langle \mu_k, \pi_k \rangle = \mathbb{E}_{p(\cdot | \mathbf{z}_{1:k})}[\pi_k]$. $\square$

The choice of bandwidth h is critical to the performance of the RPF. If h is too small, K_h concentrates around the Dirac measure, and the regularisation has no practical effect on particle diversity. Conversely, the kernel density estimate over-smooths the posterior when h is too large, artificially spreading particles across regions of low probability, and degrading filter accuracy. In this section, we present two classical choices for K_h :

- **Epanechnikov kernel.** This kernel minimises the mean integrated squared error (MISE) among all regularisation kernels [48], and the associated bandwidth is optimal when the density is Gaussian with identity covariance. It is defined by:

$$\begin{cases} K_{\text{Epa}} : x \mapsto \frac{d_x + 2}{2c_{d_x}} (1 - \|x\|^2) \mathbb{1}_{\|x\| \leq 1}(x) \\ h_{\text{Epa}} = \left(\frac{8(d_x + 4)(2\sqrt{\pi})^{d_x}}{c_{d_x}} \right)^{\frac{1}{d_x + 4}} N^{-\frac{1}{d_x + 4}} \end{cases} \quad (2.123)$$

where:

$$c_{d_x} \triangleq \frac{\pi^{d_x/2}}{\Gamma(d_x/2 + 1)} \quad (2.124)$$

is the volume of the $\mathbb{R}^{d_x}$ unit ball, and Γ is the Gamma function.

- **Gaussian kernel.** While suboptimal with respect to the Epanechnikov kernel, it is simpler to compute [49]. Its expression is given by:

$$\begin{cases} K_{\text{Gauss}} : x \mapsto (2\pi)^{-d_x/2} \exp\left(-\frac{\|x\|^2}{2}\right) \\ h_{\text{Gauss}} = \left(\frac{4}{d_x + 2} \right)^{\frac{1}{d_x + 4}} N^{-\frac{1}{d_x + 4}} \end{cases} \quad (2.125)$$

It must be noted that the assumptions for the Epanechnikov, and consequently the Gaussian, regularisation are not usually met in practice, since MC methods are precisely used when the PDF is not Gaussian.

However, the two conditions do not play the same role. The Gaussian hypothesis is only used to compute the optimal bandwidth, whose expression would otherwise depend on an unknown PDF. It is thus a calibration convention rather than a modelling assumption. In particular, it tends to overestimate h when the PDF is multimodal.

On the other hand, the unit covariance is a scaling requirement, and can be enforced exactly by a whitening step before the regularisation. This amounts to replacing every particle by $A_k^{-1} x_k^{(i)}$, where A_k is a Cholesky factor of $\hat{P}_k$:

$$\hat{P}_k = A_k A_k^\top \quad (2.126)$$

which ensures that the covariance matrix of the new particle set is the identity. While seemingly computationally heavy, this reduces to using the following regularisation kernel on the standard particle set:

$$\tilde{K}_h : x \mapsto \frac{\det(A_k)^{-1}}{h^{d_x}} K\left(\frac{1}{h} A_k^{-1} x\right) \quad (2.127)$$

Moreover, the bandwidth h is multiplied by a modulation factor $\alpha_{\text{reg}} \in [0, 1]$ to avoid over-smoothing in cases where the PDF is multimodal. The pseudo-code for the RPF is given by Algorithm 2.8.

Algorithm 2.8 Regularised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q\left(\cdot x_{k-1}^{(i)}, z_k\right)$	$\triangleright$ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}\left(z_k x_k^{(i)}\right) \tilde{p}\left(x_k^{(i)} x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)} \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}\right)}$	$\triangleright$ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	$\triangleright$ Normalisation
$\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N \left(w_k^{(i)}\right)^2$	$\triangleright$ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ then	
$\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$	$\triangleright$ Whitening
$\hat{P}_k = \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right) \left(x_k^{(i)} - \hat{x}_k\right)^\top$	
$A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$	
$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^N\right)$	$\triangleright$ Resampling
$\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$	$\triangleright$ Regularisation
$\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$	
end if	

2.5.3 Auxiliary Particle Filter

The auxiliary particle filter (APF) [50, 51] addresses the degeneracy problem at its root by exploiting a different weakness of the SIR. Since particles are propagated

before the new observation z_k is taken into account, particles that will ultimately receive low likelihood waste computational effort. The APF thus introduces a look-ahead step by using a prediction of their future likelihood to resample promising parent particles before propagation.

Formally, the APF targets the joint distribution $p(x_k, i | \mathbf{z}_{1:k})$, where $i \in \llbracket 1, N \rrbracket$ indexes the parent particle at time $k-1$. Using Bayes' theorem and the chain rule gives:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k, i, \mathbf{z}_{1:k-1}) p(x_k | i, \mathbf{z}_{1:k-1}) p(i | \mathbf{z}_{1:k-1}) \quad (2.128)$$

Each term of this decomposition can be simplified as follows:

- Assumption 2.2 states that the current observation depends only on the current state:

$$p(z_k | x_k, i, \mathbf{z}_{1:k-1}) = p(z_k | x_k) \quad (2.129)$$

- Since the parent index uniquely determines the ancestor trajectory, conditioning on i is equivalent to conditioning on $\mathbf{x}_{0:k-1}^{(i)}$. Moreover, Assumption 2.1 establishes that $(x_k)_{k \geq 0}$ is a Markov chain, and thus:

$$p(x_k | i, \mathbf{z}_{1:k-1}) = p\left(x_k | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k-1}\right) = p\left(x_k | x_{k-1}^{(i)}\right) \quad (2.130)$$

- Finally, assuming that the particle approximation is exact gives:

$$p(i | \mathbf{z}_{1:k-1}) \propto w_{k-1}^{(i)} \quad (2.131)$$

Combining these equations reduces (2.128) to:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k) p\left(x_k | x_{k-1}^{(i)}\right) w_{k-1}^{(i)} \quad (2.132)$$

However, $p(z_k | x_k)$ cannot be evaluated before x_k is drawn. It is thus approximated by $p\left(z_k | \mu_k^{(i)}\right)$, where:

$$\mu_k^{(i)} \sim p\left(\cdot | x_{k-1}^{(i)}\right) \quad (2.133)$$

is an auxiliary particle drawn from the transition density, and used to assess the consistency of the future particle $x_k^{(i)}$ with the new observation z_k . Alternatively, $\mu_k^{(i)}$ can be chosen as the mean of this density, a deterministic choice which avoids an additional draw.

In any case, the importance density is therefore:

$$q(x_k, i | \mathbf{z}_{1:k}) \propto p\left(z_k | \mu_k^{(i)}\right) p\left(x_k | x_{k-1}^{(i)}\right) w_{k-1}^{(i)} \quad (2.134)$$

Since they are drawn from the prior distribution, (2.98) ensures that the auxiliary weights $\lambda_k^{(i)}$ follow:

$$\lambda_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}\left(z_k | \mu_k^{(i)}\right) \quad (2.135)$$

In the hope that the evolution noise is sufficiently weak for the auxiliary particles to correctly represent the future particle set, the parent indices $\{i^{(j)}\}_{j=1}^N$ are resampled according to the auxiliary weights.

The new particle set is then propagated through a standard SIS algorithm using the transition density:

$$x_k^{(j)} \sim p\left(\cdot | x_{k-1}^{(i^{(j)})}\right) \quad (2.136)$$

with their weights updating as:

$$\begin{aligned} w_k^{(j)} &\propto \frac{\tilde{p}(x_k, i^{(j)} | \mathbf{z}_{1:k})}{\tilde{q}(x_k, i^{(j)} | \mathbf{z}_{1:k})} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(j)}) \tilde{p}(x_k^{(j)} | x_{k-1}^{(i^{(j)})}) w_{k-1}^{(i^{(j)})}}{\tilde{p}(z_k | \mu_k^{(i^{(j)})}) \tilde{p}(x_k^{(j)} | x_{k-1}^{(i^{(j)})}) w_{k-1}^{(i^{(j)})}} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(j)})}{\tilde{p}(z_k | \mu_k^{(i^{(j)})})} \end{aligned} \quad (2.137)$$

The full APF algorithm is given in algorithm 2.9.

Algorithm 2.9 Auxiliary Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, \mu_k^{(i)} \sim p(\cdot x_{k-1}^{(i)})$	▷ Auxiliary propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{\lambda}_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k \mu_k^{(i)})$	▷ Auxiliary update
$\forall i \in \llbracket 1, N \rrbracket, \lambda_k^{(i)} = \tilde{\lambda}_k^{(i)} / \sum_{j=1}^N \tilde{\lambda}_k^{(j)}$	▷ Normalisation
$\{-, -, i^{(j)}\}_{j=1}^N = \text{Resampling}\left(\left\{\mu_k^{(i)}, \lambda_k^{(i)}\right\}_{i=1}^N\right)$	▷ Resampling
$\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \sim p(\cdot x_{k-1}^{(i^{(j)})})$	▷ Particle propagation
$\forall j \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(j)} \propto \frac{\tilde{p}(z_k x_k^{(j)})}{\tilde{p}(z_k \mu_k^{(i^{(j)})})}$	▷ Weight update
$\forall j \in \llbracket 1, N \rrbracket, w_k^{(j)} = \tilde{w}_k^{(j)} / \sum_{i=1}^N \tilde{w}_k^{(i)}$	▷ Normalisation

Remark 2.11 (Comparison between the RPF and the APF). *Both the RPF and the APF aim to alleviate the degeneracy and impoverishment problems of the standard PF, but they target different sources:*

- *The RPF restores diversity through kernel density estimation, addressing the severe impoverishment caused by near-deterministic dynamics. It is therefore suited to low evolution noise scenarios.*
- *The APF instead pre-selects promising parents from the incoming observation. It is preferred when the observation noise is small, since a peaked likelihood triggers an important single-step weight degeneracy.*

These preferences are rules of thumb rather than theoretical results, and the APF's pre-selection also degrades under strong evolution noise. Acting at different stages, both filters are ultimately complementary rather than exclusive.

2.5.4 Rao-Blackwellised Particle Filter

The marginalised particle filter [52], also known as the Rao-Blackwellised particle filter (RBPF) [53, 54], reduces the dimension of the state-space explored by the particle filter by exploiting a partial linear-Gaussian structure in the state-space model. Suppose that the state can be decomposed as:

$$x_k = [x_k^n, x_k^l] \quad (2.138)$$

such that (2.1) is linear and Gaussian in x_k^l conditionally on x_k^n . The navigation system can be rewritten as:

$$(S) : \begin{cases} x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \\ x_k^l = F_k^l(x_{k-1}^n) x_{k-1}^l + f_k^l(x_{k-1}^n) + w_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \quad (2.139)$$

with:

$$w_k = \begin{bmatrix} w_k^n \\ w_k^l \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} Q_k^n & (Q_k^{nl})^\top \\ Q_k^{nl} & Q_k^l \end{bmatrix} \right), \quad n_k \sim \mathcal{N}(0, R_k) \quad (2.140)$$

x_0^n of known distribution, and x_0^l Gaussian.

The idea behind the RBPF is to factorise the PDF as:

$$p(\mathbf{x}_{0:k}^l, \mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) = p(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k}) p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) \quad (2.141)$$

where $p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k})$ is estimated via particle filtering techniques, and $p(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k})$ is analytically solved by a Kalman filter. Here, $\mathbf{x}_{0:k}^l$ and $\mathbf{x}_{0:k}^n$ are defined similarly to $\mathbf{x}_{0:k}$.

When $Q_k^{nl} \neq 0$, the noises w_k^l and w_k^n are correlated, and the Kalman filter cannot be applied directly. Proposition 2.10 decouples them by removing the component of w_k^l correlated with w_k^n .

Proposition 2.10 (Noise decorrelation). *The decorrelated noise:*

$$\bar{w}_k^l \triangleq w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n \quad (2.142)$$

is centred, uncorrelated with w_k^n , and has covariance:

$$\bar{Q}_k^l \triangleq \mathbb{V}[\bar{w}_k^l] = Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \quad (2.143)$$

Proof. See section B.2. □

Let:

$$\bar{F}_k^l(x_{k-1}^n) \triangleq F_k^l(x_{k-1}^n) - Q_k^{nl} (Q_k^n)^{-1} F_k^n(x_{k-1}^n) \quad (2.144)$$

$$\bar{f}_k^l(x_{k-1}^n) \triangleq f_k^l(x_{k-1}^n) - Q_k^{nl} (Q_k^n)^{-1} f_k^n(x_{k-1}^n) \quad (2.145)$$

The system (2.139) can be rewritten as:

$$(\bar{S}) : \begin{cases} x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \\ x_k^l = \bar{F}_k^l(x_{k-1}^n) x_{k-1}^l + \bar{f}_k^l(x_{k-1}^n) + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \bar{w}_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \quad (2.146)$$

which can in turn be decomposed as:

$$\begin{aligned} (S_1) : & \begin{cases} x_{k-1}^l = x_{k-1}^l \\ x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \end{cases} \\ (S_2) : & \begin{cases} x_k^l = \bar{F}_k^l(x_{k-1}^n) x_{k-1}^l + \bar{f}_k^l(x_{k-1}^n) + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \bar{w}_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \end{aligned} \quad (2.147)$$

Note the similarities between each of the systems (S_1) and (S_2) , and the navigation equations under Kalman hypotheses (2.35). Indeed, (S_1) can be treated as if the propagation of the non-linear state x_k^n were an observation of x_{k-1}^l . Its absence of dynamics leads to only performing the (pre-)correction step of the Kalman filter. On the other hand, (S_2) can be handled by the usual Kalman prediction/correction.

In full generality, a standard RBPF is computed in five steps:

- **Particle propagation.** The non-linear state is propagated according to the proposal density:

$$x_k^{n(i)} \sim q(\cdot | x_{k-1}^{n(i)}, z_k) \quad (2.148)$$

- **Kalman pre-correction.** The previous linear state is corrected according to the Kalman equations applied to (S_1) :

$$\tilde{F}_k^{n(i)} = F_k^n(x_{k-1}^{n(i)}) \quad (2.149)$$

$$y_k^{n(i)} = x_k^{n(i)} - \left(\tilde{F}_k^{n(i)} \hat{x}_{k-1|k-1}^{l(i)} + f_k^n(x_{k-1}^{n(i)}) \right) \quad (2.150)$$

$$S_k^{n(i)} = \tilde{F}_k^{n(i)} P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{n(i)} \right)^\top + Q_k^n \quad (2.151)$$

$$K_k^{n(i)} = P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{n(i)} \right)^\top \left(S_k^{n(i)} \right)^{-1} \quad (2.152)$$

$$\hat{x}_{k-1|k-1}^{l(i)} \leftarrow \hat{x}_{k-1|k-1}^{l(i)} + K_k^{n(i)} y_k^{n(i)} \quad (2.153)$$

$$P_{k-1|k-1}^{l(i)} \leftarrow \left(I - K_k^{n(i)} \tilde{F}_k^{n(i)} \right) P_{k-1|k-1}^{l(i)} \quad (2.154)$$

- **Kalman prediction.** The current linear state is then predicted via the dy-

namics of (S_2) :

$$\tilde{F}_k^{l(i)} = \bar{F}_k^l \left(x_{k-1}^{n(i)} \right) \quad (2.155)$$

$$\hat{x}_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)} \hat{x}_{k-1|k-1}^{l(i)} + \bar{f}_k^l \left(x_{k-1}^{n(i)} \right) + Q_k^{nl} (Q_k^n)^{-1} x_k^{n(i)} \quad (2.156)$$

$$P_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)} P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{l(i)} \right)^\top + \bar{Q}_k^l \quad (2.157)$$

$$\tilde{H}_k^{(i)} = H_k \left(x_k^{n(i)} \right) \quad (2.158)$$

$$y_k^{l(i)} = z_k - \left(\tilde{H}_k^{(i)} \hat{x}_{k|k-1}^{l(i)} + h_k \left(x_k^{n(i)} \right) \right) \quad (2.159)$$

$$S_k^{l(i)} = \tilde{H}_k^{(i)} P_{k|k-1}^{l(i)} \left(\tilde{H}_k^{(i)} \right)^\top + R_k \quad (2.160)$$

- **Weight update.** The weights corresponding to each particle are updated with respect to the new observation:

$$\tilde{w}_k^{n(i)} = w_{k-1}^{n(i)} \frac{\tilde{p} \left(z_k | x_k^{n(i)} \right) \tilde{p} \left(x_k^{n(i)} | x_{k-1}^{n(i)} \right)}{\tilde{q} \left(x_k^{n(i)} | x_{k-1}^{n(i)}, z_k \right)} \quad (2.161)$$

$$w_k^{n(i)} = \tilde{w}_k^{n(i)} / \sum_{j=1}^N \tilde{w}_k^{n(j)} \quad (2.162)$$

- **Kalman correction.** The linear state is corrected through the standard Kalman correction for (S_2) :

$$K_k^{l(i)} = P_{k|k-1}^{l(i)} \left(\tilde{H}_k^{(i)} \right)^\top \left(S_k^{l(i)} \right)^{-1} \quad (2.163)$$

$$\hat{x}_{k|k}^{l(i)} = \hat{x}_{k|k-1}^{l(i)} + K_k^{l(i)} y_k^{l(i)} \quad (2.164)$$

$$P_{k|k}^{l(i)} = \left(I - K_k^{l(i)} \tilde{H}_k^{(i)} \right) P_{k|k-1}^{l(i)} \quad (2.165)$$

In the RBPF, only the non-linear state is sampled, while the conditional distribution of the linear substate is propagated analytically. The particle approximation is thus the mixture [55]:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{n(i)} \delta_{x_k^{n(i)}}(dx_k^n) \mathcal{N} \left(\hat{x}_{k|k}^{l(i)}, P_{k|k}^{l(i)} \right) (dx_k^l) \quad (2.166)$$

The MMSE estimator of section 2.2 can then be approximated by:

$$\hat{x}_k = \sum_{i=1}^N w_k^{n(i)} x_k^{(i)} \quad (2.167)$$

where

$$x_k^{(i)} \triangleq \begin{bmatrix} x_k^{n(i)} \\ \hat{x}_{k|k}^{l(i)} \end{bmatrix} = \mathbb{E} \left[x_k | \mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k} \right] \quad (2.168)$$

Furthermore, since the non-linear state is deterministic when conditioned on the sampled trajectory, we have:

$$\mathbb{V} \left[x_k | \mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k} \right] = \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \quad (2.169)$$

Applying the law of total covariance finally gives the associated empirical covariance matrix:

$$\hat{P}_k = \sum_{i=1}^N w_k^{n(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top + \sum_{i=1}^N w_k^{n(i)} \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \quad (2.170)$$

where the first term is the covariance of the conditional means, and the second is the average conditional covariance.

The full RBPF algorithm is summarised in Algorithm 2.10.

Algorithm 2.10 Rao-Blackwellised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket$, propagate $x_k^{n(i)}$	using (2.148)
$\forall i \in \llbracket 1, N \rrbracket$, update $\hat{x}_{k-1 k-1}^{l(i)}$ and $P_{k-1 k-1}^{l(i)}$	using (2.149) through (2.154)
$\forall i \in \llbracket 1, N \rrbracket$, predict $\hat{x}_{k k-1}^{l(i)}$ and $P_{k k-1}^{l(i)}$	using (2.155) through (2.160)
$\forall i \in \llbracket 1, N \rrbracket$, update $w_k^{n(i)}$	using (2.161) through (2.162)
$\forall i \in \llbracket 1, N \rrbracket$, compute $\hat{x}_{k k}^{l(i)}$ and $P_{k k}^{l(i)}$	using (2.163) through (2.165)

It must be noted that the version presented by Algorithm 2.10 uses a simple SIS to handle the particles, but any more advanced algorithm presented in this section can be used instead. Furthermore, the likelihood $\tilde{p}(z_k | x_k^{n(i)})$ and the transition $\tilde{p}(x_k^{n(i)} | x_{k-1}^{n(i)})$ in the weight update are white Gaussian densities, with covariances $S_k^{l(i)}$ and $S_k^{n(i)}$ respectively. This is why $S_k^{l(i)}$ is computed within the Kalman prediction step, before the weight update. However, the correction step itself must be performed after the weight update in situations where a resampling step is involved.

Remark 2.12 (Single Kalman filter). *In general, one Kalman filter is required per particle, since $\tilde{F}_k^{n(i)}$, $\tilde{F}_k^{l(i)}$ and $\tilde{H}_k^{(i)}$ all depend on $x_{k-1}^{n(i)}$ and $x_k^{n(i)}$. However, if these matrices do not depend on the non-linear state, then the gains $K_k^{n(i)}$ and $K_k^{l(i)}$, and the covariances $P_{k|k}^{l(i)}$ are identical for all particles. In this scenario, a single Kalman filter can be used for the whole particle set, which substantially reduces the computational cost of the RBPF.*

Remark 2.13 (Triangular model). *If (2.139) is such that $F_k^n = 0$, the system (S) is called triangular. In this scenario, x_k^n does not depend on x_{k-1}^l , which decouples the two filters entirely. The Kalman pre-correction step can thus be omitted, and the RBPF becomes a simple interleaving between a PF and a collection of Kalman filters.*

CHAPTER 3

GAUSSIAN PROCESSES

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3.1 Introduction

The analytical and Monte Carlo methods of chapter 2 all assume the observation function h_k to be fully known. However, many navigation scenarios depend on a physical field that is either unavailable or known only through scarce and noisy measurements.

A first way to fill this gap is to reconstruct the field directly from the measurements: **TODO: j'avais oublié, mais à reprendre + présenter des cas d'application au krigeage**

- Deterministic interpolation, through nearest-neighbour selection, multilinear interpolation over a regular grid, or spline fitting;
- Linear regression on a fixed basis, such as polynomials or sinusoids;
- More expressive models, such as neural networks.

All of them return a single value at the queried location, without quantifying how much it can be trusted. Their use in the filters of chapter 2, however, requires the observation likelihood $p(z_k|x_k)$, a full distribution rather than a point prediction. The reconstruction error must then be absorbed into the observation noise, usually as a covariance held constant over the whole domain, although this error is almost zero next to a measurement and grows in sparsely sampled regions.

This chapter instead models such fields as a Gaussian process (GP), an adaptable framework that encodes both a prior expectation of the field and its spatial correlation structure. Together with its uncertainty, the resulting estimator can then serve as a surrogate observation model [56, 57] for the filters of chapter 2.

The rest of this chapter is organised as follows:

- Section 3.2 defines GPs through their mean functions and covariance kernels. It discusses the choice of kernel.
- Section 3.3 then introduces kriging as a map reconstruction tool in situations where we are given preexisting data samples.
- Section 3.4 addresses the cost of inverting large Gram matrices in kriging by approximating k with a reduced-rank kernel, therefore lowering the effective complexity of kriging methods.
- Finally, section 3.5 derives two representations of the mean function over a dictionary of functions, used to reconstruct the field when no prior knowledge is available.

3.2 Definition and Properties

Formally, a stochastic process generalises the notion of random variable to a random function, *i.e.* a family $(f(x))_{x \in \mathcal{U}}$ of random variables indexed by a set $\mathcal{U}$, which may represent time $\mathbb{R}_+$, physical space $\mathbb{R}^2$ or $\mathbb{R}^3$, or any other domain. As shown by Definition 3.1, a GP is a special case of stochastic process [58].

Definition 3.1 (Gaussian process). *For $d_z \in \mathbb{N}^*$, a stochastic process $(f(x))_{x \in \mathcal{U}}$ taking values in $\mathbb{R}^{d_z}$ is a GP if every finite linear combination of its values follows a Gaussian distribution. Formally, for all $n \in \mathbb{N}^*$, all $x_1, \dots, x_n \in \mathcal{U}$ and all $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, there exist $m \in \mathbb{R}^{d_z}$ and $P \in \mathcal{S}_{d_z}^+(\mathbb{R})$ such that:*

$$\sum_{i=1}^n a_i^\top f(x_i) \sim \mathcal{N}(m, P) \quad (3.1)$$

The GP is said to be non-degenerate if $P \succ 0$ whenever the $x_1, \dots, x_n$ are pairwise distinct and the coefficients $a_1, \dots, a_n$ are not all zero.

A GP is associated with a mean function:

$$m : \begin{cases} \mathcal{U} \rightarrow \mathbb{R}^{d_z} \\ x \mapsto \mathbb{E}[f(x)] \end{cases} \quad (3.2)$$

and a covariance kernel:

$$k : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))^\top] \end{cases} \quad (3.3)$$

3.2.1 Basic Properties

As shown by Proposition 3.1, m and k entirely characterise every finite-dimensional marginal of f . By Kolmogorov's extension theorem, they also uniquely determine the law of f . We thus write $f \sim \mathcal{GP}(m, k)$ for a GP with mean function m and covariance kernel k .

Proposition 3.1 (Finite-dimensional marginals). *For $n \in \mathbb{N}^*$, let $x_1, \dots, x_n$ be pairwise distinct elements of $\mathcal{U}$. The joint distribution of $F \triangleq [f(x_1)^\top, \dots, f(x_n)^\top]^\top$ is the multivariate Gaussian distribution:*

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \right) \quad (3.4)$$

where each block $k(x_i, x_j) \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$.

Proof. According to Definition 3.1, every linear combination:

$$a^\top F = \sum_{i=1}^n a_i^\top f(x_i), \quad a \triangleq \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^{nd_z} \quad (3.5)$$

is Gaussian. By the Cramér-Wold theorem [59] F is thus also a Gaussian vector of $\mathbb{R}^{nd_z}$.

Such a vector is entirely determined by its mean and covariance, which we compute. In particular, we have:

$$\mathbb{E}[F] = \begin{bmatrix} \mathbb{E}[f(x_1)] \\ \vdots \\ \mathbb{E}[f(x_n)] \end{bmatrix} = \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \quad (3.6)$$

By definition of k , it comes that:

$$\begin{aligned} \mathbb{V}[F] &= \left[\mathbb{E} \left[(f(x_i) - m(x_i))(f(x_j) - m(x_j))^\top \right] \right]_{i,j=1}^n \\ &= \left[k(x_i, x_j) \right]_{i,j=1}^n \end{aligned} \quad (3.7)$$

which is exactly (3.4). $\square$

For two tuples of inputs $\mathbf{x} \triangleq (x_1, \dots, x_n) \in \mathcal{U}^n$ and $\mathbf{x}' \triangleq (x'_1, \dots, x'_p) \in \mathcal{U}^p$ of sizes $n, p \in \mathbb{N}^*$, we introduce the following vectorised notation:

$$f(\mathbf{x}) \triangleq \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \quad (3.8)$$

$$m(\mathbf{x}) \triangleq \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \quad (3.9)$$

$$k(\mathbf{x}, \mathbf{x}') \triangleq \begin{bmatrix} k(x_1, x'_1) & \cdots & k(x_1, x'_p) \\ \vdots & \ddots & \vdots \\ k(x_n, x'_1) & \cdots & k(x_n, x'_p) \end{bmatrix} \quad (3.10)$$

Using these notations, Proposition 3.1 reads compactly:

$$\forall n \in \mathbb{N}^*, \forall \mathbf{x} \in \mathcal{U}^n \text{ pairwise distinct, } f(\mathbf{x}) \sim \mathcal{N}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x})) \quad (3.11)$$

A GP is thus a consistent generalisation of the multivariate Gaussian distribution to an infinite and potentially continuous index set. Both of its parameters admit a direct interpretation. The mean function m encodes prior knowledge about the average behaviour of f , while the kernel k captures the correlation structure. The diagonal blocks $k(x, x)$ represent the covariance of $f(x)$, while the off-diagonal blocks $k(x, x')$, for $x \neq x'$, describe the cross-covariance between $f(x)$ and $f(x')$.

In order for the GP to be non-degenerate, the matrix $k(\mathbf{x}, \mathbf{x})$ must be SPD in the sense of Definition 2.1 for every finite tuple $\mathbf{x}$ of pairwise distinct inputs. Definition 3.2 recasts this as a condition on k alone [60], and Proposition 3.2 proves the equivalence between the two.

Definition 3.2 (Symmetric positive definite kernel). *A kernel $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is symmetric positive semi-definite (SPSD) if it is:*

- **Symmetric:**

$$\forall x, x' \in \mathcal{U}, k(x', x) = k(x, x')^\top \quad (3.12)$$

- **Positive semi-definite:**

$$\begin{aligned} \forall n \in \mathbb{N}^*, \forall x_1, \dots, x_n \in \mathcal{U}, \forall a_1, \dots, a_n \in \mathbb{R}^{d_z}, \\ \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j \geq 0 \end{aligned} \quad (3.13)$$

If the inequality is strict for every choice of pairwise distinct points $x_1, \dots, x_n$ and coefficients $a_1, \dots, a_n$ not all zero, then k is symmetric positive definite (SPD).

Proposition 3.2 (Definite kernel/matrix equivalence). *The following equivalences hold:*

- $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is a SPSP kernel if and only if, for every $n \in \mathbb{N}^*$ and every $\mathbf{x} \in \mathcal{U}^n$, $k(\mathbf{x}, \mathbf{x}) \in \mathcal{M}_{nd_z, nd_z}(\mathbb{R})$ is a SPSP matrix.
- k is a SPD kernel if and only if, for every such tuple $\mathbf{x}$ of pairwise distinct points, $k(\mathbf{x}, \mathbf{x})$ is a SPD matrix.

Proof. See section B.3. □

Remark 3.1 (Necessity of distinct points). *While pairwise distinctness does not intervene in the computation of the strict version above, it is necessary for the SPD condition to be satisfiable. Indeed, at coinciding inputs $x_1 = x_2$, the choice $a_1 = -a_2 = 1$, with the other coefficients set to zero, cancels the quadratic form for non-zero coefficients, so no kernel could be SPD.*

Remark 3.2 (Non-diagonal blocks). *While Proposition 3.2 equates the SPDness of k with that of $k(\mathbf{x}, \mathbf{x})$ at distinct points, it does not make the off-diagonal blocks $k(x, x')$ SPD themselves. For $x \neq x'$, these may be non-positive or even non-symmetric.*

Throughout the rest of this document, we assume that the covariance kernel k is SPD and that the points of every collection $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{U}^n$ are pairwise distinct. This ensures that we only work with non-degenerate GPs.

3.2.2 Common Kernels

Assume in this section that $\mathcal{U}$ is a finite-dimensional normed vector space, *i.e.* $\mathcal{U} = \mathbb{R}^{d_x}$ for some $d_x \in \mathbb{N}^*$, up to isomorphism. A natural way to build kernels on $\mathcal{U}$ is to let the correlation between the field values at two points x, x' depend only on the distance $\|x - x'\|$ between them. Kernels of this form are called radial kernels [61], and are formally defined by Definition 3.3.

Since f typically represents a physical field, observations tend to become less correlated as the distance between them increases. Definition 3.4 introduces a common family of radial kernels modelling this behaviour, called the Matérn kernels [62]. Proposition 3.3 then establishes their continuity and SPDness.

Definition 3.3 (Radial kernel). *A kernel $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is radial if there exists a function $\kappa : \mathbb{R}_+ \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ and a norm $\|\cdot\|$ on $\mathbb{R}^{d_x}$ such that:*

$$\forall x, x' \in \mathbb{R}^{d_x}, k(x, x') = \kappa(\|x - x'\|) \quad (3.14)$$

To simplify notations, we identify κ with k in the remainder of this document, such that we write:

$$\forall x, x' \in \mathbb{R}^{d_x}, k(x, x') = k(\|x - x'\|) \quad (3.15)$$

Definition 3.4 (Matérn kernel). Assume that $d_z = 1$, and let $\nu, l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel of smoothness ν , correlation length l , and characteristic variance σ_f^2 is the radial kernel defined as:

$$k_\nu : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{l} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{l} \right) \end{cases} \quad (3.16)$$

where Γ is the Gamma function, and K_ν is the modified Bessel function of the second kind [63].

This family extends to $\nu = +\infty$ by taking the pointwise limit, yielding the squared exponential (SE) kernel:

$$k_{\text{SE}} \triangleq k_{+\infty} : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \exp \left(-\frac{r^2}{2l^2} \right) \end{cases} \quad (3.17)$$

Proposition 3.3 (Properties of Matérn kernels). Let $\nu \in \mathbb{R}_+^* \cup \{+\infty\}$ and $l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel associated with ν , l and σ_f^2 is a continuous SPD radial kernel.

Proof. See section B.4. □

The parameter ν controls the regularity. Indeed, for $p \in \mathbb{N}$, it can be shown [64] that a GP equipped with k_ν is p times mean-square differentiable if and only if $\nu > p$. For half-integer $\nu = p + \frac{1}{2}$, Matérn kernels can be written as the product of an exponential and a polynomial function of degree p :

$$k_{p+1/2}(r) = \sigma_f^2 \exp \left(-\frac{\sqrt{2p+1}r}{l} \right) \frac{\Gamma(p+1)}{\Gamma(2p+1)} \sum_{i=0}^p \frac{(p+i)!}{i!(p-i)!} \left(\frac{2\sqrt{2p+1}r}{l} \right)^{p-i} \quad (3.18)$$

In particular, the following kernels correspond to usual choices of ν , the SE kernel being recalled here for comparison:

- **Exponential kernel.**

$$k_{1/2}(r) = \sigma_f^2 \exp \left(-\frac{r}{l} \right) \quad (3.19)$$

- **Once-differentiable Matérn kernel.**

$$k_{3/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3}r}{l} \right) \exp \left(-\frac{\sqrt{3}r}{l} \right) \quad (3.20)$$

- **Twice-differentiable Matérn kernel.**

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{l} + \frac{5r^2}{3l^2} \right) \exp \left(-\frac{\sqrt{5}r}{l} \right) \quad (3.21)$$

- **Squared exponential kernel.**

$$k_{+\infty}(r) = \sigma_f^2 \exp \left(-\frac{r^2}{2l^2} \right) \quad (3.22)$$

Figure 3.1 illustrates the radial decay of these kernels, evaluated with unit variance $\sigma_f^2 = 1$ and correlation length $l = 1$ to facilitate comparison.

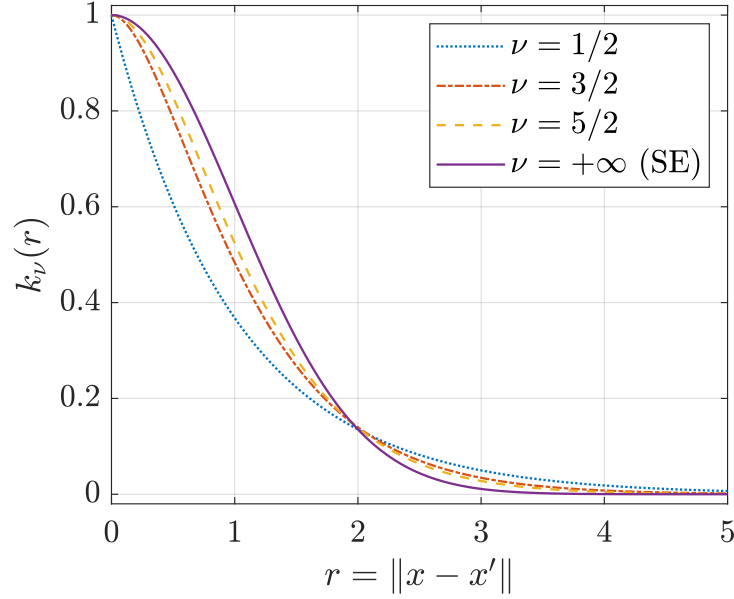


Figure 3.1: Radial profile of Matérn kernels with increasing smoothness, together with their squared exponential limit. Larger values of ν produce smoother kernels, with flatter behaviours near the origin and faster-decaying tails.

Remark 3.3 (Anisotropic kernel). *The norm used in Definition 3.3 does not necessarily need to be the canonical norm of $\mathbb{R}^{d_x}$. In particular, for $M \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$, the Mahalanobis norm:*

$$x \in \mathbb{R}^{d_x} \mapsto \|x\|_M \triangleq \sqrt{x^\top M^{-1}x} \quad (3.23)$$

may produce anisotropic radial kernels in situations where M is not proportional to the identity matrix.

The main limitation of Matérn kernels is that they are scalar, while our observation model takes values in $\mathbb{R}^{d_z}$. To alleviate this issue, Proposition 3.4 presents a method to create a vector-valued kernel from a scalar one.

Proposition 3.4 (Separable kernel). *Let $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ be a scalar SPD kernel, and let $\Sigma \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$. Then:*

$$k_{\text{sep}} : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \Sigma k(x, x') \end{cases} \quad (3.24)$$

is a valid SPD kernel for an $\mathbb{R}^{d_z}$ -valued GP.

Furthermore, when k is continuous, k_{sep} is also continuous.

Proof. The symmetry and potential continuity of k_{sep} are direct from that of k and Σ . The positive definiteness can be established by letting A be a Cholesky factor of the SPD matrix Σ , and considering, for every $n \in \mathbb{N}^*$ and every not all zero coefficients $a_1, \dots, a_n \in \mathbb{R}^{d_z}$:

$$\forall i \in \llbracket 1, n \rrbracket, b_i \triangleq A^\top a_i \quad (3.25)$$

Since A is invertible, these coefficients are not all zero, and thus:

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^n a_i^\top k_{\text{sep}}(x_i, x_j) a_j &= \sum_{i=1}^n \sum_{j=1}^n a_i^\top A A^\top a_j k(x_i, x_j) \\
&= \sum_{i=1}^n \sum_{j=1}^n b_i^\top b_j k(x_i, x_j) \\
&= \sum_{l=1}^{d_z} \sum_{i=1}^n \sum_{j=1}^n b_i^{(l)} b_j^{(l)} k(x_i, x_j)
\end{aligned} \tag{3.26}$$

where $b_i^{(l)}$ denotes the l -th component of b_i . Each inner double sum is a quadratic form of k , hence non-negative, and at least one of them is strictly positive since at least one of the vectors $(b_1^{(l)}, \dots, b_n^{(l)})$ is non-zero. $\square$

Remark 3.4 (Multi-scale kernels). For $n \in \mathbb{N}^*$, $\Sigma_1, \dots, \Sigma_n \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ and $k^{(1)}, \dots, k^{(n)}$ Matérn kernels of correlation lengths $l_1 < \dots < l_n \in \mathbb{R}_+^*$, the multi-scale kernel is:

$$k_{\text{MS}} : \begin{cases} \mathbb{R}_+ \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ r \mapsto \sum_{i=1}^n \Sigma_i k^{(i)}(r) \end{cases} \tag{3.27}$$

Each of its terms is separable in the sense of Proposition 3.4, hence SPD, and a sum of SPD kernels with positive coefficients is itself SPD. When the correlation lengths have different orders of magnitude, such kernels can be used to capture local fluctuations and global trends simultaneously.

Every kernel considered in this section is specified by a set of hyperparameters, such as ν , l and σ_f^2 for a Matérn kernel, Σ for its separable extension, or one such set per scale for a multi-scale model. Their estimation is a standard step in GP modelling, and is independent of bayesian estimation considerations. However, the specific characteristics of our applications can be used to improve the estimation of such hyperparameters. These aspects are thus discussed in section 4.2.2, and the hyperparameters are assumed to be fully specified for the rest of this chapter.

3.3 Gaussian Process Regression

Gaussian process regression, also known as kriging [65, 66, 67], estimates the value of a field at an unobserved location using noisy measurements of the same field at known locations. Formally, let $f \sim \mathcal{GP}(m, k) : \mathcal{U} \rightarrow \mathbb{R}^{d_z}$, and $x \in \mathcal{U}$. We wish to estimate the law of:

$$z \triangleq f(x) + \varepsilon \in \mathbb{R}^{d_z}, \quad \varepsilon \sim \mathcal{N}(0, R), \quad R \in \mathcal{S}_{d_z}^+(\mathbb{R}) \tag{3.28}$$

where ε is independent of f .

While kriging usually aims to reconstruct $f(x)$ itself, the query noise ε models the observation uncertainty, so that kriging estimates the noisy observation from (2.1). The standard formulation can nonetheless be recovered by taking $R = 0$.

Without further information regarding f , we would only be able to state that z follows a Gaussian distribution of expectation $m(x)$ and covariance $k(x, x) + R$. However, we are given $\tilde{N} \in \mathbb{N}^*$ distinct training points $\tilde{\mathbf{x}} \triangleq (\tilde{x}_1, \dots, \tilde{x}_{\tilde{N}}) \in \mathcal{U}^{\tilde{N}}$ and their corresponding noisy observations:

$$\tilde{\mathbf{z}} \triangleq \begin{bmatrix} f(\tilde{x}_1) + \varepsilon_1 \\ \vdots \\ f(\tilde{x}_{\tilde{N}}) + \varepsilon_{\tilde{N}} \end{bmatrix} \in \mathbb{R}^{\tilde{N}d_z}, \quad \varepsilon_i \sim \mathcal{N}(0, \tilde{R}), \quad \tilde{R} \in \mathcal{S}_{d_z}^+(\mathbb{R}) \quad (3.29)$$

where $\varepsilon_1, \dots, \varepsilon_{\tilde{N}}$ are mutually independent and independent of f .

The goal of this section is to derive the best linear unbiased estimator (BLUE) of z with respect to $\tilde{\mathbf{z}}$, *i.e.* the estimator:

$$\hat{z} = W^\top \tilde{\mathbf{z}} \quad (3.30)$$

parametrised by a weight matrix:

$$W \triangleq \begin{bmatrix} w_1 \\ \vdots \\ w_{\tilde{N}} \end{bmatrix} \in \mathcal{M}_{\tilde{N}d_z, d_z}(\mathbb{R}) \quad (3.31)$$

such that the error vector has an expectation of 0 and a minimum covariance. This setting is illustrated by Figure 3.2 for a $\mathbb{R}^2 \rightarrow \mathbb{R}$ field.

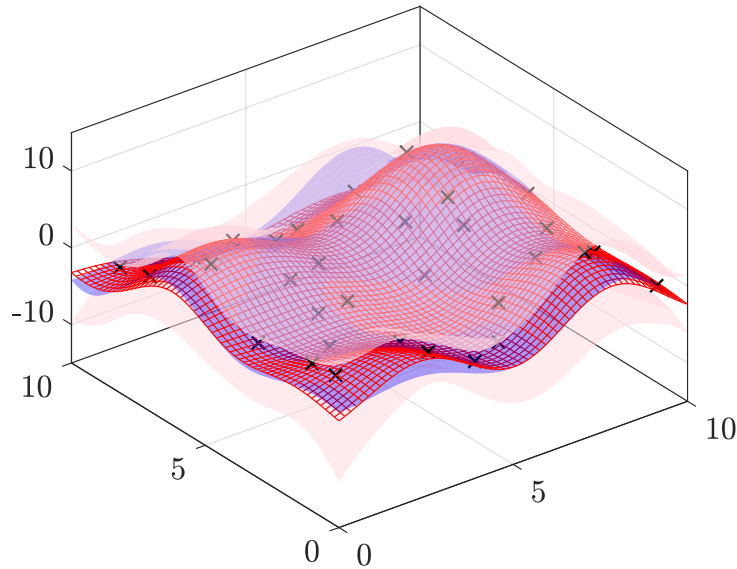


Figure 3.2: Ordinary kriging (red mesh) of an unknown field (blue surface) from noisy samples (black crosses). The confidence band (red envelope) is proportional to the square root of the interpolation covariance, and widens away from the samples.

Since $\hat{z} - z$ is a vector, its covariance is a matrix, and cannot be minimised as such. To alleviate this issue, we minimise its trace, which is exactly the mean squared error:

$$\text{tr}(\mathbb{V}[\hat{z} - z]) = \mathbb{E} [\|\hat{z} - z\|^2] \quad (3.32)$$

In the following, we present three kriging estimates, based on increasingly relaxed assumptions made on our knowledge of the underlying GP. Each method relies on the inversion of the Gram matrix:

$$\begin{aligned} G(\tilde{\mathbf{x}}) &\triangleq k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} \\ &= \begin{bmatrix} k(\tilde{x}_1, \tilde{x}_1) + \tilde{R} & \cdots & k(\tilde{x}_1, \tilde{x}_{\tilde{N}}) \\ \vdots & \ddots & \vdots \\ k(\tilde{x}_{\tilde{N}}, \tilde{x}_1) & \cdots & k(\tilde{x}_{\tilde{N}}, \tilde{x}_{\tilde{N}}) + \tilde{R} \end{bmatrix} \end{aligned} \quad (3.33)$$

of $\tilde{\mathbf{x}}$ with respect to f , where $\otimes$ is the Kronecker matrix product. Since k is SPD and the training points are distinct, it comes that:

$$G(\tilde{\mathbf{x}}) \in \mathcal{S}_{\tilde{N}d_z}^{++}(\mathbb{R}) \quad (3.34)$$

and so $G(\tilde{\mathbf{x}})$ is invertible, even in the noiseless case $\tilde{R} = 0$.

We also assume that the query point x differs from the training points $\tilde{\mathbf{x}}$, so that the joint law of $(z, \tilde{\mathbf{z}})$ is non-degenerate. This hypothesis is not restrictive and is only needed for the proofs. Indeed, the estimators are continuous in x and thus remain valid at the training points as soon as m and k are continuous, which we assume.

Moreover, $\mathbb{V}[\hat{z} - z]$ is a covariance matrix, and therefore SPSPD whatever the weight matrix W . It is furthermore positive definite when x is not one of the training points, since $\hat{z} - z$ is a non-degenerate linear combination of z and $\tilde{\mathbf{z}}$.

3.3.1 Simple Kriging

Simple kriging (SK) is the simplest form of kriging, and assumes that the mean function m is known. Proposition 3.5 gives a closed-form expression of the SK estimator and its associated covariance matrix, while Proposition 3.7 shows that it coincides with the conditional distribution of z given $\tilde{\mathbf{z}}$.

Proposition 3.5 (Simple kriging estimator). *Using the notations introduced in section 3.3, the SK estimator of z and its error covariance are:*

$$\begin{cases} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = m(x) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\ R_{\text{SK}}(x, \tilde{\mathbf{x}}) = k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (3.35)$$

Proof. Since m is known, we may assume it to be identically equal to 0 by working with $f - m$ instead of f , without any loss of generality. In the centred case, (3.32) can be developed as follows:

$$\mathbb{V}[W^\top \tilde{\mathbf{z}} - z] = W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \quad (3.36)$$

By symmetry of $G(\tilde{\mathbf{x}})$, differentiating (3.32) with respect to W yields:

$$\begin{aligned} &\frac{\partial}{\partial W} (\text{tr} (W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R)) = 0 \\ \iff &\frac{\partial}{\partial W} (\text{tr} (W^\top G(\tilde{\mathbf{x}}) W)) - 2 \frac{\partial}{\partial W} (\text{tr} (W^\top k(\tilde{\mathbf{x}}, x))) = 0 \\ \iff &2G(\tilde{\mathbf{x}}) W - 2k(\tilde{\mathbf{x}}, x) = 0 \\ \iff &W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (3.37)$$

The SK estimator is thus:

$$\hat{z}_{\text{SK}} = W^\top \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \quad (3.38)$$

and its corresponding covariance:

$$\begin{aligned} R_{\text{SK}} &= W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\ &= k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} G(\tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &\quad - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) + k(x, x) + R \\ &= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (3.39)$$

The general case is obtained by adding m back into f . $\square$

Remark 3.5 (Additive query noise). *In Proposition 3.5, the term R appears because the noisy query z is predicted rather than the field value $f(x)$. The two share the same estimator, and differ in covariance only by an additive R term. Kriging the field can thus be done by subtracting R , or equivalently by taking $R = 0$ as mentioned in the introduction to kriging.*

Remark 3.6 (Error covariance minimisation). *While the SK weight matrix $W = G^{-1}k(\tilde{\mathbf{x}}, x)$ was obtained by minimising the trace of $\mathbb{V}[\hat{z} - z]$, it minimises the whole covariance in the SPSSD sense. Indeed:*

$$\begin{aligned} \mathbb{V}[\hat{z} - z] - R_{\text{SK}} &= W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &= (W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x))^\top G(\tilde{\mathbf{x}}) (W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)) \\ &\succeq 0 \end{aligned} \quad (3.40)$$

with equality if and only if $W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)$ since $G(\tilde{\mathbf{x}}) \succ 0$.

Proposition 3.6 (Exact interpolation in the noiseless case). *Assume that $\tilde{R} = 0$, and let $i \in \llbracket 1, \tilde{N} \rrbracket$. The SK estimator interpolates the data at $\tilde{x}_i$ exactly:*

$$\begin{cases} \hat{z}_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = \tilde{z}_i \\ R_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}) = R \end{cases} \quad (3.41)$$

In particular, the error covariance reduces to the query noise R alone, and vanishes when $R = 0$. That degenerate case is the one set aside at the start of this section, and is reached by continuity in x . On the contrary, a positive training noise $\tilde{R} \succ 0$ turns kriging into a smoother rather than an interpolator [68].

Proof. Since $\tilde{R} = 0$, the Gram matrix reduces to $G(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})$. Let $e_i \triangleq [0, \dots, 0, 1, 0, \dots, 0]^\top$ be the i -th canonical basis vector of $\mathbb{R}^{\tilde{N}}$. The matrix $e_i^\top \otimes I_{d_z}$ extracts the i -th block of a stacked matrix, such that:

$$(e_i^\top \otimes I_{d_z}) G(\tilde{\mathbf{x}}) = (e_i^\top \otimes I_{d_z}) k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) = k(\tilde{x}_i, \tilde{\mathbf{x}}) \quad (3.42)$$

The kriging weights thus reduce to this selector:

$$k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} = (e_i^\top \otimes I_{d_z}) G(\tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} = e_i^\top \otimes I_{d_z} \quad (3.43)$$

Evaluating (3.35) at $x = \tilde{x}_i$ by the continuity yields:

$$\begin{aligned}
\hat{z}_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) &= m(\tilde{x}_i) + k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\
&= m(\tilde{x}_i) + (e_i^\top \otimes I_{d_z}) (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\
&= m(\tilde{x}_i) + \tilde{z}_i - m(\tilde{x}_i) \\
&= \tilde{z}_i
\end{aligned} \tag{3.44}$$

Similarly:

$$\begin{aligned}
R_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}) &= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, \tilde{x}_i) \\
&= k(\tilde{x}_i, \tilde{x}_i) + R - (e_i^\top \otimes I_{d_z}) k(\tilde{\mathbf{x}}, \tilde{x}_i) \\
&= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{x}_i) \\
&= R
\end{aligned} \tag{3.45}$$

□

Proposition 3.7 (Gaussian conditioning interpretation). *The conditional distribution of z knowing $\tilde{\mathbf{z}}$ is Gaussian, parametrised by the SK estimator and its error covariance matrix:*

$$z | \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m), R_{\text{SK}}(x, \tilde{\mathbf{x}})) \tag{3.46}$$

Proof. Since they are evaluations of the same GP under an additive Gaussian white noise, z and $\tilde{\mathbf{z}}$ are jointly Gaussian. Their joint distribution is:

$$\begin{bmatrix} z \\ \tilde{\mathbf{z}} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x) \\ m(\tilde{\mathbf{x}}) \end{bmatrix}, \begin{bmatrix} k(x, x) + R & k(x, \tilde{\mathbf{x}}) \\ k(\tilde{\mathbf{x}}, x) & G(\tilde{\mathbf{x}}) \end{bmatrix} \right) \tag{3.47}$$

As noted at the start of this section, this joint covariance is SPD. The Gaussian conditioning formula (see Proposition A.1) can thus be used to give the desired result. □

This result is a special case of the more general result established in Proposition 2.1. Since the unbiased hypothesis of the BLUE is automatically verified by the SK estimator, the minimisation criterion (3.32) amounts to finding the MMSE estimator of z given $\tilde{\mathbf{z}}$, which is always the conditional expectation.

Remark 3.7 (Simultaneous estimation). *The simultaneous estimation of f at several query points $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{U}^n$ can be done by directly replacing x with $\mathbf{x}$ in (3.35). In particular, the $\mathcal{O}(\tilde{N}^3)$ cost of inverting $G(\tilde{\mathbf{x}})$ can be shared across the n estimations whether they are performed successively or simultaneously.*

However, computing only the n diagonal blocks of $R_{\text{SK}}(\mathbf{x}, \tilde{\mathbf{x}})$ costs $\mathcal{O}(\tilde{N}^3 + n\tilde{N}^2)$, whereas forming the full matrix before discarding its off-diagonal blocks costs $\mathcal{O}(\tilde{N}^3 + n\tilde{N}^2 + n^2\tilde{N})$ due to the extra off-diagonal matrix products. The off-diagonal blocks describe the correlation between the estimation errors made at distinct query points, and should only be computed when that correlation is needed.

3.3.2 Ordinary Kriging

Assuming the mean function to be known is a strong assumption, and can be unrealistic in practical scenarios. While an approximated model could be used instead of the true m , this would cause the kriging estimator to be overly confident in its estimation. Ordinary kriging (OK) extends its simple counterpart by assuming that the mean is an unknown constant $m \in \mathbb{R}^{d_z}$, and estimating it alongside the weights.

The unbiased assumption of the OK estimator yields:

$$\begin{aligned}
 & \mathbb{E}[\hat{z} - z] = 0 \\
 \iff & W^\top \mathbb{E}[\tilde{\mathbf{z}}] - \mathbb{E}[z] = 0 \\
 \iff & \left(\sum_{i=1}^{\tilde{N}} w_i^\top - I_{d_z} \right) m = 0 \\
 \iff & C^\top W - I_{d_z} = 0
 \end{aligned} \tag{3.48}$$

with:

$$C \triangleq \mathbf{1}_{\tilde{N}} \otimes I_{d_z} \in \mathcal{M}_{\tilde{N}d_z, d_z}(\mathbb{R}), \quad \mathbf{1}_{\tilde{N}} \triangleq \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^{\tilde{N}} \tag{3.49}$$

The OK weights thus minimise (3.32) under the constraint (3.48). Using the Lagrange multiplier method [69], Proposition 3.8 derives the corresponding estimator and covariance matrix, and Remark 3.8 establishes the link between ordinary and simple kriging.

Proposition 3.8 (Ordinary kriging estimator). *The OK estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_{\text{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - C \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ R_{\text{OK}}(x, \tilde{\mathbf{x}}) = R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \tag{3.50}$$

with:

$$\begin{cases} \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = (C^\top G(\tilde{\mathbf{x}})^{-1} C)^{-1} C^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) = (I_{d_z} - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} C) (C^\top G(\tilde{\mathbf{x}})^{-1} C)^{-1} \\ \quad (I_{d_z} - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} C)^\top \end{cases} \tag{3.51}$$

and $R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) \succeq 0$.

Proof. See section B.5. □

Remark 3.8 (Comparison between simple and ordinary kriging). *The formulas obtained in (3.50) closely resemble those of simple kriging. In fact, they can be rewritten as follows:*

$$\begin{cases} \hat{z}_{\text{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ R_{\text{OK}}(x, \tilde{\mathbf{x}}) = R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) + R_{\text{SK}}(x, \tilde{\mathbf{x}}) \end{cases} \tag{3.52}$$

where, by abuse of notation, $\hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ denotes either the estimated mean vector or its replication for each the training point $\tilde{\mathbf{x}}$, namely:

$$\underbrace{\begin{bmatrix} \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ \vdots \\ \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \end{bmatrix}}_{\tilde{N} \text{ times}} = C \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \quad (3.53)$$

Ordinary kriging can therefore be interpreted as simple kriging in which the mean m is concurrently estimated from the observations. The uncertainty associated with this second estimation is captured by the covariance term $R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) \succeq 0$, and is added to the SK covariance.

3.3.3 Universal Kriging

Universal kriging (UK) [70] generalises the constant mean of ordinary kriging to a linear combination of $r_{\text{UK}} \in \mathbb{N}^*$ known trend functions $g_1, \dots, g_{r_{\text{UK}}} : \mathcal{U} \rightarrow \mathbb{R}$:

$$m(x) = g(x) \beta = \sum_{i=1}^{r_{\text{UK}}} g_i(x) \beta_i \quad (3.54)$$

where $g(x)$ is the trend matrix:

$$g(x) \triangleq \begin{bmatrix} g_1(x) & \cdots & g_{r_{\text{UK}}}(x) \end{bmatrix} \otimes I_{d_z} \in \mathcal{M}_{d_z, r_{\text{UK}} d_z}(\mathbb{R}) \quad (3.55)$$

and β gathers the unknown coefficients $\beta_i \in \mathbb{R}^{d_z}$:

$$\beta \triangleq \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_{r_{\text{UK}}} \end{bmatrix} \in \mathbb{R}^{r_{\text{UK}} d_z} \quad (3.56)$$

The trend matrix at the training points stacks these evaluations:

$$g(\tilde{\mathbf{x}}) = \begin{bmatrix} g(\tilde{x}_1) \\ \vdots \\ g(\tilde{x}_{\tilde{N}}) \end{bmatrix} \in \mathcal{M}_{\tilde{N} d_z, r_{\text{UK}} d_z}(\mathbb{R}) \quad (3.57)$$

so that $m(\tilde{\mathbf{x}}) = g(\tilde{\mathbf{x}}) \beta$ lies in $\mathbb{R}^{\tilde{N} d_z}$, as does $\tilde{\mathbf{z}}$.

This matrix is assumed to have full column rank, which requires at least $\tilde{N} \geq r_{\text{UK}}$ training points, and the trend functions to be linearly independent once restricted to the training points. Proposition 3.9 generalises Proposition 3.8 accordingly.

Proposition 3.9 (Universal kriging estimator). *Assume that $g(\tilde{\mathbf{x}})$ has full column rank. The UK estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = g(x) \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}}) \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)) \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) = R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (3.58)$$

with:

$$\begin{cases} \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = (g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^{-1} g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) = (g(x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})) (g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^{-1} \\ \quad (g(x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^\top \end{cases} \quad (3.59)$$

and $R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) \succeq 0$.

Proof. See section B.6. □

Remark 3.9 (Link with ordinary and simple kriging). *As mentioned in the beginning of this section, ordinary kriging is a special case of universal kriging, which arises when $r_{\text{UK}} = 1$ and $g_1 \equiv 1$. In this scenario, the trend matrices reduce to $g(x) = I_{d_z}$ and $g(\tilde{\mathbf{x}}) = C$. This, in turn, yields:*

$$\begin{cases} \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 1) = \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, 1) = R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.60)$$

and thus:

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 1) = \hat{z}_{\text{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}, 1) = R_{\text{OK}}(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.61)$$

Similarly to Remark 3.8, universal kriging can be seen as the simultaneous simple kriging from the training points and mean estimation from the trend functions. Indeed, comparing (3.58) with Proposition 3.5 shows that the following relation:

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)) \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) = R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) + R_{\text{SK}}(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.62)$$

Remark 3.8 is the special case $r_{\text{UK}} = 1$, $g_1 \equiv 1$ of this reading, where the estimated mean function is constant.

Remark 3.10 (Kriging with a coarse model). *Universal kriging is primarily relevant when a coarse model of the field is available, such as a polynomial trend [71] or a simplified reference model [72]. In the simplest case, setting $r_{\text{UK}} = 1$, and choosing g_1 as this model amounts to kriging the discrepancy. The GP then corrects the approximation rather than reconstructing the field from scratch.*

Proposition 3.7 reads simple kriging as a Gaussian conditioning, which ordinary and universal kriging do not admit as such, their estimators being BLUEs of an unknown mean rather than posterior expectations. This interpretation can nonetheless be recovered by using a Gaussian prior on the trend coefficients, as proven by Proposition 3.10.

Proposition 3.10 (Conditioning under a vague trend prior). *Consider the augmented kernel:*

$$k_B(x, x') \triangleq k(x, x') + g(x) B g(x')^\top \quad (3.63)$$

Universal kriging under k is equivalent to simple kriging under k_B . Letting $B^{-1} \rightarrow 0$ leaves that prior flat and returns (3.58) [66]. In this limit, $\mathcal{N}(\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g), R_{\text{UK}}(x, \tilde{\mathbf{x}}, g))$ is the conditional law of z knowing $\tilde{\mathbf{z}}$, and (3.50) follows as the case $r_{\text{UK}} = 1$, $g_1 \equiv 1$.

Proof. See section B.7. □

TODO: à vérifier — trois modifications dans cette sous-section. La remarque de complexité ci-dessous est nouvelle. Proposition 3.10 est passée de remarque à proposition, et son premier paragraphe, qui doublonnait le second, a été supprimé — reste à décider si son corps doit être reformulé en énoncé.

Remark 3.11 (Cost of estimating the mean). *Ordinary and universal kriging both add the estimation of a mean to their simple counterpart, at a cost that (3.58) makes explicit. Besides the $\mathcal{O}(\tilde{N}^3)$ factorisation of $G(\tilde{\mathbf{x}})$, they need $G(\tilde{\mathbf{x}})^{-1}g(\tilde{\mathbf{x}})$, which reuses that factorisation for r_{UK} additional right-hand sides and costs $\mathcal{O}(r_{\text{UK}}\tilde{N}^2)$, then the assembly and the inversion of $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1}g(\tilde{\mathbf{x}})$, for $\mathcal{O}(r_{\text{UK}}^2\tilde{N} + r_{\text{UK}}^3)$. Like the factorisation itself, these are paid once and shared across query points, each of which then carries an extra $\mathcal{O}(r_{\text{UK}}\tilde{N})$.*

Estimating the mean is therefore free at the dominant order as long as $r_{\text{UK}} \ll \tilde{N}$, ordinary kriging being the extreme case $r_{\text{UK}} = 1$ of one additional right-hand side and a $d_z \times d_z$ inversion. The same ratio governs the conditioning of $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1}g(\tilde{\mathbf{x}})$, which degrades as r_{UK} approaches $\tilde{N}$, so the arithmetic cost and the numerical one are bounded together.

3.4 Reduced-Rank Kernel Approximation

Exact GP regression requires inverting the $\tilde{N}d_z \times \tilde{N}d_z$ Gram matrix $G(\tilde{\mathbf{x}})$, which scales as $\mathcal{O}(\tilde{N}^3)$. When the number of training points is large, this operation becomes prohibitive. Reduced-rank methods [73, 74] circumvent this issue by approximating the kernel as a linear combination of $r_{\text{KL}} \in \mathbb{N}^*$ basis functions, with $r_{\text{KL}} < \tilde{N}$. The inversion cost is reduced to $\mathcal{O}(r_{\text{KL}}^3)$, bringing down the overall kriging cost to $\mathcal{O}(r_{\text{KL}}^2\tilde{N})$.

Mercer’s theorem [75], presented in Proposition 3.12, establishes the existence of such a decomposition as soon as k is a continuous SPD kernel, which Matérn kernels satisfy, as proven by Proposition 3.3, and $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. The Woodbury inversion formula is then used to rewrite the kriging estimator in a computationally efficient expression.

3.4.1 Mercer’s Theorem

We start by generalising Definition 3.2 to functional operators on $\mathcal{L}^2(\mathcal{U})$.

Definition 3.5 (Self-adjoint positive semi-definite operator). *A functional operator T on $\mathcal{L}^2(\mathcal{U})$ is self-adjoint positive semi-definite (SPSD) if:*

$$\begin{cases} \forall \phi, \phi' \in \mathcal{L}^2(\mathcal{U}), \langle T\phi | \phi' \rangle = \langle \phi | T\phi' \rangle & \text{self-adjoint} \\ \forall \phi \in \mathcal{L}^2(\mathcal{U}), \langle \phi | T\phi \rangle \geq 0 & \text{positive semi-definite} \end{cases} \quad (3.64)$$

where $\langle \cdot | \cdot \rangle$ is the following scalar product:

$$\forall \phi, \phi' \in \mathcal{L}^2(\mathcal{U}) \mapsto \langle \phi | \phi' \rangle \triangleq \int_{\mathcal{U}} \phi(x)^\top \phi'(x) dx \quad (3.65)$$

Proposition 3.11 (Mercer operator). *Assume that $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$ and k is a continuous SPD kernel. The Mercer operator:*

$$\mathcal{K} : \begin{cases} \mathcal{L}^2(\mathcal{U}) \rightarrow \mathcal{L}^2(\mathcal{U}) \\ \phi \mapsto \left(x \mapsto \int_{\mathcal{U}} k(x, y) \phi(y) dy \right) \end{cases} \quad (3.66)$$

is a compact SPSP functional operator.

Proof. See section B.8. □

Proposition 3.12 (Mercer's theorem). *Under the assumptions of Proposition 3.11, there exists a non-increasing sequence $(\lambda_i)_{i \in \mathbb{N}^*}$ of strictly positive eigenvalues of $\mathcal{K}$ converging to 0, and an associated orthonormal family $(\psi_i)_{i \in \mathbb{N}^*}$ of continuous $\mathbb{R}^{d_z}$ -valued eigenfunctions, such that:*

$$\forall x, x' \in \mathcal{U}, k(x, x') = \sum_{i=1}^{+\infty} \lambda_i \psi_i(x) \psi_i(x')^\top \quad (3.67)$$

Furthermore, this series converges absolutely and uniformly on $\mathcal{U} \times \mathcal{U}$.

Proof. See section B.9. □

Remark 3.12 (Definiteness without invertibility). *Since all the eigenvalues λ_i are strictly positive, the quadratic form:*

$$\phi \in \mathcal{L}^2(\mathcal{U}) \mapsto \langle \phi | \mathcal{K} \phi \rangle = \sum_{i=1}^{+\infty} \lambda_i \langle \phi | \psi_i \rangle^2 \quad (3.68)$$

is strictly positive for every $\phi \neq 0$, hence positive definite. For a symmetric matrix, definiteness amounts to invertibility, but this equivalence fails in infinite dimension. In particular, since $\mathcal{K}$ is compact with $\lambda_i \rightarrow 0$, it admits no positive lower bound on the unit sphere, so 0 belongs to its spectrum. $\mathcal{K}^{-1}$ is thus unbounded, and so the Mercer operator is positive definite yet not invertible.

In finite dimension, strict positivity and invertibility coincide and jointly characterise SPD matrices, whereas this is not the case in infinite dimension. This is not restrictive, however, as only the finite Gram matrices induced by k are inverted throughout this document.

3.4.2 Efficient Kriging Estimation

Proposition 3.13 (Reduced-rank approximation). *For a continuous SPD kernel k , we define the rank of k as the dimension of the image of $\mathcal{L}^2(\mathcal{U})$ by its associated Mercer operator:*

$$\text{rank}(k) \triangleq \dim(\mathcal{K}(\mathcal{L}^2(\mathcal{U}))) \quad (3.69)$$

Under the assumptions of Proposition 3.11, the truncated kernel:

$$k_{r_{\text{KL}}} : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \sum_{i=1}^{r_{\text{KL}}} \lambda_i \psi_i(x) \psi_i(x')^\top \end{cases} \quad (3.70)$$

is a continuous SPSD kernel of rank at most $r_{\text{KL}} \in \mathbb{N}^*$. It converges uniformly towards k as $r_{\text{KL}} \rightarrow +\infty$, and minimises $\|k - k_{r_{\text{KL}}}\|_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}$ among all kernels of rank at most r_{KL} .

Proof. Let $r_{\text{KL}} \in \mathbb{N}^*$ and $\phi \in \mathcal{L}^2(\mathcal{U})$. We have:

$$\mathcal{K}_{r_{\text{KL}}} \phi = \sum_{i=1}^{r_{\text{KL}}} \lambda_i \langle \phi | \psi_i \rangle \psi_i \quad (3.71)$$

where $\mathcal{K}_{r_{\text{KL}}}$ is the Mercer operator associated with $k_{r_{\text{KL}}}$. Every $\mathcal{K}_{r_{\text{KL}}} \phi$ thus belongs to the subspace of continuous functions spanned by $\psi_1, \dots, \psi_{r_{\text{KL}}}$. The orthonormality of the basis ensures that this subspace is of dimension r_{KL} , and thus the rank of $k_{r_{\text{KL}}}$ is at most r_{KL} . The uniform convergence of $(k_{r_{\text{KL}}})_{r_{\text{KL}} \in \mathbb{N}^*}$ has already been proven by Proposition 3.12.

Finally, applying the Eckart-Young-Mirsky theorem [76, 77] to $\mathcal{K}$ and using the orthonormality of $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ yields:

$$\min_{\substack{\tilde{k} \in \mathcal{L}^2(\mathcal{U} \times \mathcal{U}) \\ \text{rank}(\tilde{k}) \leq r_{\text{KL}}}} \|k - \tilde{k}\|_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}^2 = \sum_{j > r} \lambda_j^2 = \|k - k_{r_{\text{KL}}}\|_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}^2 \quad (3.72)$$

□

In compact form, the eigenfunctions are collected into:

$$\Psi(x) \triangleq \begin{bmatrix} \psi_1(x) & \cdots & \psi_{r_{\text{KL}}}(x) \end{bmatrix} \in \mathcal{M}_{d_z, r_{\text{KL}}}(\mathbb{R}) \quad (3.73)$$

and the corresponding eigenvalues are gathered as:

$$\Lambda \triangleq \begin{bmatrix} \lambda_1 & & (0) \\ & \ddots & \\ (0) & & \lambda_{r_{\text{KL}}} \end{bmatrix} \in \mathcal{M}_{r_{\text{KL}}, r_{\text{KL}}}(\mathbb{R}) \quad (3.74)$$

By Proposition 3.12, the eigenvalues λ_i are strictly positive, so that $\Lambda \succ 0$ and Λ^{-1} is well-defined. Proposition 3.13 can thus be rewritten as:

$$k(x, x') \approx \Psi(x) \Lambda \Psi(x')^\top \quad (3.75)$$

and then:

$$G(\tilde{\mathbf{x}}) \approx \Psi(\tilde{\mathbf{x}}) \Lambda \Psi(\tilde{\mathbf{x}})^\top + I_{\tilde{N}} \otimes \tilde{R}, \quad \Psi(\tilde{\mathbf{x}}) = \begin{bmatrix} \Psi(\tilde{x}_1) \\ \vdots \\ \Psi(\tilde{x}_{\tilde{N}}) \end{bmatrix} \quad (3.76)$$

Proposition 3.14 (Reduced-rank simple kriging). *Let:*

$$\tilde{G}(\tilde{\mathbf{x}}) \triangleq \Lambda^{-1} + \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{N}} \otimes \tilde{R} \right)^{-1} \Psi(\tilde{\mathbf{x}}) \in \mathcal{M}_{r_{\text{KL}}, r_{\text{KL}}}(\mathbb{R}) \quad (3.77)$$

be the reduced-rank Gram matrix. Under the reduced-rank approximation of (3.76), the centred SK estimator of Proposition 3.5 and its error covariance matrix reduce to:

$$\begin{cases} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{N}} \otimes \tilde{R} \right)^{-1} \tilde{\mathbf{z}} \\ R_{\text{SK}}(x, \tilde{\mathbf{x}}) \approx R + \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{cases} \quad (3.78)$$

which require only the inversion of $\tilde{G}(\tilde{\mathbf{x}})$.

When there exists $\sigma > 0$ such that $\tilde{R} = \sigma^2 I$, (3.78) further simplify to:

$$\begin{cases} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \Psi(\tilde{\mathbf{x}})^\top \tilde{\mathbf{z}} \\ R_{\text{SK}}(x, \tilde{\mathbf{x}}) \approx R + \sigma^2 \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \Psi(x)^\top \end{cases} \quad (3.79)$$

These expressions remain valid at $\sigma = 0$ provided $\Psi(\tilde{\mathbf{x}})$ has full column rank.

Proof. See section B.10. □

Remark 3.13 (Construction of the eigenfunctions). While Mercer's theorem establishes the existence of the decomposition presented by (3.67), it does not provide a closed-form expression for $k_{r_{\text{KL}}}$. In practice [78], additional assumptions on k , such as stationarity or specific spectral properties, are usually made in order to derive an explicit approximation of the form of Proposition 3.13.

3.4.3 Karhunen-Loève Decomposition

TODO [79]

3.5 Reproducing Kernel Hilbert Spaces

Beyond regression, the covariance kernel also induces a functional Hilbert space, known as reproducing kernel Hilbert space (RKHS) [80, 81]. The RKHS representation is complementary to the probabilistic viewpoint described throughout this chapter. Indeed, while the GP describes a probabilistic distribution over functions, the RKHS associated with its covariance kernel provides a deterministic functional space in which such functions can be represented, manipulated, or approximated. For the sake of simplicity, we assume that $d_z = 1$, so that the reproducing property can be written in its most natural form.

3.5.1 Reproducing Property and Definition

Let $\mathcal{U}$ be an arbitrary set, and let $(\mathcal{H}, \langle \cdot | \cdot \rangle_{\mathcal{H}})$ denote a Hilbert space of functions from $\mathcal{U}$ to $\mathbb{R}$. Definition 3.6 and proposition 3.15 describe the conditions for this space to be an RKHS.

Definition 3.6 (Reproducing kernel Hilbert space). *A Hilbert space $(\mathcal{H}, \langle \cdot | \cdot \rangle_{\mathcal{H}})$ of functions from $\mathcal{U}$ to $\mathbb{R}$ is an RKHS if there exists a function $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ such that:*

$$\begin{cases} \forall x \in \mathcal{U}, k(x, \cdot) \in \mathcal{H} \\ \forall x \in \mathcal{U}, \forall f \in \mathcal{H}, \langle f | k(x, \cdot) \rangle_{\mathcal{H}} = f(x) \end{cases} \quad (3.80)$$

This second equation is the reproducing property of $\mathcal{H}$.

Proposition 3.15 (Reproducing kernel). *When such a k exists, it is symmetric and unique. It is called the reproducing kernel of $\mathcal{H}$.*

Proof. Let k_1 satisfy Definition 3.6, and let $x, x' \in \mathcal{U}$. Then:

$$\begin{aligned} k_1(x, x') &= (k_1(x, \cdot))(x') \\ &= \langle k_1(x, \cdot) | k_1(x', \cdot) \rangle_{\mathcal{H}} \\ &= \langle k_1(x', \cdot) | k_1(x, \cdot) \rangle_{\mathcal{H}} \\ &= k_1(x', x) \end{aligned} \quad (3.81)$$

so k_1 is symmetric.

Assume that there exists another function k_2 that verifies Definition 3.6. It comes that:

$$\begin{aligned} k_2(x, x') &= (k_2(x, \cdot))(x') \\ &= \langle k_2(x, \cdot) | k_1(x', \cdot) \rangle_{\mathcal{H}} \\ &= \langle k_1(x', \cdot) | k_2(x, \cdot) \rangle_{\mathcal{H}} \\ &= k_1(x', x) \\ &= k_1(x, x') \end{aligned} \quad (3.82)$$

and so $k_2 = k_1$. □

It can be shown [82] that $\mathcal{H}$ is an RKHS if, and only if, the evaluation functional at every point of the domain:

$$e_x : f \in \mathcal{H} \mapsto f(x) \quad (3.83)$$

is continuous on $\mathcal{H}$ for every $x \in \mathcal{U}$. Its reproducing kernel is necessarily SPSP. Conversely, every SPSP kernel k admits a unique RKHS $\mathcal{H}_k$ for which k is the reproducing kernel. Such an RKHS admits an explicit construction, as detailed in Proposition 3.16.

Proposition 3.16 (Native RKHS). *Let k be an SPSP kernel, let $\mathcal{H}_k$ be the completion of the set spanned by $\{k(x, \cdot), x \in \mathcal{U}\}$, and let $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ be defined on the generators by:*

$$\forall x, x' \in \mathcal{U}, \langle k(x, \cdot) | k(x', \cdot) \rangle_{\mathcal{H}_k} \triangleq k(x, x') \quad (3.84)$$

Then $(\mathcal{H}_k, \langle \cdot | \cdot \rangle_{\mathcal{H}_k})$ is a Hilbert space and an RKHS. Its reproducing kernel is k , and it is the only RKHS for which k is the reproducing kernel. It is called the native RKHS of k .

Proof. See section B.11. □

TODO: reprendre ici

Let $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ be a scalar SPSP kernel on an arbitrary set $\mathcal{U}$. Definition 3.6 formally defines the RKHS induced by k .

Definition 3.7 (Reproducing kernel Hilbert space). *The reproducing kernel Hilbert space (RKHS) $\mathcal{H}_k$ associated with k is the completion of the span of $\{k(\cdot, x) \mid x \in \mathcal{U}\}$ for the inner product defined on the generators by:*

It is characterised by the reproducing property. For all $\psi \in \mathcal{H}_k$ and $x \in \mathcal{U}$,

$$\psi(x) = \langle \psi \mid k(\cdot, x) \rangle_{\mathcal{H}_k} \quad (3.85)$$

The reproducing property (3.85) states that point evaluation amounts to applying a continuous linear functional on $\mathcal{H}_k$, which endows the space with a regularity entirely dictated by k . This space provides a variational reading of kriging [83].

Proposition 3.17 (Kriging as minimum-norm interpolation). *In the centred case, the SK estimator (3.35) is the solution of the regularised least-squares problem in $\mathcal{H}_k$:*

$$z_s^* = \operatorname{argmin}_{\psi \in \mathcal{H}_k} \left(\frac{1}{\tilde{R}} \sum_{i=1}^{\tilde{N}} (\psi(\tilde{x}_i) - \tilde{z}_i)^2 + \|\psi\|_{\mathcal{H}_k}^2 \right) \quad (3.86)$$

which, in the noise-free limit $\tilde{R} \rightarrow 0$, reduces to the minimum-norm interpolant $\operatorname{argmin}\{\|\psi\|_{\mathcal{H}_k} \mid \psi(\tilde{x}_i) = \tilde{z}_i\}$.

Proof. By the representer theorem [84, 85], a minimiser of (3.86) lies in the finite-dimensional span of $\{k(\cdot, \tilde{x}_i)\}_{i=1}^{\tilde{N}}$, i.e. $\psi = \sum_i \alpha_i k(\cdot, \tilde{x}_i)$. The reproducing property gives $\psi(\tilde{x}_j) = \sum_i \alpha_i k(\tilde{x}_j, \tilde{x}_i)$ and $\|\psi\|_{\mathcal{H}_k}^2 = \alpha^\top k(\tilde{X}, \tilde{X}) \alpha$, so (3.86) becomes the quadratic problem $\operatorname{argmin}_\alpha \tilde{R}^{-1} \|k(\tilde{X}, \tilde{X})\alpha - \tilde{Z}\|^2 + \alpha^\top k(\tilde{X}, \tilde{X})\alpha$. Cancelling its gradient gives $\alpha = G^{-1} \tilde{Z}$, hence $\psi(x) = k(x, \tilde{X}) G^{-1} \tilde{Z}$, which is (3.35) for $m = 0$. $\square$

Remark 3.14 (GP and RKHS correspondence). *The two viewpoints are complementary [86]. The GP model gives a probabilistic reading of kriging (posterior mean and variance under a Gaussian prior), while the RKHS gives a deterministic one (minimum-norm interpolant in a function space). Writing the Mercer decomposition of the kernel over its eigenpairs $(\lambda_j, \phi_j)_{j \geq 1}$ as $k(x, x') = \sum_{j \geq 1} \lambda_j \phi_j(x) \phi_j(x')$, a function $\psi = \sum_{j \geq 1} c_j \phi_j$ belongs to $\mathcal{H}_k$ with squared norm $\|\psi\|_{\mathcal{H}_k}^2 = \sum_{j \geq 1} c_j^2 / \lambda_j$; the norm thus penalises most the components carried by low-eigenvalue, rapidly oscillating eigenfunctions, so the decay of the eigenvalues λ_j sets the smoothness enforced by k . Choosing a kernel therefore amounts to choosing a regularity class for the unknown field. A smooth kernel such as the SE kernel yields infinitely differentiable interpolants, whereas a rougher Matérn kernel admits functions of limited smoothness.*

Espaces natifs : [87]

Part II

Navigating on Gaussian Processes

CHAPTER 4

KRIGING-BASED NAVIGATION

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4.1 Introduction

Assume a navigation scenario similar to that of chapter 2:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \quad (4.1)$$

with the exception that h_k is unknown, and the observation noise is additive for the sake of simplicity. Since the likelihood density cannot be evaluated without an observation model, Bayesian data fusion is no longer applicable, and we may only rely on inertial navigation to estimate the state x_k .

In our scenarios, the observation function describes the measurement of a physical field, which does not change over the course of the mission, by an embedded sensor at each time step. Furthermore, we assume that the environment has been sampled before the mission, whether during a previous reconnaissance mission, or using a known but deteriorated map. This is formally modelled by assumption 4.1.

Assumption 4.1 (Observation model). *The observation function is a time-invariant GP, and its noise is an additive Gaussian white noise of covariance matrix $R_k \succeq 0$:*

$$z_k = h(x_k) + n_k, \quad n_k \sim \mathcal{N}(0, R_k) \quad (4.2)$$

where $m : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$ is the potentially unknown mean of the GP, and $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$ is its chosen covariance kernel. Here, n_k plays the role of the query noise ε of section 3.3, and R_k is its associated covariance R .

We also have access to $\tilde{N} \in \mathbb{N}^*$ distinct training inputs $\tilde{\mathbf{x}}$ and their noisy observations $\tilde{\mathbf{z}}$, degraded by a noise of covariance $\tilde{R} \succeq 0$, in the sense of section 3.3.

Remark 4.1 (Time-varying observation function). *In full generality, the observed physical field could vary over time, and so the underlying GP would need to include a temporal input. While the kriging theory perfectly works with this additional parameter, it would require the training data to be sampled over time as well as space.*

This chapter is organised as follows:

- Section 4.2 combines chapters 2 and 3 to create a kriging-based framework for autonomous navigation in a poorly-known environment.
- Section 4.3 discusses non-stationary observation fields, and the necessity of training data selection for kriging.
- Section 4.4 improves upon this framework by clustering the particle cloud to better capture the non-stationarities of the field.
- Finally, section 4.6 quantifies the estimation error committed by subsampling the training data used for kriging.

4.2 Kriging the Observation Model

Using assumption 4.1, the likelihood follows:

$$z_k | x_k, \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_K(x_k, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g), R_K(x_k, \tilde{\mathbf{x}}; g)) \quad (4.3)$$

where $\hat{z}_K$ denotes any of the three kriging estimators, R_K is the corresponding covariance matrix, and m and g are optional arguments depending on the chosen kriging method. In particular, R_K already carries the query noise R_k , so it must not be added a second time.

TODO: à reprendre

Remark 4.2 (Correlated map error). *The filtering recursion does not call for (4.3), but for $p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}, \tilde{\mathbf{z}})$. Marginalising over the field, then using that the observation noise is white once h is given:*

$$\begin{aligned} p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}, \tilde{\mathbf{z}}) &= \int p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}, \tilde{\mathbf{z}}, h) p(h | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}, \tilde{\mathbf{z}}) dh \\ &= \int p(z_k | x_k, h) p(h | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}, \tilde{\mathbf{z}}) dh \end{aligned} \quad (4.4)$$

The first factor is the one (4.3) uses as well. The second is not: $\mathbf{z}_{1:k-1}$ are noisy evaluations of h at known inputs, of the same nature as $\tilde{\mathbf{z}}$, and they update its posterior. Integrating against that posterior is kriging over the augmented training set, in the sense of `th:kriging_update`:

$$z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}, \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_K(x_k, [\tilde{\mathbf{x}}, \mathbf{x}_{0:k-1}], [\tilde{\mathbf{z}}, \mathbf{z}_{1:k-1}]; m, g), R_K(x_k, [\tilde{\mathbf{x}}, \mathbf{x}_{0:k-1}]; g)) \quad (4.5)$$

Using (4.3) therefore freezes the training set at $\tilde{\mathbf{x}}$ instead of appending to it what the mission measures. Both laws coincide at the first time step, and again when the survey determines the field. Appending otherwise requires the states at which the observations were acquired, which are precisely the quantities being estimated.

The filter then counts as independent evidence two observations that carry almost the same reconstruction error, so its covariance shrinks faster than it should and a reconstruction bias reinforces itself instead of averaging out. The severity of the approximation is measured by α_{corr} , the number of consecutive steps spent within one correlation length of that error, that is, the ratio of this length to the distance $v \Delta_t$ covered in one time step. It stays close to 1 as long as a time step covers more than the spacing of the training data. None of the usual guards is applied here, neither inflating R_k , nor discarding observations at the rate at which the field decorrelates, nor modelling the error as a Gauss-Markov process along the trajectory and augmenting the state accordingly [88].

4.2.1 Choosing the Kriging Method

Section 3.3 leaves the choice between the three kriging methods open. Table 4.1 presents a $\mathbb{R} \rightarrow \mathbb{R}$ kriging scenario, from which every kriging estimator attempts to reconstruct the true function. A numerical comparison of the estimators is presented at the end of this section.

Five kriging estimators, two simple, one ordinary, and two universal, are evaluated on this scenario, as illustrated by Figures 4.1 to 4.3 respectively. The continuous blue line draws the true function, while the training points are represented by the black crosses. The kriged function is in dashed red, and its associated confidence interval is the pale red envelope. Since h is scalar, the confidence interval is computed as:

$$\left[\hat{z}_K(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) - 3\sqrt{R_K(x, \tilde{\mathbf{x}}; g)}, \hat{z}_K(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) + 3\sqrt{R_K(x, \tilde{\mathbf{x}}; g)} \right] \quad (4.6)$$

True function	$h : x \in \mathbb{R} \mapsto x \sin(x) + 10 \in \mathbb{R}$
GP mean	$m : x \mapsto 10$ or $m : x \mapsto 0$
GP covariance	$k : x, x' \in \mathbb{R} \mapsto 5^2 \exp\left(-\frac{\ x - x'\ _2^2}{2 \times 1.2^2}\right)$
Training data	$\tilde{\mathbf{x}} = \{0.5, 1.5\} \cup \{5.5, 6.5, \dots, 9.5\}$
Training noise	$\tilde{R} = 1$
Query noise	$R = 0$
Trend functions	$g : x \mapsto \begin{bmatrix} 1 & x \end{bmatrix}$ or $g : x \mapsto \begin{bmatrix} 1 & x & \dots & x^5 \end{bmatrix}$
Prediction domain	$[-3, 13]$

Table 4.1: An example kriging scenario. The SK and UK estimators are given either mean function and trend bases to illustrate their reliance on prior knowledge, as shown in their respective figures.

and contains $h(x)$ with probability $\text{erf}(3/\sqrt{2}) \approx 99.73\%$, where erf is the Gauss error function.

While it is the computationally simplest estimator, the SK estimator relies heavily on the prior knowledge of the mean function m , and poorly represents the true function in regions with a lower density of training points when that knowledge is wrong. This is particularly apparent in the gap within the training points distribution, and outside the sampled region.

As motivated by section 3.3, ordinary kriging tackles this issue by directly estimating a constant mean function instead of relying on a given prior. Its behaviour is similar to that of the SK estimator with the correct mean function, at the cost of a slightly increased confidence interval. This is due to the additional uncertainty associated with estimating the mean, and is quantified latter in this section.

For universal kriging, we considered polynomial means of degree 1 and 5. The affine basis behaves much like ordinary kriging, which it extends by a single coefficient. The affine UK envelope is barely wider than that of OK near the samples but grows faster outside the interpolation domain.

On the other hand, the degree 5 model overfits the seven training points with six coefficients, which causes the corresponding estimator to diverge from the true function farther away from the training data. This is reflected in the UK covariance, which indicates high uncertainty in poorly sampled regions.

Since the training observations are noisy, a single draw of $\tilde{\mathbf{z}}$ cannot accurately represent the performance of the estimators above. To alleviate this issue, the training observations are redrawn over $n_{\text{MC}} = 1000$ Monte Carlo simulations, in order to compute the root mean squared error (RMSE):

$$\text{RMSE}_{\hat{z}_K} : x \mapsto \sqrt{\frac{1}{n_{\text{MC}}} \sum_{i=1}^{n_{\text{MC}}} \|\hat{z}_K(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}^{(i)}; m, g) - h(x)\|_2^2} \quad (4.7)$$

of the estimators, where $\tilde{\mathbf{z}}^{(i)}$ denotes the i -th draw of the training observations.

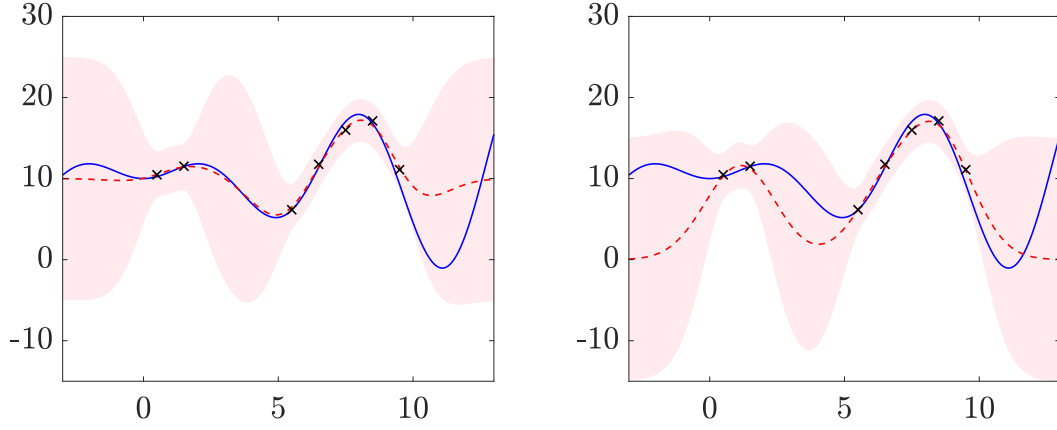


Figure 4.1: Simple kriging under a correct mean function, on the left, and under a wrong one, on the right. Since R_{SK} does not depend on m in (3.35), both shaded areas have the same width along the x -axis.

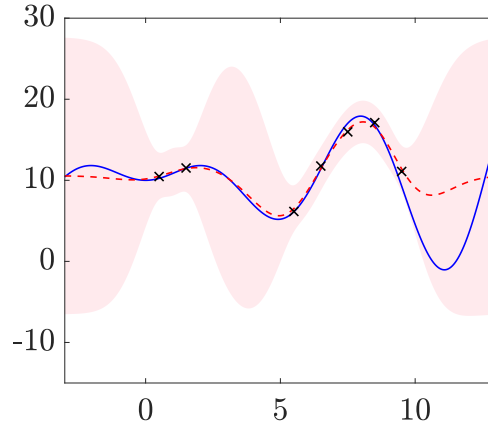


Figure 4.2: Simultaneous kriging of the training data and mean estimation by ordinary kriging. The constant mean is reconstructed here at $\hat{m}_{\text{OK}} = 10.56$.

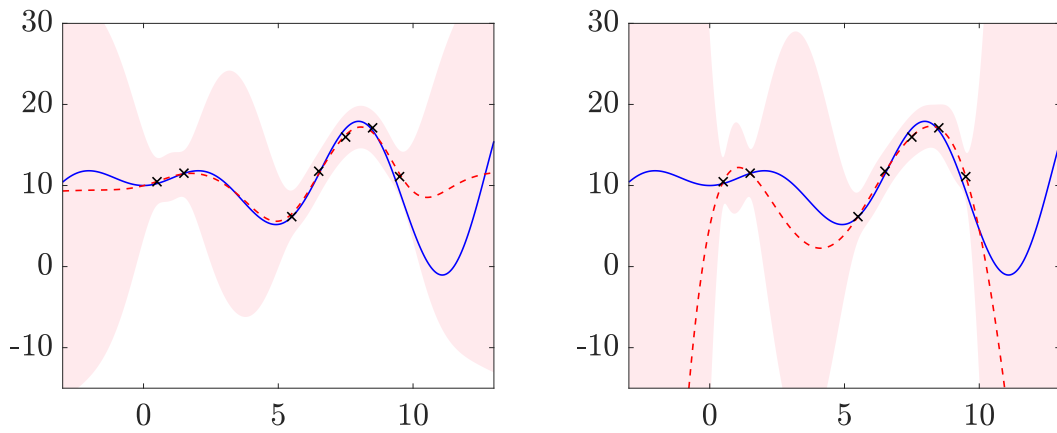


Figure 4.3: Universal kriging over an affine trend basis, on the left, and over a degree five monomial basis, on the right. The trend coefficients are estimated here at $\hat{m}_{\text{UK}} = [9.78, 0.15]^\top$ and $\hat{m}_{\text{UK}} = [4.95, 15.85, -10.82, 2.40, -0.20, 0.01]^\top$ respectively.

Their average RMSE (ARMSE):

$$\text{ARMSE}_{\hat{z}_K}(E) \triangleq \frac{\int_E \text{RMSE}_{\hat{z}_K}(x) dx}{\int_E dx} \quad (4.8)$$

is then computed over three subsets:

$$\begin{cases} E_{\text{in}} \triangleq [0.5, 1.5] \cup [5.5, 9.5] \\ E_{\text{gap}} \triangleq [1.5, 5.5] \\ E_{\text{out}} \triangleq [-3, 0.5] \cup [9.5, 13] \end{cases} \quad (4.9)$$

of the prediction domain, as presented in Table 4.2.

Estimator	ARMSE		
	in	gap	out
Simple, correct mean	0.85	1.15	3.23
Simple, wrong mean	0.87	3.69	5.88
Ordinary	0.85	1.26	3.25
Universal, affine	0.85	1.25	3.54
Universal, degree 5	0.98	4.02	70.88

Table 4.2: ARMSE of the five estimators over the three subsets E_{in} , E_{gap} , and E_{out} of the prediction domain.

All five estimators perform similarly in the densely sampled region, their ARMSE values ranging from 0.85 to 0.87, with the exception of the overfitted UK estimator. In this region, the high density of training data limits the influence of the mean on the estimation, and is thus not informative for assessing the individual impact of each estimator.

In the gap region, the choice of mean becomes more influential due to the sparsity of training points. The SK estimator with the correct mean achieves an ARMSE of 1.15, while using an incorrect one increases the error to 3.69. In contrast, both the OK and affine UK estimators remain close to the correctly informed SK estimator, with an increase of less than 10% in ARMSE. The degree 5 UK estimator performs worst, its ARMSE representing a 250% increase over that of the correct SK estimator.

This tendency suggests that estimating the mean from the available observations generally makes the estimator more robust, provided that the mean model does not overfit the data. It is even more pronounced in the extrapolation region, where the SK estimator with an incorrect mean, and especially the high-order UK estimator, show a substantial increase in ARMSE compared with the three other estimators. However, this robustness comes at a cost, as neither the OK nor the affine UK estimator ever outperforms a correctly informed simple kriging.

Although the previous results highlight the differences in point-estimation accuracy between the five estimators, a key advantage of kriging is its ability to quantify

Estimator	Confidence half-width			$R_{\hat{m}}/R_K$		
	in	gap	out	in	gap	out
Simple, correct mean	2.71	9.05	11.58	0.0%	0.0%	0.0%
Simple, wrong mean	2.71	9.05	11.58	0.0%	0.0%	0.0%
Ordinary	2.71	9.55	12.73	0.4%	8.9%	15.0%
Universal, affine	2.72	9.74	16.55	0.8%	11.9%	43.0%
Universal, degree 5	3.22	15.70	320.03	20.5%	59.6%	94.5%

Table 4.3: Average confidence interval half-width and average share of the covariance caused by the estimation of the mean for the five estimators over each region. The share is computed as the pointwise ratio of $R_{\hat{m}}$ over $R_K = R_{\hat{m}} + R_{SK}$.

the corresponding uncertainty. Table 4.3 compares the confidence intervals and the contribution of the mean estimation to the overall covariance for the five estimators.

This table confirms the trend observed in the ARMSE results, as the average confidence intervals also increase with distance from the training points. As shown by the first row, the kriging covariance grows on its own, since the lack of close samples causes the estimator to rely more on its mean. In such regions, the estimated mean becomes more influential, and thus accounts for an increasing share of the total uncertainty.

Furthermore, it must be noted that both SK confidence intervals are the same, showing that simple kriging is unable to account for an incorrectly specified prior mean. Both present a half-width of 9.05 in the gap, well inside the sampled range, although their ARMSE differ by a factor of three. This issue is particularly relevant in autonomous navigation scenarios, as a poorly informed SK estimator will be overconfident, resulting in skewed likelihood peaks. This, in turn, may lead to relevant particles being discarded while incorrect ones are retained, and so simple kriging is set aside.

Conversely, the degree 5 UK estimator fails by trying to fit six trend coefficients to only seven training points. Outside the sampled range, the mean becomes highly uncertain, and accounts for an average of 94.5% of the total covariance. Unlike simple kriging, the UK estimator propagates this uncertainty to its confidence interval, whose half-width reaches 320.03. From a computational point of view, universal kriging requires $\tilde{N} \geq r_{UK}$, and causes the matrix $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})$ to become ill-conditioned when the training set is too small relative to the number of trend functions.

Furthermore, every estimator requires the $\mathcal{O}(\tilde{N}^3)$ inversion of $G(\tilde{\mathbf{x}})$ once, after which, solving any related system costs $\mathcal{O}(\tilde{N}^2)$. Simple kriging thus requires an additional $\mathcal{O}(\tilde{N}^2)$ at each query point. Estimating the mean amounts to solving $r_{UK}d_z$ equations, which costs $\mathcal{O}(r_{UK}\tilde{N}^2)$ once and $\mathcal{O}(r_{UK}\tilde{N})$ per query. The inversion of $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}) \in \mathcal{S}_{r_{UK}d_z}^{++}(\mathbb{R})$ does not depend on $\tilde{N}$, and is negligible as soon as $r_{UK}d_z \ll \tilde{N}$. Table 4.4 gives the empirical cost of each estimator in floating-point operations (flop), for $\tilde{N} \in \{7, 100, 1600\}$.

From 7 to 1600 training points, the setup cost of simple kriging rises from 114 flops to 1.37 Gflop, and its per-query cost from 49 to 2.56 Mflop. In our scenarios, every particle is a query point at each time step, so real-time constraints rule out

Estimator	Setup (flops)			Per query (flops)		
$\tilde{N}$	7	100	1600	7	100	1600
Simple	114	333k	1.37G	49	10k	2.56M
Ordinary	+112	+20.2k	+5.12M	+14	+200	+3.2k
Universal, affine	+255	+40.8k	+10.3M	+28	+400	+6.4k
Universal, degree 5	+1.16k	+127k	+30.8M	+84	+1.2k	+19.2k

Table 4.4: Computational cost of the five estimators, split into the setup and the query costs. The simple kriging row is the cost of reconstructing the field, paid by every estimator, and the following ones are what estimating the mean adds to it. Since their cost is independent of the mean they were given the two simple estimators share the same row.

a training set of that size. The other end of the table is no more comfortable: on seven points, the affine basis costs twice the reconstruction of the field, 255 flops against 114, and the degree 5 basis ten times.

Ordinary kriging is thus retained for the remainder of this chapter. Estimating a constant mean from the observations is robust to misinformation, its ARMSE staying within 10% of a correctly informed simple kriging everywhere, when a wrong mean triples the error in the gap. It also reports that estimation, the term $R_{\hat{m}_{\text{OK}}}$ of remark 3.8 rising from 0.4% of the covariance near the samples to 8.9% in the gap, so its confidence interval widens exactly where its accuracy degrades. Finally, the additional $d_z \times d_z$ inversion it requires is the smallest trend a kriging estimator can carry, and remains affordable at both ends of table 4.4.

4.2.2 Hyperparameter Estimation

Recall that section 3.2.2 considered every kernel subsequently used for kriging to be fully known, with both their form and hyperparameters specified. This assumption is relaxed here by considering only the form to be known, while the hyperparameters θ must be estimated from the available data.

TODO: à vérifier, sous-section rédigée par Claude

Chapter 3 derives every result conditionally on the kernel, and section 4.2.1 compares its estimators with that kernel held fixed. This assumption is discharged here. A kernel is specified by a vector of hyperparameters θ whose components are of two kinds. Some describe the class of fields the GP is assumed to belong to, such as a regularity or a set of active input directions, and are posited by the modeller alongside the kernel family itself. The others describe the particular field being surveyed, such as the amplitude of its variations or the distance over which they decorrelate, and are read from the training data of assumption 4.1. Only the latter are estimated below.

No closed form is available for them. The mean enters the kriging equations linearly, which is what lets proposition 3.8 recover it by a generalised least squares regression, whereas θ acts through the kernel, hence through the inverse of a matrix it parameterises. Estimating it is therefore an optimisation problem, and the criterion retained is the likelihood the model assigns to the observations already collected, a procedure known as evidence maximisation [89].

That likelihood is available in closed form. Proposition 3.1 gives the joint law of any finite collection of field values, and the training observations are such a collection degraded by an independent Gaussian noise, so they are themselves jointly Gaussian. Writing $G_\theta \triangleq k_\theta(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R}$ for the noise-augmented Gram matrix, whose argument is dropped to keep the hyperparameters visible, and $\tilde{\mathbf{r}} \triangleq \tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})$ for the residual to the GP mean, their density reads:

$$p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \theta) = \frac{1}{\sqrt{(2\pi)^{\tilde{N}d_z} \det(G_\theta)}} \exp\left(-\frac{1}{2}\tilde{\mathbf{r}}^\top G_\theta^{-1}\tilde{\mathbf{r}}\right) \quad (4.10)$$

Proposition 4.1 (Maximum likelihood hyperparameters). *A hyperparameter vector θ maximising (4.10) satisfies, for each component θ_i :*

$$\text{tr}\left[\left((G_\theta^{-1}\tilde{\mathbf{r}})(G_\theta^{-1}\tilde{\mathbf{r}})^\top - G_\theta^{-1}\right) \frac{\partial G_\theta}{\partial \theta_i}\right] = 0 \quad (4.11)$$

Proof. Maximising (4.10) is equivalent to maximising its logarithm:

$$\ell(\theta) \triangleq \log p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \theta) = -\frac{1}{2}(\tilde{\mathbf{r}}^\top G_\theta^{-1}\tilde{\mathbf{r}} + \log \det(G_\theta)) - \frac{\tilde{N}d_z}{2} \log(2\pi) \quad (4.12)$$

A maximiser cancels every partial derivative. Using the differential of the matrix inverse, $\frac{\partial}{\partial \theta_i} G_\theta^{-1} = -G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} G_\theta^{-1}$, and Jacobi's formula, $\frac{\partial}{\partial \theta_i} \log \det(G_\theta) = \text{tr}\left(G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i}\right)$, it comes that:

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_i} = 0 &\iff \tilde{\mathbf{r}}^\top G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} G_\theta^{-1} \tilde{\mathbf{r}} - \text{tr}\left(G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i}\right) = 0 \\ &\iff \text{tr}\left[\left((G_\theta^{-1}\tilde{\mathbf{r}})(G_\theta^{-1}\tilde{\mathbf{r}})^\top - G_\theta^{-1}\right) \frac{\partial G_\theta}{\partial \theta_i}\right] = 0 \end{aligned} \quad (4.13)$$

using the cyclic property of the trace to rewrite the scalar quadratic form

$\tilde{\mathbf{r}}^\top G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} G_\theta^{-1} \tilde{\mathbf{r}}$ as $\text{tr}\left((G_\theta^{-1}\tilde{\mathbf{r}})(G_\theta^{-1}\tilde{\mathbf{r}})^\top \frac{\partial G_\theta}{\partial \theta_i}\right)$, since G_θ^{-1} is symmetric. $\square$

Remark 4.3 (Profiled mean). *The density (4.10) is written for a known mean, whereas the estimator retained in section 4.2.1 estimates it instead. Maximising ℓ over a constant mean at fixed θ cancels $C^\top G_\theta^{-1}\tilde{\mathbf{r}}$, and thus returns the very estimate $\hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ of proposition 3.8, evaluated under the current θ . The two estimations nest rather than compete, the residual $\tilde{\mathbf{r}}$ being taken to a mean recomputed in closed form at every evaluation of (4.10).*

The two terms of ℓ pull in opposite directions, which is what makes its maximum informative rather than degenerate. The quadratic form rewards a covariance ample enough to explain the observed residual, and keeps improving as the field is allowed to vary more freely, while the log-determinant penalises exactly that freedom, measuring the volume of observation space over which the model spreads its mass. Maximising their sum thus retains the least flexible field that remains consistent with the survey.

Remark 4.4 (Non-convexity). *The log-marginal-likelihood is generally non-convex in θ , so (4.11) is solved numerically by gradient-based optimisation [90], and may admit several local maxima corresponding to different interpretations of the data, e.g. a low-noise rapidly-varying field versus a high-noise slowly-varying one.*

The hyperparameters are thus obtained by ascending ℓ numerically, one step of which is given by Algorithm 4.1. Two implementation choices are worth stating. First, G_θ is never inverted, its Cholesky factor L such that $G_\theta = LL^\top$ yielding both the quadratic form and the log-determinant. This $\mathcal{O}(\tilde{N}^3)$ factorisation is the one table 4.4 already charges to the reconstruction of the field, except that the ascent pays it once per step instead of once per mission, after which each component of the gradient costs $\mathcal{O}(\tilde{N}^2)$. Second, the ascent is carried out on $\eta \triangleq \log \theta$ rather than on θ itself, which enforces the positivity of the hyperparameters and turns each step into a multiplicative update, matching the scale invariance of quantities carrying a physical unit. The chain rule then gives $\frac{\partial G_\theta}{\partial \eta_i} = \theta_i \frac{\partial G_\theta}{\partial \theta_i}$, so that the kernel family enters the ascent through the single quantity $\frac{\partial G_\theta}{\partial \theta_i}$, and nowhere else.

Algorithm 4.1 Hyperparameter estimation by evidence maximisation

$G = k_{\text{exp}(\eta^{(t-1)})}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R}$	▷ Noise-augmented Gram matrix
$L \text{ s.t. } G = LL^\top$	▷ Cholesky factorisation
$\alpha = L^{-\top} (L^{-1} \tilde{\mathbf{r}})$	▷ Two triangular solves
$\ell = -\frac{1}{2} \tilde{\mathbf{r}}^\top \alpha - \sum_{j=1}^{\tilde{N}d_z} \log L_{jj} - \frac{\tilde{N}d_z}{2} \log(2\pi)$	▷ Log-evidence
$W = \alpha\alpha^\top - L^{-\top} L^{-1}$	
$\forall i, \nabla_i = \frac{1}{2} \text{tr} \left(W \frac{\partial G}{\partial \eta_i} \right)$	▷ Gradient of ℓ
$\Delta\eta = \text{Direction}(\nabla)$	▷ Gradient, conjugate gradient or quasi-Newton
$\eta^{(t)} = \eta^{(t-1)} + \rho \Delta\eta$	▷ Line search on the step length ρ

The step is iterated from $\eta^{(0)} = \log \theta^{(0)}$ until the gradient falls below a chosen tolerance, the estimate being $\hat{\theta} = \exp(\eta^{(t)})$ at the last iterate. Its starting point $\theta^{(0)}$ is a free input, and the non-convexity of ℓ makes it a meaningful one, since the ascent only reaches the maximum of the basin it is started in. It is therefore read from the survey rather than fixed arbitrarily, each hyperparameter being initialised by the empirical quantity it describes, an amplitude by the sample variance of $\tilde{\mathbf{z}}$ less the known noise variance, a correlation length by the mean distance between neighbouring training inputs, which is the shortest length the survey can resolve. Should several starting points be drawn around these values, the one reaching the highest ℓ is retained.

4.2.3 Simulation Scenario

Every navigation result of this chapter is produced by the same scenario, as described in this section. The observation field represented by h is a magnetic anomaly map of a $200\text{km} \times 200\text{km}$ region above Mexico, and extracted from [91]. The map is sampled on a regular grid of 201×201 points with a resolution of $1\text{km} \times 1\text{km}$, and is read by a bilinear interpolation between the points.

From this map are extracted $\tilde{N}$ training samples $(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$, where $\tilde{\mathbf{x}}$ is sampled from a regular grid, each node being independently jittered by up to 30% of the grid resolution along both axes. The grid resolution is varied from 2.5km to 7.5km, with increments of 1km. This corresponds to $\tilde{N} \in \{79^2, 56^2, 44^2, 36^2, 30^2, 26^2\}$. The corresponding observations $\tilde{\mathbf{z}}$ are corrupted by a Gaussian white noise of variance

$\tilde{R} = 10^2 \text{nT}^2$. The irregular grid models navigation errors during the prior survey mission, while the white noise represents the imperfect measurements of a real sensor.

The vehicle flies over the map at a constant speed of 500m/s along a mower pattern consisting of three parallel branches, each 80km long, and connected by half-turns with a radius of 25.5km. The trajectory is sampled every $\Delta_t = 5\text{s}$, and lasts $n_{\text{steps}} = 150$ steps. A single time step thus covers 2.50km.

The vehicle state consists of its horizontal position and velocity:

$$x_k = \begin{bmatrix} p_k^{(x)} & p_k^{(y)} & v_k^{(x)} & v_k^{(y)} \end{bmatrix}^\top, \quad d_x = 4 \quad (4.14)$$

and its propagation is degraded with a Gaussian white noise of covariance matrix:

$$Q_k = \begin{bmatrix} 100^2 & 0 & 0 & 0 \\ 0 & 100^2 & 0 & 0 \\ 0 & 0 & 1^2 & 0 \\ 0 & 0 & 0 & 1^2 \end{bmatrix} \quad (4.15)$$

whose position and velocity entries are respectively expressed in m^2 and m^2s^{-2} .

The observation function is the scalar ($d_z = 1$) OK estimation of the magnetic anomaly field, conditioned on the $\tilde{N}$ training points, and degraded by a Gaussian white noise of covariance $R_k = 5^2 \text{nT}^2$. The chosen kernel is a Matérn kernel with smoothness $\nu = 3/2$, which corresponds to the smallest half-integer for which the field and the kriging mean are differentiable, and its hyperparameters are estimated as described in section 4.2.2. A bilinear observation function on the same samples is also used as comparison.

The filter is an RPF, initialised on a cloud of $N = 5000$ particles, drawn around a point itself drawn from $\mathcal{N}(x_0, P_0)$, with:

$$P_0 = \begin{bmatrix} 8000^2 & 0 & 0 & 0 \\ 0 & 8000^2 & 0 & 0 \\ 0 & 0 & 2^2 & 0 \\ 0 & 0 & 0 & 2^2 \end{bmatrix} \quad (4.16)$$

in the same units, to prevent the particle cloud from being centred on the true initial state. The resampling threshold is fixed to $N_{\text{th}} = 0.5N$, and the regularisation coefficient is $\alpha_{\text{reg}} = 0.3$. This scenario is summarised by Table 4.5, and illustrated by Figure 4.4.

Each configuration is run over $n_{\text{MC}} = 1000$ Monte Carlo realisations, redrawing the initial state and white noises. The training data are also redrawn every 10 runs so that the metrics are averaged over the reconstruction as well as the trajectory. The chosen estimator is the MMSE estimator from (2.83). The performance of each one is read through two quantities:

- The estimation RMSE, defined as:

$$\text{RMSE}_{\hat{x}} : k \mapsto \sqrt{\frac{1}{n_{\text{MC}}} \sum_{i=1}^{n_{\text{MC}}} \left\| \varphi_{\text{pos}} \left(\hat{x}_k^{(i)} \right) - \varphi_{\text{pos}}(x_k) \right\|_2^2} \quad (4.17)$$

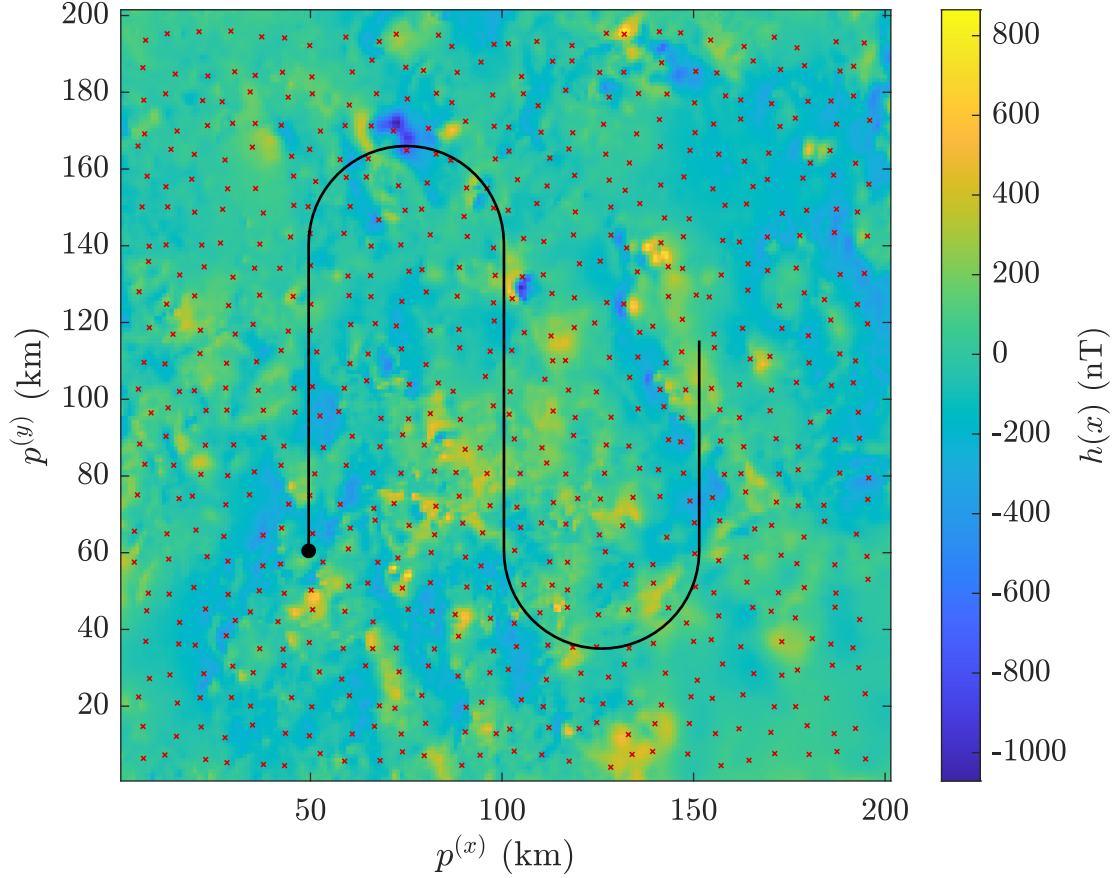


Figure 4.4: The magnetic field anomaly used throughout this chapter. The black line represents the overflow trajectory, and the red crosses correspond to the training points on a 7.5km-resolution grid.

where $\hat{x}_k^{(i)}$ is the state estimated by the i -th filter at time k , x_k is the corresponding true state, and φ_{pos} is the projection onto the positions: **TODO: définir une matrice à la place**

$$\varphi_{\text{pos}} : \begin{bmatrix} p^{(x)} & p^{(y)} & v^{(x)} & v^{(y)} \end{bmatrix}^\top \mapsto \begin{bmatrix} p^{(x)} & p^{(y)} \end{bmatrix}^\top \quad (4.18)$$

This metric gives an absolute value for the average error made by the filter.

- The normalised estimation error squared (NEES):

$$\text{NEES}_{\hat{x}} : k \mapsto \frac{1}{n_{\text{MC}} d_x} \sum_{i=1}^{n_{\text{MC}}} \left\| \hat{x}_k^{(i)} - x_k \right\|_{\hat{P}_k^{(i)}}^2 \quad (4.19)$$

where $\|\cdot\|_{\hat{P}_k^{(i)}}$ is the Mahalanobis norm associated with the error covariance:

$$\left\| \hat{x}_k^{(i)} - x_k \right\|_{\hat{P}_k^{(i)}}^2 \triangleq \left(\hat{x}_k^{(i)} - x_k \right)^\top \left(\hat{P}_k^{(i)} \right)^{-1} \left(\hat{x}_k^{(i)} - x_k \right) \quad (4.20)$$

This metric detects overconfident filters by using the Mahalanobis norm associated with the error covariance matrix rather than the standard Euclidean distance.

Map	Surface	200km $\times$ 200km
	Resolution	1km $\times$ 1km
Kriging	Samples	Jittered grid
	Jitter	0.3 $\times$ Resolution
	Resolution	$\in \{2.5, 3.5, \dots, 7.5\}$ km
	$\tilde{N}$	$\in \{79^2, 56^2, 44^2, 36^2, 30^2, 26^2\}$
	$\tilde{R}$	10^2nT^2
Trajectory	Form	3-branch mower
	Branch length	80km
	Turn radius	25.5km
	Speed	500m/s
	Δ_t	5s
	n_{steps}	150
State	d_x	4
	x_k	$\left[p_k^{(x)}, p_k^{(y)}, v_k^{(x)}, v_k^{(y)} \right]^\top$
	Q_k	$100^2 \text{m}^2, 1^2 \text{m}^2 \text{s}^{-2}$
Observation	d_z	1
	Function	OK or bilinear
	Kriging kernel	Matérn, $\nu = 3/2$
	R_k	5^2nT^2
Filter	Filter	RPF
	N	5000
	N_{th}	$0.5N$
	α_{reg}	0.3
	P_0	$8000^2 \text{m}^2, 2^2 \text{m}^2 \text{s}^{-2}$

Table 4.5: Simulation parameters for the navigation scenario. For improved readability, covariance matrices are represented by their diagonal coefficients only.

Similarly, a run is declared convergent under two different criteria, as each describes a different behaviour:

- A fixed threshold of 2000m, under which the median over the last ten steps of a filter must be. This metric is simple to read and ensures that the estimation is relatively close to the true state. However, the threshold value is arbitrary.
- A Mahalanobis criterion, where the median of (4.20) over the last ten steps must remain under the 99% quantile of a χ^2 law with d_x degrees of freedom, written $\chi_{d_x}^2(99\%)$. While this criterion follows the property of a Gaussian error, it primarily assesses the consistency between the estimation error and the uncertainty reported by the filter. Consequently, a filter with a large

estimation error may still satisfy this criterion if it reports a sufficiently large uncertainty.

4.2.4 Numerical Results

In this section, we compare the performance of the filter using 13 different observation functions:

- The first configuration corresponds to the known map, and serves as a reference case.
- The six next use ordinary kriging at different grid resolutions.
- The final six instead use bilinear interpolation on the same grid.

Before analysing the filters themselves, it is important to assess the quality of the reconstructed maps. Figure 4.5 presents the reconstruction ARMSE along the true trajectory for ordinary kriging and bilinear interpolation across the different grid resolutions, in the sense of (4.8).

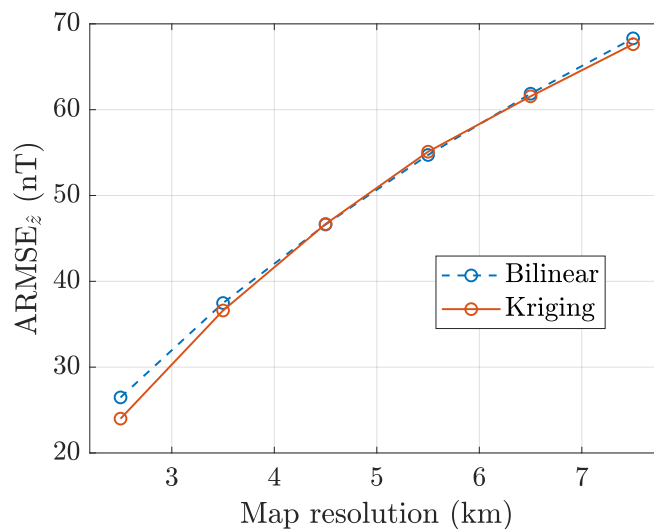


Figure 4.5: Reconstruction ARMSE of the field over the trajectory corridor for ordinary kriging and bilinear interpolation as a function of the training grid resolution. The domain E considered is a rectangular corridor around the true trajectory, with a $28km$ margin. It covers 74% of the map.

As one could have expected, denser grids lead to better reconstructions, the ARMSE evolving as the logarithm of the grid resolution. However, kriging and bilinear interpolation exhibit similar errors, within 1nT of each other at every resolution but the finest. Thus, the advantage of kriging cannot be primarily attributed to the reconstruction accuracy alone.

Focusing now on the filter performance, Figures 4.6 and 4.7 respectively compare the RMSE and NEES of the filter using the true map, kriging and bilinear interpolation on each of the six grid resolutions. For the sake of readability, both metrics are only computed on the converged runs with respect to the threshold criterion.

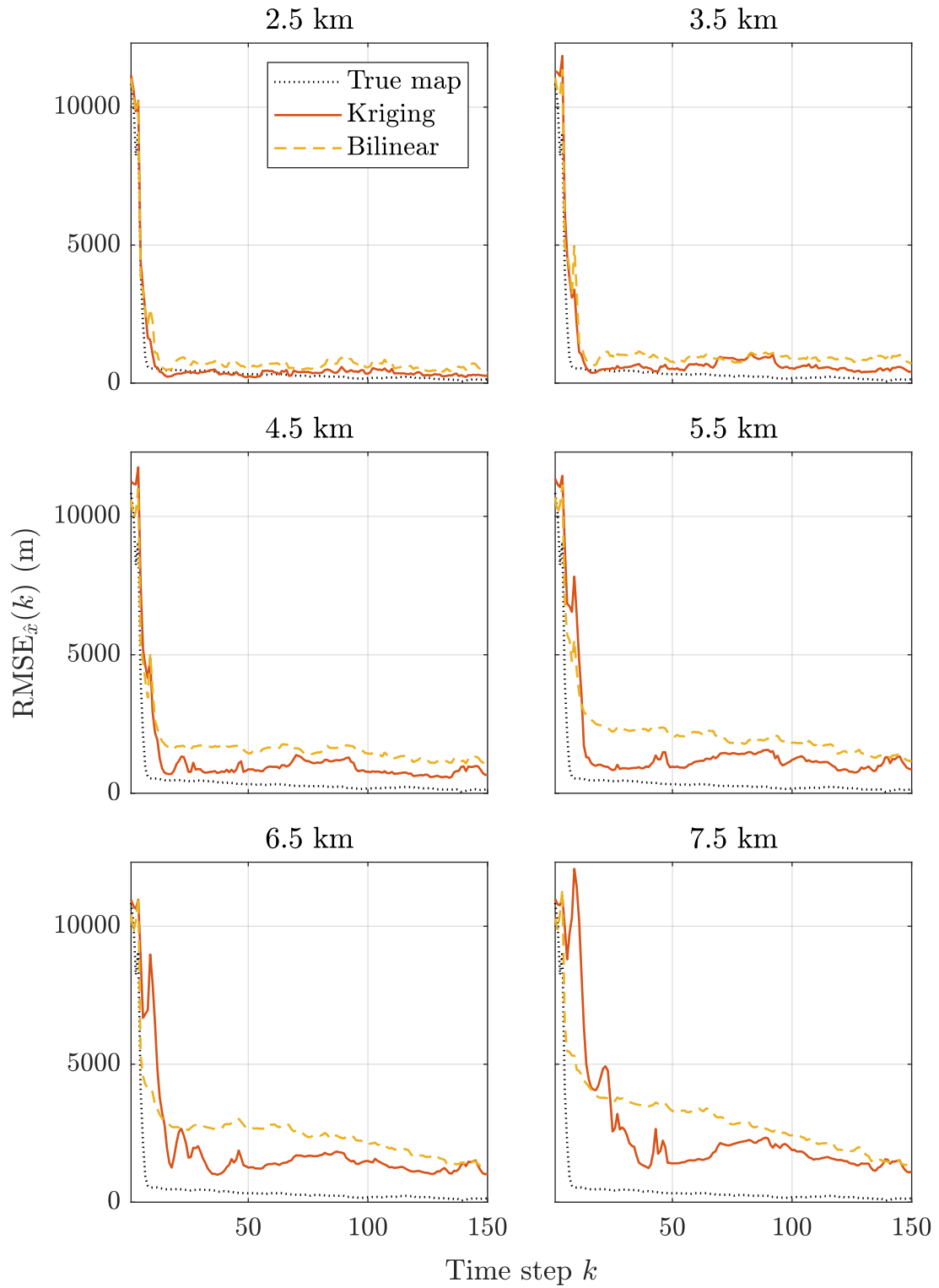


Figure 4.6: Estimation RMSE for the 13 configurations. Each panel corresponds to a different grid resolution, and the true map result is repeated for comparison.

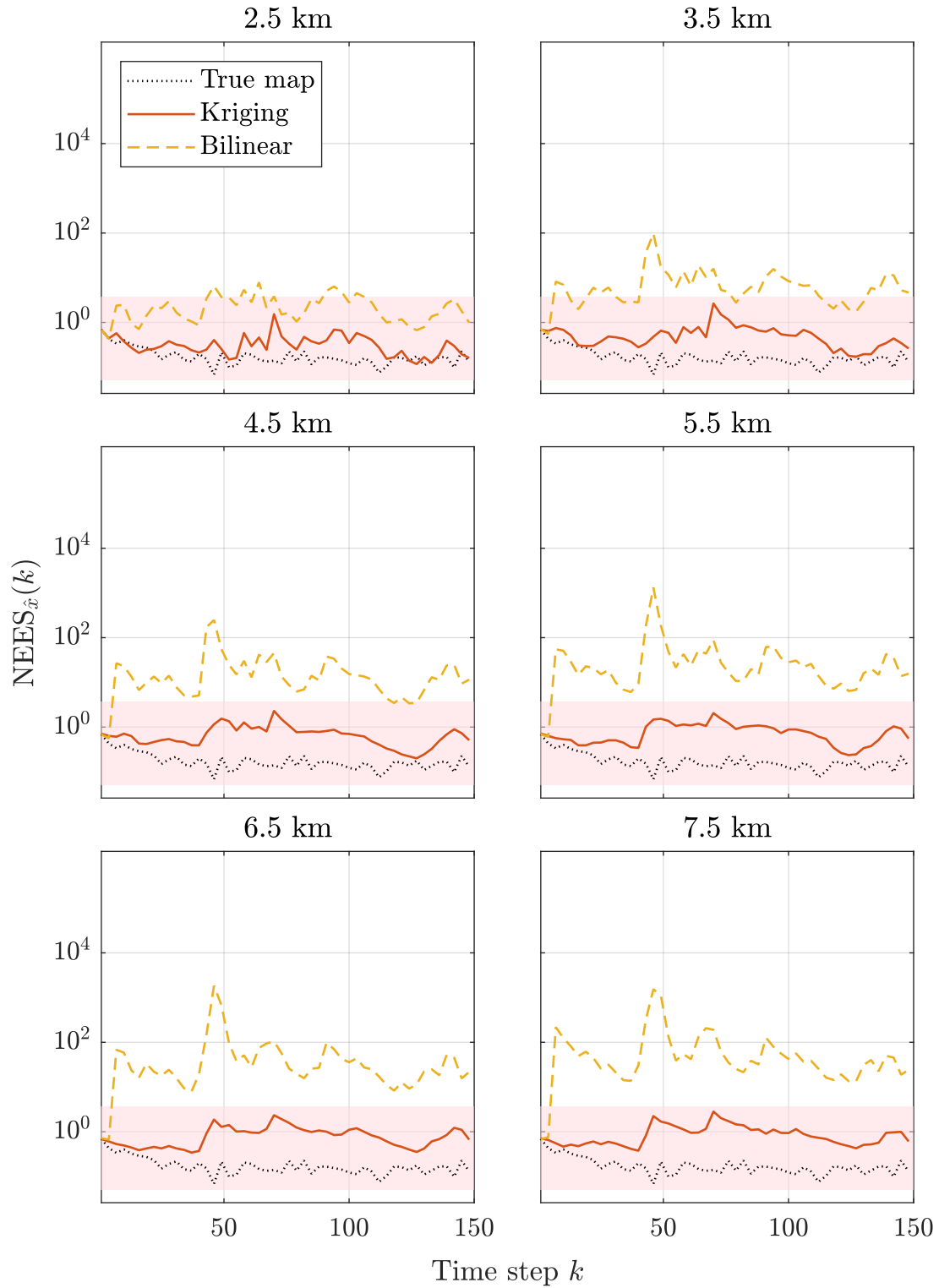


Figure 4.7: Estimation NEES for the 13 configurations. The shaded area delimits the quantiles of a χ^2 law with d_x degrees of freedom, within which the NEES of a consistent filter falls with probability 0.99.

Three main results can be drawn from the RMSE curves. First, filters with access to more training points from a denser grid achieve better performance, with the limit case of the filter using the true map outperforming every other configuration. Second, kriging-based filters consistently achieve lower RMSEs than their bilinear counterparts despite having similar map reconstruction accuracy. This result, combined with Figure 4.5, indicates that the advantage of kriging lies in its ability to quantify reconstruction uncertainty through its covariance matrix. Finally, the time required to reach a stable estimation error increases from 13 to 25 steps as the grid resolution decreases from 2.5 to 7.5 km.

The NEES curves further highlight the superiority of kriging methods by taking the estimated error covariance matrix into account. Kriging-based filters stay within the consistency band at every resolution, whereas bilinear interpolation lies above it even on the densest map. This shows that kriging not only leads to more accurate state estimates than bilinear interpolation, but also provides a better characterisation of their associated uncertainty.

In terms of convergence rates, Figure 4.8 starts by extracting the relevant the median errors from the previous curves, and Figure 4.9 computes the associated convergence rates using the two metrics introduced at the beginning of this section.

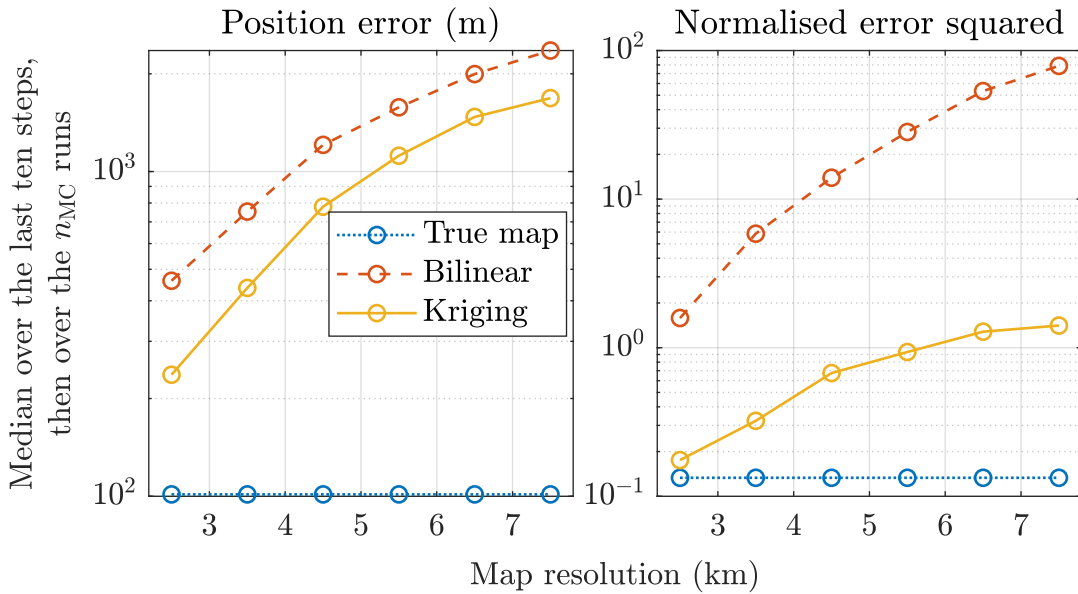


Figure 4.8: Average median RMSE and NEES at the end of the flight for each grid resolution. These metrics are used to determine whether a run has converged.

The first two figures show that the bilinear to kriging RMSE ratio ranges from 1.4 to 1.9, and increases to 9 to 56 for the NEES. Furthermore, while the RMSE ratio tends to decrease at lower grid resolutions, the NEES ratio follows the opposite trend. Bilinear interpolation thus completely fails to measure its accuracy at lower resolutions, while kriging degrades in step with its own accuracy.

The convergence rates follow the same pattern. Under the threshold criterion, the rate for kriging decreases from 100% to 60% as the grid resolution decreases, while that of bilinear interpolation decreases from 99 to 42%. The two thus remain comparable. For the Mahalanobis criterion, however, kriging only falls to 81% while bilinear interpolation collapses to 1%, having lost every run whose reported

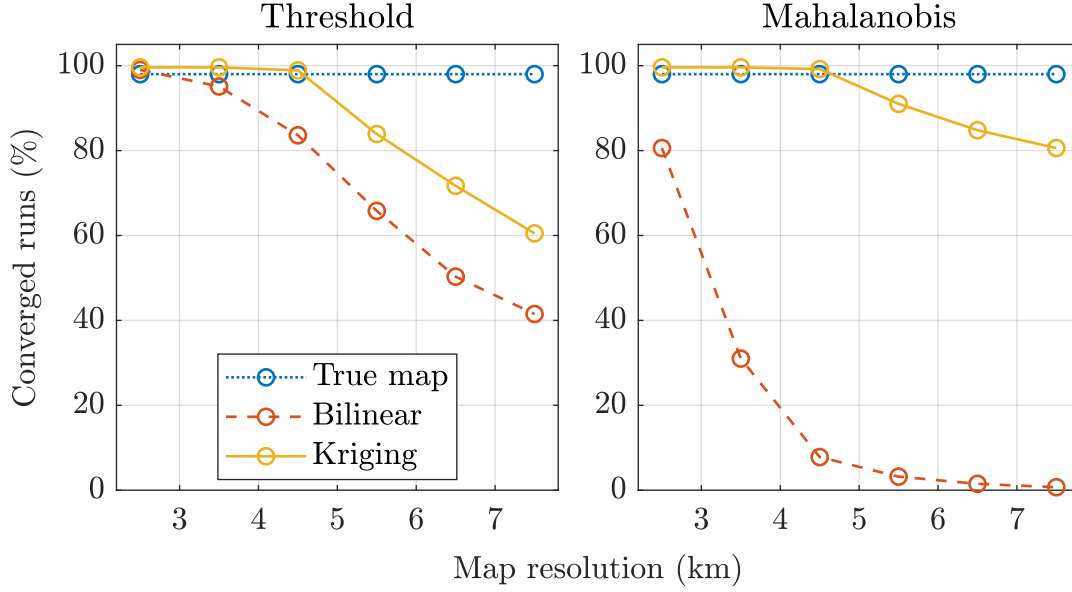


Figure 4.9: Percentage of converged runs against the grid resolution for the two metrics. The true map does not depend on the resolution, and has a convergence rate of 98% under both criteria.

uncertainty is consistent with its error.

However, the advantages of kriging come at an increased computational and memory costs, as described by Table 4.6.

Observation model	Setup (flops)	Queries (flops)	Memory (bytes)
True map	–	6M	323k
Bilinear	–	6M	150k
Kriging, 2.5km	81G	29.2T	312M
Kriging, 3.5km	10.3G	7.38T	78.8M
Kriging, 4.5km	2.42G	2.81T	30M
Kriging, 5.5km	726M	1.26T	13.5M
Kriging, 6.5km	243M	608G	6.5M
Kriging, 7.5km	103M	343G	3.67M

Table 4.6:

While the setup cost can be paid before the mission, the vehicle must still be able to store between 3.67 and 312MB of data, and sustain an average computational rate of 0.46 to 38.9Gflops/s over the navigation period. These requirements remain affordable at low grid resolutions, but these configurations result in poor navigation performance, as shown above. Increasing the grid resolution improves navigation accuracy, but quickly leads to prohibitive storage and computational requirements.

4.3 Adaptive Kriging

In this section, we propose to use training data subsampling, *i.e.* selecting a subset $\tilde{\mathbf{x}}^r$ of size $\tilde{N}^r$ of the training points and only using them to perform the regression, as a way to solve the computational issues raised above. Since the computational cost of kriging scales cubically with the number of training points, quadratically for memory, discarding half the data reduces the computational load by a factor of 8 and requires 4 times less storage.

However, this method comes at a cost. In the following, we describe two subsampling strategies, each requiring to perform a subsampling step at each time step. This means that no precomputation can be done before the mission, and in particular that inverting $G(\tilde{\mathbf{x}}^r) \in \mathcal{S}_{\tilde{N}^r, d_z}^{++}(\mathbb{R})$ and $C^\top G(\tilde{\mathbf{x}}^r)^{-1} C \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ must be done in real time.

4.3.1 Random Subsampling

The simplest idea to reduce the number of training points is to let $p \in [0, 1]$, and draw $s^{(i)}$ from a Bernoulli distribution of success probability p . Any training sample $\tilde{x}_i$ for which the corresponding $s^{(i)}$ is a success is retained to form $\tilde{\mathbf{x}}^r$, while the others are discarded.

By linearity of the expectation, the number of retained data is, on average:

$$\tilde{N}^r = p \times \tilde{N} \quad (4.21)$$

TODO: numerical results here

4.3.2 Adaptive Subsampling Window

The above results show that the computational limits described in section 4.2.4 are not the only reasons to forgo using static kriging-based navigation algorithms. Indeed, the estimators introduced in section 3.3 assume the use of a single kernel over the entire domain, and assumption 4.1 inherits that hypothesis. In section 4.2.2, we estimated the hyperparameters of the kernel once before the mission, meaning that they are assumed to be independent of the inputs:

$$k_{3/2} : x, x' \mapsto \sigma_f^2 \left(1 + \frac{\sqrt{3} \|x - x'\|_2}{l} \right) \exp \left(-\frac{\sqrt{3} \|x - x'\|_2}{l} \right) \quad (4.22)$$

The considered kernel is thus radial, and the resulting GP is therefore stationary. This is unrealistic for a physical field, especially at large spatial scales. However, stationarity cannot be simply dropped, as the hyperparameters $\theta \triangleq (\sigma_f, l)$ of such a kernel would have to be fitted at every point of the domain, where a single sample is available. We therefore assume the field to be non-stationary, but its hyperparameters to vary over a scale L_θ much larger than the correlation length itself. Under this hypothesis, a neighbourhood of radius r_w such that:

$$l \ll r_w \ll L_\theta \quad (4.23)$$

is large enough to enable the accurate estimation of σ_f and l , while small enough to avoid averaging those values over distances on which they change.

In this section, we propose an adaptive subsampling strategy to optimise the kriging performance of the filter according to three criteria:

- They must cover a region whose characteristic size is of the order of r_w , so that the local hyperparameters can be accurately estimated without averaging them over a region where they vary significantly.
- The retained training points must cover an area large enough to avoid extrapolation at the particle locations, which degrades the reconstruction of the observation function, as quantified in section 4.2.1.
- Finally, the number of retained training points must remain small enough to ensure that the filter can be computed within real-time constraints.

The first constraint naturally gives the shape of retained subset. Let $\hat{x}_{k|k-1}$ be the predicted estimation of the filter at time k :

$$\hat{x}_{k|k-1} \triangleq \sum_{i=1}^N w_{k-1}^{(i)} x_k^{(i)} \approx \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k-1})}[x_k] \quad (4.24)$$

and $\hat{P}_{k|k-1}$ denote the corresponding covariance matrix:

$$\hat{P}_{k|k-1} \triangleq \sum_{i=1}^N w_{k-1}^{(i)} \left(x_k^{(i)} - \hat{x}_{k|k-1} \right) \left(x_k^{(i)} - \hat{x}_{k|k-1} \right)^\top \quad (4.25)$$

TODO: à relire These two quantities can be computed after the prediction step at time k , but before the following weight update. In particular, they represent the state of the particle set at time k before kriging the environment.

The window itself is a region of the domain the map is defined on, and not of the state space: $\hat{P}_{k|k-1}$ mixes positions and velocities, so its eigenvalues carry no single unit and a radius drawn from them would be meaningless. The relevant quantity is the position block of the predicted covariance:

$$\hat{P}_{k|k-1}^{\text{pos}} \triangleq \varphi_{\text{pos}} \hat{P}_{k|k-1} \varphi_{\text{pos}}^\top \quad (4.26)$$

Indeed, if the particles follow a Gaussian distribution of mean $\hat{x}_{k|k-1}$ and covariance matrix $\hat{P}_{k|k-1}$, then each particle $x_k^{(i)}$ has a q chance that:

$$\left\| \varphi_{\text{pos}} \left(x_k^{(i)} - \hat{x}_{k|k-1} \right) \right\|_{\hat{P}_{k|k-1}^{\text{pos}}}^2 \leq \chi_2^2(q) \quad (4.27)$$

where $\chi_d^2(q)$ is the q quantile of a χ^2 distribution with d degrees of freedom, and $q \triangleq 99\%$ is the confidence level already used by the convergence criterion of the previous section.

That region is an ellipse, which is an inconvenient shape to select training data with. Bounding it by a disc costs a single spectral inequality: since $\left(\hat{P}_{k|k-1}^{\text{pos}} \right)^{-1} \succeq \lambda_{\max}^{-1} \left(\hat{P}_{k|k-1}^{\text{pos}} \right) I_2$, it comes that:

$$\begin{aligned} \left\| \varphi_{\text{pos}} \left(x_k^{(i)} - \hat{x}_{k|k-1} \right) \right\|_2 &\leq \sqrt{\lambda_{\max} \left(\hat{P}_{k|k-1}^{\text{pos}} \right)} \times \left\| \varphi_{\text{pos}} \left(x_k^{(i)} - \hat{x}_{k|k-1} \right) \right\|_{\hat{P}_{k|k-1}^{\text{pos}}} \\ &\leq c(q) \sqrt{\lambda_{\max} \left(\hat{P}_{k|k-1}^{\text{pos}} \right)} \end{aligned} \quad (4.28)$$

with a probability of at least q , the coefficient being available in closed form since a χ^2 distribution with two degrees of freedom is an exponential one:

$$c(q) \triangleq \sqrt{\chi_2^2(q)} = \sqrt{-2\ln(1-q)} \quad (4.29)$$

The probability is bounded from below rather than attained, the disc containing the ellipse it was built from, with equality only where $\hat{P}_{k|k-1}^{\text{pos}}$ is isotropic. At the retained confidence level $c(99\%) = 3.03$, and the value $c = 3$ used in the remainder is the radius of the disc holding 98.9% of the cloud.

TODO: il faut dire que "naïvement" on pourrait prendre la boule centrée sur l'état prédit et dont le rayon vient d'être calculé, mais il manque 3 choses:

- on n'est qu'à 99% de chance de tomber dedans et encore on pourrait être à la limite
- on n'a pas de taille/de nombre de données minimale
- si c'est immense, ce qui est possible, alors on a besoin de garder la fenêtre grande car il ne faut pas extrapoler en les particules, mais on peut quand-même n'estimer les HP que sur certaines données d'entraînement pour gagner du temps

Algorithm 4.2 Adaptive Kriging Regularised Particle Filter

$$\begin{aligned}
& \forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot | x_{k-1}^{(i)}, z_k) &> \text{Particle propagation} \\
& \forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})} &> \text{Weight update} \\
& \forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)} &> \text{Normalisation} \\
& \hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N (w_k^{(i)})^2 &> \text{Resampling criterion} \\
& \text{if } \hat{N}_{\text{eff}} < N_{\text{th}} \text{ then} \\
& \quad \hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)} &> \text{Whitening} \\
& \quad \hat{P}_k = \sum_{i=1}^N w_k^{(i)} (x_k^{(i)} - \hat{x}_k) (x_k^{(i)} - \hat{x}_k)^\top \\
& \quad A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top \\
& \quad \left\{ x_k^{(j)}, w_k^{(j)}, - \right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right) &> \text{Resampling} \\
& \quad \forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K &> \text{Regularisation} \\
& \quad \forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)} \\
& \text{end if}
\end{aligned}$$

4.3.3 Numerical Results

As for static kriging, this section evaluates the performance of the AK-PF across the six grid resolutions. The static kriging RPF performance are also given as a

reference. Along the RMSE, the NEES, an other metric is introduced to help explain the variations in computational and memory costs between each configuration. This metric is the number of retained training points, *i.e.* the value of $\tilde{N}^r$, at each time step. Since the

TODO: nouvelle métrique The third is the number of survey samples read per query, written $\tilde{N}_w$, together with the number of hyperparameter fits, both of which measure cost in a way that a wall clock cannot since they do not depend on the machine.

4.3.4 Sensitivity to the Window Parameters

4.4 Clustered Adaptive Kriging

While it outperforms static kriging, the Ak-RPF window can still

4.5 Ancienne intro

TODO: à incorporer aux bons endroits

Chapter 3 established kriging as a surrogate observation model, able to return both a prediction of the field and the uncertainty attached to it. That construction, however, assumes that the whole training set is available to every regression and that a single kernel describes the field over the entire domain. Neither assumption survives a navigation scenario.

Three obstacles arise when the kriged model is placed inside the recursion of a particle filter:

- the Gram matrix must be inverted at every time step, at a cost cubic in the number of training points, whereas the filter runs in real time;
- physical fields such as terrain elevation or magnetic anomaly are not stationary over an operating domain of several hundred kilometres, so a single hyperparameter vector cannot describe them everywhere;
- at a given step, the filter only queries the field where its particles are, so the vast majority of the training points contribute nothing to the likelihood.

The last point turns the first two into an opportunity. If only a neighbourhood of the particle cloud is ever queried, then only the training data lying in that neighbourhood need to be used, and the hyperparameters need only describe the field locally. The particle cloud, which the filter already maintains, thus provides a natural criterion to select both.

bonne phrase mais à mettre ici As mentioned in section 4.4, a simple interpolation of h through $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$ would only return a single value per queried state, whereas the Bayesian correction step needs the entire likelihood density $p(z_k|x_k)$.

The rest of this chapter is organised as follows:

- Section 4.6 quantifies the error committed when a kriging estimator is built on a subset of its training data, independently of any navigation scenario.
- Section 4.4 then builds the proposed filter [92, 93] one component at a time, and measures each of them as it is introduced.

4.6 Kriging Subsampling Error

Discarding training points lowers the cost of the regression and loses accuracy, and the heuristics used to choose which points to keep usually leave that loss unquantified. This section makes it explicit [94], by rewriting the discrepancy between the full and the subsampled estimators as a function of the information carried by the discarded data alone.

décision préalable de publication. Le préprint [94] n'est pas encore accepté, mais la section est écrite ici comme si elle tenait – elle est autonome et ne dépend d'aucun résultat de navigation. Si l'article n'aboutit pas, la structure reste valable et seules les bornes de la section 4.6.3 sont à raccourcir.

harmonisation de notations avec le section 3.3, à faire avant d'écrire les preuves.

- le préprint est écrit à sortie scalaire, moyenne nulle et bruit $\tilde{r}$ scalaire, alors que le manuscrit travaille à valeurs dans $\mathbb{R}^{d_z}$ avec $I_{\tilde{N}} \otimes \tilde{R}$. Soit on généralise les bornes (les bornes entrées par entrée de la section 4.6.3 supposent $\tilde{r}$ scalaire), soit on annonce explicitement que la section se place dans le cas scalaire centré. Ma recommandation est la seconde, avec une remarque en fin de section – la généralisation vectorielle n'apporte rien à l'argument et alourdirait toutes les preuves ;
- le préprint note φ et ψ les deux quantités résiduelles, mais φ désigne déjà l'application de l'état dans le manuscrit et ψ les estimateurs du section 2.2. La proposition retenue ci-dessous est $k_{f|c}$ et $\tilde{\mathbf{z}}_{f|c}$, qui se lisent « écarté sachant retenu » et se raccordent à la convention $\cdot_{m|n}$ de la liste des symboles.

4.6.1 Subsampling Metrics

Subsampling splits the training set into the points that are kept, usually close to the query location, and those that are discarded:

$$\tilde{\mathbf{x}} = \begin{bmatrix} \tilde{\mathbf{x}}_c & \tilde{\mathbf{x}}_f \end{bmatrix}, \quad \tilde{\mathbf{z}} = \begin{bmatrix} \tilde{\mathbf{z}}_c & \tilde{\mathbf{z}}_f \end{bmatrix} \quad (4.30)$$

of respective sizes $\tilde{N}_c$ and $\tilde{N}_f$. This decomposition isolates the predictive contribution of the discarded points from that of the retained ones, and every quantity below is built on it.

définir les deux métriques, chacune capturant un aspect distinct de l'approximation.

- l'erreur de variance e_r , dans $\mathcal{L}^2$, qui mesure l'écart quadratique moyen entre les deux estimateurs et ne dépend donc pas des observations ;
- l'erreur d'interpolation e_f , dans $\mathcal{L}^1$ – conditionnellement à $\tilde{\mathbf{z}}$, l'estimateur krigé est une fonction déterministe, et l'écart absolu entre les deux valeurs prédites est alors la mesure naturelle.

expliquer pourquoi les deux sont nécessaires plutôt que de n'en garder qu'une – e_r borne la dégradation de la covariance transmise au filtre, e_f borne le déplacement du pic de vraisemblance. Les deux effets sont ceux qui font diverger un filtre particulier, et ils ne se déduisent pas l'un de l'autre, ce que la section 4.6.3 précisera.

4.6.2 Residual Process

c'est la partie technique de la section. Enchaînement du préprint, à reprendre tel quel.

partitionnement par blocs de la matrice de Gram et matrice de Gram évidée $G_h(\tilde{\mathbf{x}}_c)$, obtenue en ne gardant que le bloc des données retenues et en annulant le reste. Proposition – l'estimateur sous-échantillonné s'écrit avec le pseudo-inverse de Moore-Penrose de $G_h(\tilde{\mathbf{x}}_c)$ à la place de $G(\tilde{\mathbf{x}})^{-1}$, ce qui met les deux estimateurs sous une forme commune et rend leur différence lisible.

définir la différence d'inverses de Gram $D_G \triangleq G(\tilde{\mathbf{x}})^{-1} - G_h(\tilde{\mathbf{x}}_c)^\dagger$, puis la proposition qui exprime les deux métriques à travers elle. La preuve repose sur un argument de projection orthogonale – l'estimateur krigé étant l'espérance a posteriori, il est la projection de $f(x)$ sur l'espace engendré par les observations bruitées, l'erreur du krigeage complet est orthogonale à cet espace, et il vient que e_r^2 est exactement la différence des deux variances de krigeage. C'est le point le plus élégant du préprint, à ne pas expédier.

décomposition de D_G par la formule d'inversion par blocs de Woodbury (proposition A.5 de l'annexe), faisant apparaître le complément de Schur S de $G(\tilde{\mathbf{x}})$ par rapport à $G(\tilde{\mathbf{x}}_c)$.

Two residual quantities emerge from this decomposition, both measuring what the retained set fails to explain about the discarded one:

$$\begin{cases} k_{f|c}(x) \triangleq k(\tilde{\mathbf{x}}_f, x) - k(\tilde{\mathbf{x}}_f, \tilde{\mathbf{x}}_c) G(\tilde{\mathbf{x}}_c)^{-1} k(\tilde{\mathbf{x}}_c, x) \\ \tilde{\mathbf{z}}_{f|c} \triangleq \tilde{\mathbf{z}}_f - k(\tilde{\mathbf{x}}_f, \tilde{\mathbf{x}}_c) G(\tilde{\mathbf{x}}_c)^{-1} \tilde{\mathbf{z}}_c \end{cases} \quad (4.31)$$

The residual covariance function $k_{f|c}$ is the part of the covariance between the query point and the discarded set that the retained set does not already account for, while the residual observation vector $\tilde{\mathbf{z}}_{f|c}$ is the information carried by the discarded observations and not predictable from the retained ones.

lemme des identités résiduelles, puis proposition donnant la forme résiduelle des deux erreurs – elles ne dépendent plus que de (4.31) et de S . Insister sur la lecture : si les données écartées sont bien prédites par celles qu'on garde, les deux résidus s'annulent et l'erreur avec eux, ce qui est exactement le critère qu'une heuristique de sélection devrait viser.

4.6.3 Relation Between the Errors

corollaire central – munir $\mathbb{R}^{\tilde{N}_f}$ du produit scalaire de Mahalanobis associé à S^{-1} ramène les deux métriques à des quantités géométriques dans un même espace euclidien, e_r étant la norme du résidu de covariance et e_f son produit scalaire avec le résidu d'observation. Les deux erreurs cessent alors d'être deux calculs séparés.

c'est de ce corollaire que sort la relation entre les deux erreurs, par Cauchy-Schwarz – l'erreur d'interpolation est majorée par le produit de l'erreur de variance et de la norme du résidu d'observation. Autrement dit, e_r contrôle e_f à un facteur près qui ne dépend que des observations écartées et pas du point de requête. Le dire explicitement, c'est la réponse à la question « faut-il borner les deux ? ».

bornes proprement dites, à partir d'hypothèses entrée par entrée sur le noyau – κ pour la covariance intra-ensemble, α pour la covariance croisée entre données retenues et écartées, ε pour la covariance entre le point de requête et les données

écartées, dans le régime visé $0 < \varepsilon, \alpha \ll \kappa \leq k_{\max}$ qui traduit exactement la stratégie de sélection par proximité. Deux familles :

- bornes entrée par entrée, via un lemme sur les normes des deux résidus et sur la norme d'opérateur de S^{-1} , puis Cauchy-Schwarz ;
- bornes au pire cas, obtenues en supposant les majorations atteintes, plus fines que les précédentes puisqu'elles évitent l'inégalité triangulaire et font intervenir les plus grandes valeurs propres des matrices de Gram au lieu des plus petites.

équivalents asymptotiques selon le régime spectral, avec le cas dégénéré à bruit dominant (les deux estimateurs sont également mauvais, les erreurs tendent vers 0 – test de cohérence des bornes) et le cas général. Conclusion à retenir : c'est α qui commande, et $\tilde{\mathbf{x}}_f$ ne peut être écarté sans dommage que si sa covariance croisée avec $\tilde{\mathbf{x}}_c$ reste faible.

choix de découpage à trancher – les bornes entrée par entrée, les bornes au pire cas et les équivalents asymptotiques font trois blocs de résultats dans une seule sous-section, ce qui est lourd. Soit on ouvre une quatrième sous-section « Error Bounds », soit on découpe en **subsubsection**. Je recommande la quatrième sous-section, en gardant ici la seule relation entre e_r et e_f , qui est ce que le titre annonce.

fermer la section sur ce que les bornes apportent en pratique – elles relient l'erreur tolérée au nombre de points qu'on peut écarter, dans les deux sens (un niveau d'erreur donne un taux de sous-échantillonnage admissible, une stratégie de sélection donne une erreur chiffrée). Validation numérique sur du krigeage seul, sans filtre, avec **Exemple de carte krigée.png** et **Exemple de krigeage 2D.png** en illustrations candidates.

Part III

Appendices

APPENDIX A

USEFUL MATRIX IDENTITIES

Contents

A.1	Gaussian Conditioning Formula	96
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A.1 Gaussian Conditioning Formula

Proposition A.1 (Gaussian conditioning [95]). *For $n, p \in \mathbb{N}$, let $U \in \mathbb{R}^n$ and $V \in \mathbb{R}^p$ be jointly Gaussian random variables, of respective means μ_U, μ_V and covariance matrices $A \succ 0$ and $D \succ 0$. There exists $B \in \mathcal{M}_{n,p}(\mathbb{R})$ such that:*

$$\begin{bmatrix} U \\ V \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_U \\ \mu_V \end{bmatrix}, \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \right) \quad (\text{A.1})$$

We assume B to be such that this joint covariance is SPD in the sense of Definition 2.1. The conditional distribution of U given $V = v$ is then:

$$U|V = v \sim \mathcal{N}(\mu_U + BD^{-1}(v - \mu_V), P_{U|V}) \quad (\text{A.2})$$

with:

$$P_{U|V} \triangleq A - BD^{-1}B^\top \succ 0 \quad (\text{A.3})$$

Proof. The existence of B is direct by definition of jointly Gaussian random variables. As $D \succ 0$ is invertible, $S \triangleq A - BD^{-1}B^\top$ the Schur complement with respect to D is well-defined.

Let $u \neq 0$:

$$\begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix}^\top \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix} = u^\top (A - BD^{-1}B^\top) u > 0 \quad (\text{A.4})$$

by SPDness of the joint covariance, the left-hand vector being non-zero. S is thus also invertible, and we can apply the inferior right Schur inversion formula (see Proposition A.2) to obtain:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \triangleq \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}^{-1} \quad (\text{A.5})$$

For some realisation u of U and v of V , we define $\tilde{u} \triangleq u - \mu_U$ and $\tilde{v} \triangleq v - \mu_V$, and we compute:

$$\begin{aligned} \alpha &\triangleq \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix}^\top \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix} \\ &= \tilde{u}^\top \bar{A} \tilde{u} + \tilde{u}^\top \bar{B} \tilde{v} + \tilde{v}^\top \bar{C} \tilde{u} + \tilde{v}^\top \bar{D} \tilde{v} \\ &= \tilde{u}^\top S^{-1} \tilde{u} + \tilde{u}^\top S^{-1} (-BD^{-1} \tilde{v}) + (-BD^{-1} \tilde{v})^\top S^{-1} \tilde{u} + \\ &\quad (-BD^{-1} \tilde{v})^\top S^{-1} (-BD^{-1} \tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \\ &= (\tilde{u} - BD^{-1} \tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1} \tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \end{aligned} \quad (\text{A.6})$$

The conditional density is thus:

$$\begin{aligned} p(\tilde{u}|\tilde{v}) &\propto \frac{p(\tilde{u}, \tilde{v})}{p(\tilde{v})} \\ &\propto \frac{\exp\left(-\frac{1}{2}\alpha\right)}{\exp\left(-\frac{1}{2}\tilde{v}^\top D^{-1} \tilde{v}\right)} \\ &\propto \exp\left(-\frac{1}{2}(\tilde{u} - BD^{-1} \tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1} \tilde{v})\right) \end{aligned} \quad (\text{A.7})$$

With this formulation, it comes that $U|V = v$ is Gaussian with mean:

$$\mu_{U|V} \triangleq \mu_U + BD^{-1}(v - \mu_V) \quad (\text{A.8})$$

and covariance:

$$P_{U|V} \triangleq S = A - BD^{-1}B^\top \succ 0 \quad (\text{A.9})$$

□

A.2 Schur Inversion Formula

Proposition A.2 (Schur inversion formula [96]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$, with A invertible. We define the matrix M and its Schur complement with respect to A as:*

$$M \triangleq \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \quad M/A \triangleq D - CA^{-1}B \quad (\text{A.10})$$

If M/A is invertible, then M is invertible and:

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} & -A^{-1}B(M/A)^{-1} \\ -(M/A)^{-1}CA^{-1} & (M/A)^{-1} \end{bmatrix} \quad (\text{A.11})$$

Proof. Let $\bar{M}$ be the matrix defined by the right-hand side of the formula, and write its blocks as:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} = \bar{M} \quad (\text{A.12})$$

The identity $M\bar{M} = I$ yields:

$$\begin{cases} A\bar{A} + B\bar{C} = I \\ A\bar{B} + B\bar{D} = 0 \\ C\bar{A} + D\bar{C} = 0 \\ C\bar{B} + D\bar{D} = I \end{cases} \quad (\text{A.13})$$

In particular, the second equation can be rewritten as:

$$\bar{B} = -A^{-1}B\bar{D} \quad (\text{A.14})$$

which can be plugged into the fourth equation to give:

$$\begin{aligned} C\bar{B} + D\bar{D} &= I \\ \iff -CA^{-1}B\bar{D} + D\bar{D} &= I \\ \iff (D - CA^{-1}B)\bar{D} &= I \\ \iff (M/A)\bar{D} &= I \\ \iff \bar{D} &= (M/A)^{-1} \end{aligned} \quad (\text{A.15})$$

(A.14) can then be solved:

$$\bar{B} = -A^{-1}B(M/A)^{-1} \quad (\text{A.16})$$

A similar operation can be performed on the two remaining equations to obtain:

$$\bar{A} = A^{-1}(I - B\bar{C}) \quad (\text{A.17})$$

and:

$$\begin{aligned} & C\bar{A} + D\bar{C} = 0 \\ \iff & CA^{-1}(I - B\bar{C}) + D\bar{C} = 0 \\ \iff & CA^{-1} + (D - CA^{-1}B)\bar{C} = 0 \\ \iff & CA^{-1} + (M/A)\bar{C} = 0 \\ \iff & \bar{C} = -(M/A)^{-1}CA^{-1} \end{aligned} \quad (\text{A.18})$$

Finally:

$$\bar{A} = A^{-1}(I - B\bar{C}) = A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} \quad (\text{A.19})$$

It can be verified by direct calculations that:

$$M\bar{M} = \bar{M}M = I \quad (\text{A.20})$$

which proves that M is indeed invertible with inverse $\bar{M}$. $\square$

Remark A.1 (Inversion with respect to D). *If D is invertible, one can instead introduce the Schur complement with respect to D :*

$$M/D \triangleq A - BD^{-1}C \quad (\text{A.21})$$

When M/D is also invertible, an analogous block inversion formula holds:

$$M^{-1} = \begin{bmatrix} (M/D)^{-1} & -(M/D)^{-1}BD^{-1} \\ -D^{-1}C(M/D)^{-1} & D^{-1} + D^{-1}C(M/D)^{-1}BD^{-1} \end{bmatrix} \quad (\text{A.22})$$

Remark A.2 (Combining both formulations). *When both A and D are invertible, the Woodbury lemma proves that the two formulations give the same matrices for each block. In particular, they can be combined block by block at will.*

A.3 Woodbury Inversion Formula

Proposition A.3 (Inversion identity). *For $n \in \mathbb{N}$, let $P \in \mathcal{M}_{n,n}(\mathbb{R})$ such that $I + P$ is invertible. Then:*

$$(I + P)^{-1} = I - (I + P)^{-1}P \quad (\text{A.23})$$

Proof. This result is direct from:

$$I = (I + P)^{-1}(I + P) = (I + P)^{-1} + (I + P)^{-1}P \quad (\text{A.24})$$

$\square$

Proposition A.4 (Push-through identity [97]). *For $n, p \in \mathbb{N}$, let $P \in \mathcal{M}_{n,p}(\mathbb{R})$ and $Q \in \mathcal{M}_{p,n}(\mathbb{R})$ such that $I_n + PQ$ is invertible. Then $I_p + QP$ is also invertible and:*

$$(I_n + PQ)^{-1}P = P(I_p + QP)^{-1} \quad (\text{A.25})$$

Proof. We start by using the identity:

$$P + PQP = P(I_p + QP) = (I_n + PQ)P \quad (\text{A.26})$$

Thus:

$$\begin{aligned} P(I_p + QP) &= (I_n + PQ)P \\ \iff P &= (I_n + PQ)P(I_p + QP)^{-1} \\ \iff (I_n + PQ)^{-1}P &= P(I_p + QP)^{-1} \end{aligned} \quad (\text{A.27})$$

with $I_p + QP$ admitting a right inverse, hence being invertible. $\square$

Proposition A.5 (Generalised Woodbury formula [98]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$. We assume that A , D , and $D + CA^{-1}B$ are invertible. Then $A + BD^{-1}C$ is also invertible, and:*

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \quad (\text{A.28})$$

Proof. We first factorise:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= (A(I + A^{-1}BD^{-1}C))^{-1} \\ &= (I + A^{-1}BD^{-1}C)^{-1}A^{-1} \end{aligned} \quad (\text{A.29})$$

Applying Proposition A.3 to $A^{-1}BD^{-1}C$ yields:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= \left(I - (I + A^{-1}BD^{-1}C)^{-1}A^{-1}BD^{-1}C \right) A^{-1} \\ &= A^{-1} - (I + A^{-1}BD^{-1}C)^{-1}A^{-1}BD^{-1}CA^{-1} \end{aligned} \quad (\text{A.30})$$

Using Proposition A.4 with $P = A^{-1}B$ and $Q = D^{-1}C$ further gives:

$$(I + A^{-1}BD^{-1}C)^{-1}A^{-1}B = A^{-1}B(I + D^{-1}CA^{-1}B)^{-1} \quad (\text{A.31})$$

Therefore:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= A^{-1} - A^{-1}B(I + D^{-1}CA^{-1}B)^{-1}D^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B(D(I + D^{-1}CA^{-1}B))^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \end{aligned} \quad (\text{A.32})$$

$\square$

APPENDIX B

DEFERRED PROOFS

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B.1 Proof of Proposition 2.3

Prediction. Let $\mathbb{V}[X]$ denote the variance of the random variable X :

$$\mathbb{V}[X] \triangleq \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top] \quad (\text{B.1})$$

Since $x_k = F_k x_{k-1} + b_k^{(e)} + w_k$ is an affine transformation of two independent Gaussian random variables $x_{k-1} | \mathbf{z}_{1:k-1} \sim \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$ and $w_k \sim \mathcal{N}(0, Q_k)$, the predicted state $x_k | \mathbf{z}_{1:k-1}$ is also Gaussian, with:

$$\begin{aligned} \hat{x}_{k|k-1} &\triangleq \mathbb{E}[x_k | \mathbf{z}_{1:k-1}] \\ &= F_k \mathbb{E}[x_{k-1} | \mathbf{z}_{1:k-1}] + b_k^{(e)} + \mathbb{E}[w_k | \mathbf{z}_{1:k-1}] \\ &= F_k \hat{x}_{k-1|k-1} + b_k^{(e)} \end{aligned} \quad (\text{B.2})$$

and:

$$\begin{aligned} P_{k|k-1} &\triangleq \mathbb{V}[x_k | \mathbf{z}_{1:k-1}] \\ &= F_k \mathbb{V}[x_{k-1} | \mathbf{z}_{1:k-1}] F_k^\top + \mathbb{V}[w_k | \mathbf{z}_{1:k-1}] \\ &= F_k P_{k-1|k-1} F_k^\top + Q_k \end{aligned} \quad (\text{B.3})$$

The predicted covariance is thus SPD as the sum of an SPSD matrix and the SPD matrix Q_k .

Correction. The joint distribution of (x_k, z_k) given $\mathbf{z}_{1:k-1}$ is also Gaussian, since $z_k = H_k x_k + b_k^{(o)} + n_k$ is an affine transformation of the independent Gaussian variables $x_k | \mathbf{z}_{1:k-1} \sim \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1})$ and $n_k \sim \mathcal{N}(0, R_k)$. It comes that:

$$\begin{bmatrix} x_k \\ z_k \end{bmatrix} \Big| \mathbf{z}_{1:k-1} \sim \mathcal{N} \left(\begin{bmatrix} \hat{x}_{k|k-1} \\ H_k \hat{x}_{k|k-1} + b_k^{(o)} \end{bmatrix}, \begin{bmatrix} P_{k|k-1} & P_{k|k-1} H_k^\top \\ H_k P_{k|k-1} & S_k \end{bmatrix} \right) \quad (\text{B.4})$$

where:

$$S_k \triangleq H_k P_{k|k-1} H_k^\top + R_k \quad (\text{B.5})$$

is called the innovation covariance, and is the covariance associated with the innovation vector:

$$y_k \triangleq z_k - \left(H_k \hat{x}_{k|k-1} + b_k^{(o)} \right) \quad (\text{B.6})$$

Since $R_k \succ 0$, the independent pair (x_k, n_k) has the covariance:

$$\begin{bmatrix} P_{k|k-1} & 0 \\ 0 & R_k \end{bmatrix} \succ 0 \quad (\text{B.7})$$

Its image under the invertible linear map $(x, n) \mapsto (x, H_k x + n)$ therefore has an SPD covariance as well. The Gaussian conditioning formula (see Proposition A.1) can thus be applied to give the posterior distribution $x_k | \mathbf{z}_{1:k}$ is Gaussian, with:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k \quad (\text{B.8})$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (\text{B.9})$$

where:

$$K_k \triangleq P_{k|k-1} H_k^\top S_k^{-1} \quad (\text{B.10})$$

is the Kalman gain.

Finally, Proposition A.1 proves that the Schur complement of the SPD joint covariance:

$$P_{k|k-1} - (P_{k|k-1} H_k^\top) S_k^{-1} (H_k P_{k|k-1}) = P_{k|k-1} - K_k H_k P_{k|k-1} = P_{k|k} \quad (\text{B.11})$$

is itself SPD, so $P_{k|k} \succ 0$.

B.2 Proof of Proposition 2.10

As a linear combination of the centred noises w_k^l and w_k^n , $\bar{w}_k^l$ is centred. Its covariance develops as:

$$\begin{aligned} \bar{Q}_k^l &= \mathbb{E} [\bar{w}_k^l \bar{w}_k^{l\top}] \\ &= \mathbb{E} [(w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] \\ &= \mathbb{E} [w_k^l (w_k^l)^\top] - \mathbb{E} [w_k^l (Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] - \mathbb{E} [(Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^l)^\top] \\ &\quad + \mathbb{E} [(Q_k^{nl} (Q_k^n)^{-1} w_k^n) (Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] \\ &= \mathbb{E} [w_k^l (w_k^l)^\top] - \mathbb{E} [w_k^l (w_k^n)^\top] (Q_k^n)^{-\top} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^l)^\top] \\ &\quad + Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^n)^\top] (Q_k^n)^{-\top} (Q_k^{nl})^\top \\ &= Q_k^l - Q_k^{nl} (Q_k^n)^{-\top} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top + Q_k^{nl} (Q_k^n)^{-1} Q_k^n (Q_k^n)^{-\top} (Q_k^{nl})^\top \\ &= Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top + Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \\ &= Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \end{aligned} \quad (\text{B.12})$$

Finally, computing:

$$\begin{aligned} \mathbb{E} [\bar{w}_k^l (w_k^n)^\top] &= \mathbb{E} [(w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^n)^\top] \\ &= \mathbb{E} [w_k^l (w_k^n)^\top] - Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^n)^\top] \\ &= Q_k^{nl} - Q_k^{nl} (Q_k^n)^{-1} Q_k^n \\ &= 0 \end{aligned} \quad (\text{B.13})$$

shows that $\bar{w}_k^l$ and w_k^n are uncorrelated.

B.3 Proof of Proposition 3.2

Forward implication. Let $n \in \mathbb{N}^*$ and $\mathbf{x} \in \mathcal{U}^n$. It comes from the symmetry of k that:

$$\begin{aligned}
 k(\mathbf{x}, \mathbf{x}) &= \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \\
 &= \begin{bmatrix} k(x_1, x_1)^\top & \cdots & k(x_n, x_1)^\top \\ \vdots & \ddots & \vdots \\ k(x_1, x_n)^\top & \cdots & k(x_n, x_n)^\top \end{bmatrix} \\
 &= \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix}^\top \\
 &= k(\mathbf{x}, \mathbf{x})^\top
 \end{aligned} \tag{B.14}$$

Let $a \in \mathbb{R}^{nd_z}$. We decompose $a = [a_1^\top, \dots, a_n^\top]^\top$ with $a_i \in \mathbb{R}^{d_z}$, $i \in \llbracket 1, n \rrbracket$ to obtain:

$$\begin{aligned}
 a^\top k(\mathbf{x}, \mathbf{x}) a &= \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix} \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\
 &= \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j \\
 &\geq 0
 \end{aligned} \tag{B.15}$$

Reverse implication. Let $x, x' \in \mathcal{U}$. By symmetry of $k((x, x'), (x, x'))$, it comes that:

$$\begin{aligned}
 \begin{bmatrix} k(x, x) & k(x, x') \\ k(x', x) & k(x', x') \end{bmatrix} &= \begin{bmatrix} k(x, x) & k(x, x') \\ k(x', x) & k(x', x') \end{bmatrix}^\top \\
 &= \begin{bmatrix} k(x, x)^\top & k(x', x)^\top \\ k(x, x')^\top & k(x', x')^\top \end{bmatrix}
 \end{aligned} \tag{B.16}$$

In particular, the upper-right block gives:

$$k(x, x') = k(x', x)^\top \tag{B.17}$$

Let $n \in \mathbb{N}^*$, $\mathbf{x} \in \mathcal{U}^n$, and $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, and set $a \triangleq [a_1^\top, \dots, a_n^\top]^\top \in \mathbb{R}^{nd_z}$. Then:

$$\begin{aligned}
 \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j &= \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix} \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\
 &= a^\top k(\mathbf{x}, \mathbf{x}) a \\
 &\geq 0
 \end{aligned} \tag{B.18}$$

Strict version. The same computations hold with strict inequalities, provided the inputs are pairwise distinct and the coefficients not all zero.

B.4 Proof of Proposition 3.3

Assume that $\nu \in \mathbb{R}_+^*$ and let k_ν denote such a function. We also let:

$$g : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} r^\nu K_\nu(r) \end{cases} \quad \text{and} \quad \tilde{r} : \begin{cases} \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}_+ \\ x, x' \mapsto \frac{\sqrt{2\nu}}{l} \|x - x'\| \end{cases} \quad (\text{B.19})$$

$\tilde{r}$ and K_ν are continuous on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$ and $\mathbb{R}_+^*$ respectively, and the latter admits the following asymptotic expansion as $r \rightarrow 0^+$:

$$K_\nu(r) = 2^{\nu-1} \Gamma(\nu) r^{-\nu} + o(r^{-\nu}) \quad (\text{B.20})$$

Therefore, f can be continuously extended to 0 by setting $f(0) = \sigma_f^2$, and k_ν is thus continuous on the entirety of $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$.

Let $x, x' \in \mathbb{R}^{d_x}$. By symmetry of the norm, $\tilde{r}(x, x') = \tilde{r}(x', x)$, and thus:

$$k_\nu = g \circ \tilde{r} \quad (\text{B.21})$$

is symmetric, with $\circ$ being the composition operator.

Furthermore, Bochner's theorem [99] states that a continuous integrable function with a strictly positive Fourier transform is SPD. It can be shown [100] that the Fourier transform of k_ν on $\mathbb{R}^{d_x}$ is:

$$S_\nu : \begin{cases} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ \omega \mapsto C \left(\frac{2\nu}{l^2} + \|\omega\|^2 \right)^{-(\nu+d_x/2)} \end{cases} \quad (\text{B.22})$$

where $C > 0$ is a normalising constant which only depends on d_x and the parameters of k_ν . Since $\nu > 0$, $l > 0$, it comes that:

$$\forall \omega \in \mathbb{R}^{d_x}, S_\nu(\omega) > 0 \quad (\text{B.23})$$

and thus k_ν is a SPD kernel. By definition, it is also a radial kernel.

The case $\nu = +\infty$ is more direct. The squared exponential kernel is continuous on the whole of $\mathbb{R}_+$ without any extension, and symmetric by the same composition argument. Its Fourier transform on $\mathbb{R}^{d_x}$ is:

$$S_\nu : \begin{cases} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ \omega \mapsto C \exp \left(-\frac{l^2 \|\omega\|^2}{2} \right) \end{cases} \quad (\text{B.24})$$

which is again strictly positive, so Bochner's theorem applies as above.

B.5 Proof of Proposition 3.8

Let:

$$\mathcal{L}(W, \xi) \triangleq \text{tr}(\mathbb{V}[W^\top \tilde{\mathbf{z}} - z]) + 2 \text{tr}(\xi^\top (C^\top W - I_{d_z})) \quad (\text{B.25})$$

be the Lagrangian associated with the optimisation problem (3.32) under the constraint (3.48), and $\xi \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$ the corresponding Lagrange multiplier.

This problem is solved by finding the minimum of $\mathcal{L}$, *i.e.* finding a zero of its derivative. Writing G for $G(\tilde{\mathbf{x}})$, the same properties used in the proof of Proposition 3.5 give:

$$\begin{aligned} & \frac{\partial \mathcal{L}}{\partial W}(W, \xi) = 0 \\ \iff & 2G W - 2k(\tilde{\mathbf{x}}, x) + 2 \frac{\partial}{\partial W} (\text{tr}(\xi^\top (C^\top W - I_{d_z}))) = 0 \\ \iff & 2G W - 2k(\tilde{\mathbf{x}}, x) + 2C\xi = 0 \\ \iff & G W + C\xi = k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (\text{B.26})$$

Together with the constraint $C^\top W = I_{d_z}$, this yields the block system:

$$\begin{bmatrix} G & C \\ C^\top & 0 \end{bmatrix} \begin{bmatrix} W \\ \xi \end{bmatrix} = \begin{bmatrix} k(\tilde{\mathbf{x}}, x) \\ I_{d_z} \end{bmatrix} \quad (\text{B.27})$$

The left matrix block can be inverted using the upper left Schur inversion formula from Proposition A.2. Its Schur complement,

$$S \triangleq 0 - C^\top G^{-1} C = -C^\top G^{-1} C, \quad (\text{B.28})$$

is invertible since $G \succ 0$ and C has full column rank. The convention of Proposition A.2 gives the four blocks of the inverse:

$$\begin{cases} \bar{A} = G^{-1} + G^{-1} C S^{-1} C^\top G^{-1} \\ \bar{B} = -G^{-1} C S^{-1} \\ \bar{C} = -S^{-1} C^\top G^{-1} \\ \bar{D} = S^{-1} \end{cases} \quad (\text{B.29})$$

Let $M \triangleq -S^{-1} = (C^\top G^{-1} C)^{-1}$. The weight matrix can be derived from the upper blocks of the inverse:

$$\begin{aligned} W &= \bar{A} k(\tilde{\mathbf{x}}, x) + \bar{B} I_{d_z} \\ &= (G^{-1} + G^{-1} C S^{-1} C^\top G^{-1}) k(\tilde{\mathbf{x}}, x) - G^{-1} C S^{-1} \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C S^{-1} (C^\top G^{-1} k(\tilde{\mathbf{x}}, x) - I_{d_z}) \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C M (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \end{aligned} \quad (\text{B.30})$$

and ξ from the lower blocks:

$$\begin{aligned} \xi &= \bar{C} k(\tilde{\mathbf{x}}, x) + \bar{D} I_{d_z} \\ &= -S^{-1} C^\top G^{-1} k(\tilde{\mathbf{x}}, x) + S^{-1} \\ &= S^{-1} (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \\ &= -M (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \end{aligned} \quad (\text{B.31})$$

Since M is symmetric, the OK estimator is:

$$\begin{aligned}\hat{z}_{\text{OK}} &= W^\top \tilde{\mathbf{z}} \\ &= k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + (I_{d_z} - k(x, \tilde{\mathbf{x}}) G^{-1} C) M C^\top G^{-1} \tilde{\mathbf{z}} \\ &= M C^\top G^{-1} \tilde{\mathbf{z}} + k(x, \tilde{\mathbf{x}}) G^{-1} (\tilde{\mathbf{z}} - C M C^\top G^{-1} \tilde{\mathbf{z}})\end{aligned}\quad (\text{B.32})$$

which is exactly (3.50), the estimated mean being $\hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = M C^\top G^{-1} \tilde{\mathbf{z}}$.

Since R_{OK} is a centred moment of the estimation error, it is insensitive to the unknown mean and (3.36) can be applied. Using (B.26) by $W^\top$ and the constraint $W^\top C = I_{d_z}$ gives the identity:

$$\begin{aligned}W^\top G W &= W^\top (k(\tilde{\mathbf{x}}, x) - C \xi) \\ &= W^\top k(\tilde{\mathbf{x}}, x) - W^\top C \xi \\ &= W^\top k(\tilde{\mathbf{x}}, x) - \xi\end{aligned}\quad (\text{B.33})$$

Let $U \triangleq I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)$ so that the expressions of W and ξ are simplified to:

$$W = G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C M U \quad \text{and} \quad \xi = -M U \quad (\text{B.34})$$

Combining these results yields:

$$\begin{aligned}R_{\text{OK}} &= W^\top G W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\ &= W^\top k(\tilde{\mathbf{x}}, x) - \xi - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\ &= -\xi - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\ &= M U - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) G^{-1} C M U + k(x, x) + R \\ &= (I_{d_z} - k(x, \tilde{\mathbf{x}}) G^{-1} C) M U + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) \\ &= U^\top M U + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x)\end{aligned}\quad (\text{B.35})$$

Finally, recognising $R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) = U^\top M U \succeq 0$, since $M \succ 0$, gives the desired expression.

B.6 Proof of Proposition 3.9

The unbiased assumption now reads:

$$g(\tilde{\mathbf{x}})^\top W - g(x)^\top = 0 \quad (\text{B.36})$$

and the Lagrange multiplier method of Proposition 3.8 can be transposed to universal kriging under the substitutions:

$$\begin{cases} C \longrightarrow g(\tilde{\mathbf{x}}) \\ I_{d_z} \longrightarrow g(x)^\top \\ \xi \in \mathcal{M}_{r_{\text{UK}} d_z, d_z}(\mathbb{R}) \end{cases} \quad (\text{B.37})$$

The full column rank of $g(\tilde{\mathbf{x}})$ ensures that S invertible, and that M is SPD. Writing G for $G(\tilde{\mathbf{x}})$, the shorthands of that proof thus become:

$$\begin{cases} M \triangleq -S^{-1} = (g(\tilde{\mathbf{x}})^\top G^{-1} g(\tilde{\mathbf{x}}))^{-1} \\ U \triangleq g(x)^\top - g(\tilde{\mathbf{x}})^\top G^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (\text{B.38})$$

so that the weight matrix and Lagrange multiplier are:

$$\begin{cases} W = G^{-1}k(\tilde{\mathbf{x}}, x) + G^{-1}g(\tilde{\mathbf{x}}) M U \\ \xi = -M U \end{cases} \quad (\text{B.39})$$

Finally, the bilinear identity (B.33) becomes:

$$W^\top G W = W^\top k(\tilde{\mathbf{x}}, x) - g(x) \xi \quad (\text{B.40})$$

The UK estimator and covariance matrices are derived exactly like for Proposition 3.8, using the corrected results.

B.7 Proof of Proposition 3.10

TODO: à relire Let $B \in \mathcal{S}_{r_{\text{UK}d_z}}^{++}(\mathbb{R})$, and give the unknown coefficients the prior $\beta \sim \mathcal{N}(0, B)$, independent of the centred process $f \sim \mathcal{GP}(0, k)$. Randomising the trend this way makes the field $f_B \triangleq f + g \beta$ a centred GP of kernel k_B by Proposition 3.1, so that kriging f_B is Proposition 3.5 read with k_B in place of k . We write $\hat{z}_B$ and R_B for the estimator and covariance it returns, G for $G(\tilde{\mathbf{x}})$ and g for $g(\tilde{\mathbf{x}})$, and reuse the shorthands M and U of section B.6. The Gram matrix of f_B is:

$$G_B \triangleq k_B(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} = G + g B g^\top \quad (\text{B.41})$$

By Proposition A.5 and $g^\top G^{-1} g = M^{-1}$, it comes that:

$$G_B^{-1} = G^{-1} - G^{-1} g M_B g^\top G^{-1}, \quad M_B \triangleq (B^{-1} + M^{-1})^{-1} \quad (\text{B.42})$$

Since $M_B (B^{-1} + M^{-1}) = I$, we have $M^{-1} M_B = I - B^{-1} M_B$, and thus the two identities:

$$B - B M^{-1} M_B = M_B, \quad M_B M^{-1} B = B - M_B \quad (\text{B.43})$$

Let $A \triangleq k_B(x, \tilde{\mathbf{x}}) = k(x, \tilde{\mathbf{x}}) + g(x) B g^\top$. Developing $A G_B^{-1}$ and collecting its terms in B through the first identity of (B.43) yields:

$$A G_B^{-1} = k(x, \tilde{\mathbf{x}}) G^{-1} + U^\top M_B g^\top G^{-1} \quad (\text{B.44})$$

whence the estimator:

$$\hat{z}_B = A G_B^{-1} \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + U^\top M_B g^\top G^{-1} \tilde{\mathbf{z}} \quad (\text{B.45})$$

For the covariance, (B.44) and the second identity of (B.43) give:

$$\begin{aligned} A G_B^{-1} A^\top &= k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) \\ &\quad + k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top + U^\top (B - M_B) g(x)^\top \end{aligned} \quad (\text{B.46})$$

and $R_B = k_B(x, x) + R - A G_B^{-1} A^\top$ with $k_B(x, x) = k(x, x) + g(x) B g(x)^\top$. The terms in B cancel, since:

$$g(x) B g(x)^\top - k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top - U^\top B g(x)^\top = U^\top B g(x)^\top - U^\top B g(x)^\top = 0 \quad (\text{B.47})$$

while the two terms in M_B assemble into the sandwich of (3.59):

$$U^\top M_B g(x)^\top - U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) = U^\top M_B U \quad (\text{B.48})$$

so that:

$$R_B = k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B U \quad (\text{B.49})$$

Both expressions are those of Proposition 3.9 with M_B substituted for M , which proves the stronger statement that kriging under k_B is universal kriging regularised by the prior. Since $M_B \rightarrow M$ as $B^{-1} \rightarrow 0$, the limit returns (3.58) and (3.59).

The coefficients follow the same course. The Gaussian linear model gives β the posterior $\mathcal{N}(M_B g^\top G^{-1} \tilde{\mathbf{z}}, M_B)$, whose limit is $\mathcal{N}(\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g), M)$. The estimator of (3.59) is therefore the posterior mean of the trend, and M its posterior covariance.

B.8 Proof of Proposition 3.11

As $\mathcal{U}$ is compact and k continuous, k is bounded on $\mathcal{U} \times \mathcal{U}$, and $\mathcal{K}$ is compact.

Let $\phi, \phi' \in \mathcal{L}^2(\mathcal{U})$. Cauchy-Schwarz inequality ensures that there exists a constant $c \in \mathbb{R}_+$ such that:

$$\iint_{\mathcal{U} \times \mathcal{U}} |\phi(y)^\top k(x, y)^\top \phi'(x)| \, dx \, dy \leq c \sqrt{\langle \phi | \phi \rangle} \sqrt{\langle \phi' | \phi' \rangle} < +\infty \quad (\text{B.50})$$

Fubini's integration theorem can thus be used to give:

$$\begin{aligned} \langle \mathcal{K}\phi | \phi' \rangle &= \int_{\mathcal{U}} (\mathcal{K}\phi)(x)^\top \phi'(x) \, dx \\ &= \int_{\mathcal{U}} \left(\int_{\mathcal{U}} k(x, y) \phi(y) \, dy \right)^\top \phi'(x) \, dx \\ &= \int_{\mathcal{U}} \int_{\mathcal{U}} \phi(y)^\top k(x, y)^\top \phi'(x) \, dy \, dx \\ &= \int_{\mathcal{U}} \int_{\mathcal{U}} \phi(y)^\top k(y, x) \phi'(x) \, dx \, dy \\ &= \int_{\mathcal{U}} \phi(y)^\top \left(\int_{\mathcal{U}} k(y, x) \phi'(x) \, dx \right) \, dy \\ &= \int_{\mathcal{U}} \phi(y)^\top (\mathcal{K}\phi')(y) \, dy \\ &= \langle \phi | \mathcal{K}\phi' \rangle \end{aligned} \quad (\text{B.51})$$

Let $\phi : \mathcal{U} \rightarrow \mathbb{R}^{d_x}$ be a continuous function, and let $n \in \mathbb{N}^*$. We define $(\mathcal{U}_i^{(n)})_{i \in \llbracket 1, n \rrbracket}$ a partition of $\mathcal{U}$ such that the diameter of each $\mathcal{U}_i^{(n)}$, $i \in \llbracket 1, n \rrbracket$ tends to 0 when $n \rightarrow +\infty$. Here, the diameter of a subset of the Euclidean space $\mathbb{R}^{d_x}$ is defined as the maximum distance between two points of this subset. For $i \in \llbracket 1, n \rrbracket$, we define $x_i^{(n)} \in \mathcal{U}_i^{(n)}$, and we let:

$$a_i^{(n)} \triangleq \left(\int_{\mathcal{U}_i^{(n)}} dx \right) \phi(x_i^{(n)}) \quad (\text{B.52})$$

By continuity of ϕ , the function:

$$(x, y) \mapsto \phi(x)^\top k(x, y) \phi(y) \quad (\text{B.53})$$

is continuous on the compact $\mathcal{U} \times \mathcal{U}$, so its Riemann sums converge:

$$\sum_{i=1}^n \sum_{j=1}^n \left(a_i^{(n)}\right)^\top k\left(x_i^{(n)}, x_j^{(n)}\right) a_j^{(n)} \xrightarrow{n \rightarrow +\infty} \iint_{\mathcal{U} \times \mathcal{U}} \phi(x)^\top k(x, y) \phi(y) dx dy = \langle \phi | \mathcal{K} \phi \rangle \quad (\text{B.54})$$

The SPSDness of k ensures that each Riemann sum is non-negative. By passing to the limit, it follows that:

$$\langle \phi | \mathcal{K} \phi \rangle \geq 0 \quad (\text{B.55})$$

for every continuous function ϕ .

Finally, recall that the set of continuous functions of $\mathcal{U}$ is dense in $\mathcal{L}^2(\mathcal{U})$. By continuity of:

$$\phi \in \mathcal{L}^2(\mathcal{U}) \mapsto \langle \phi | \mathcal{K} \phi \rangle \quad (\text{B.56})$$

the inequality (B.55) extends to all $\phi \in \mathcal{L}^2(\mathcal{U})$.

B.9 Proof of Proposition 3.12

By Proposition 3.11, $\mathcal{K}$ is a compact and self-adjoint operator on $\mathcal{L}^2(\mathcal{U})$. The spectral theorem [101] thus states that there exists an orthonormal basis $(\psi_i)_{i \in \mathbb{N}^*}$ of $\mathcal{L}^2(\mathcal{U})$ consisting of eigenfunctions of $\mathcal{K}$, with each corresponding eigenvalue $(\lambda_i)_{i \in \mathbb{N}^*}$ being real. Up to reordering, the sequence of eigenvalues can be considered to be non-increasing.

Since k is uniformly continuous on the compact $\mathcal{U} \times \mathcal{U}$, $\mathcal{K}\phi$ is continuous for every $\phi \in \mathcal{L}^2(\mathcal{U})$. In particular, for each $i \in \mathbb{N}^*$ such that $\lambda_i \neq 0$, the eigenfunction:

$$\psi_i = \lambda_i^{-1} \mathcal{K} \psi_i \quad (\text{B.57})$$

is continuous. Without loss of generality, it must be noted that the ψ_i associated with a null eigenvalue do not contribute to (3.67) and may thus be dismissed.

Since $(\psi_i)_{i \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{L}^2(\mathcal{U})$, the family $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{L}^2(\mathcal{U} \times \mathcal{U})$. Let $\langle \cdot | \cdot \rangle_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}$ denote:

$$\forall \phi, \phi' \in \mathcal{L}^2(\mathcal{U} \times \mathcal{U}) \mapsto \langle \phi | \phi' \rangle_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})} \triangleq \iint_{\mathcal{U} \times \mathcal{U}} \text{tr}(\phi(x, y)^\top \phi'(x, y)) dx dy \quad (\text{B.58})$$

Let $i, j \in \mathbb{N}^*$. We have:

$$\begin{aligned} \langle k | \psi_i \psi_j^\top \rangle_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})} &= \iint_{\mathcal{U} \times \mathcal{U}} \text{tr}(k(x, y)^\top \psi_i(x) \psi_j(y)^\top) dx dy \\ &= \iint_{\mathcal{U} \times \mathcal{U}} \psi_j(y)^\top k(x, y)^\top \psi_i(x) dx dy \\ &= \int_{\mathcal{U}} \psi_j(y)^\top \left(\int_{\mathcal{U}} k(y, x) \psi_i(x) dx \right) dy \\ &= \int_{\mathcal{U}} \psi_j(y)^\top \mathcal{K} \psi_i(y) dy \\ &= \lambda_i \int_{\mathcal{U}} \psi_j(y)^\top \psi_i(y) dy \\ &= \lambda_i \delta_{i,j} \end{aligned} \quad (\text{B.59})$$

where:

$$\delta_{i,j} \triangleq \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (\text{B.60})$$

is the Kronecker symbol.

Because $k \in \mathcal{L}^2(\mathcal{U} \times \mathcal{U})$, it follows from the completeness of $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ that:

$$k_{r_{\text{KL}}}(x, x') \triangleq \sum_{i=1}^{r_{\text{KL}}} \lambda_i \psi_i(x) \psi_i(x')^\top \quad (\text{B.61})$$

converges to k in the $\mathcal{L}^2(\mathcal{U} \times \mathcal{U})$ sense.

It must be noted that $\mathcal{L}^2$ convergence only identifies k up to a Lebesgue-null subset of $\mathcal{U} \times \mathcal{U}$, and gives no control at any specific point of the domain. In particular, it is not enough to justify substituting $k_{r_{\text{KL}}}$ for k in kriging scenarios. However, since k is continuous on $\mathcal{U} \times \mathcal{U}$, the convergence can be extended to absolute and uniform convergence on the entire domain [79]. The argument applies Dini's theorem [102] to the non-decreasing sequence $(x \mapsto \text{tr } k_{r_{\text{KL}}}(x, x))_{r_{\text{KL}} \in \mathbb{N}^*}$, whose limit is continuous on the compact $\mathcal{U}$, and extends the result off the diagonal through the Cauchy-Schwarz inequality.

Since $\mathcal{K}$ is SPSPD, every eigenvalue is non-negative. Assume that there exists a rank $r_0 \in \mathbb{N}^*$ such that $\lambda_i = 0$ as soon as $i > r_0$. Then the decomposition (3.67) would reduce to the finite sum:

$$k = \sum_{i=1}^{r_0} \lambda_i \psi_i \psi_i^\top \quad (\text{B.62})$$

whose Gram matrices have rank at most r_0 and are thus singular at any n distinct points with $nd_z > r_0$, contradicting the SPDness of k by Proposition 3.2. Consequently:

$$\lambda_1 \geq \lambda_2 \geq \dots > 0 \quad (\text{B.63})$$

B.10 Proof of Proposition 3.14

As (3.76) is a rank r_{KL} perturbation of $A \triangleq I_{\tilde{N}} \otimes \tilde{R}$, the Woodbury inversion formula (see Proposition A.5) can be used to reduce the $\tilde{N}d_z \times \tilde{N}d_z$ inversion of $G(\tilde{\mathbf{x}})$ to an $r \times r$ one:

$$G(\tilde{\mathbf{x}})^{-1} \approx A^{-1} - A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \quad (\text{B.64})$$

Computing the centred SK estimator from Proposition 3.5 using this approximation and the push-through identity (see Proposition A.4) yields:

$$\begin{aligned} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) &= k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ &\approx \Psi(x) \Lambda \Psi(\tilde{\mathbf{x}})^\top \left(I - A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \right) A^{-1} \tilde{\mathbf{z}} \\ &= \Psi(x) \Lambda \left(I - \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \right) \Psi(\tilde{\mathbf{x}})^\top A^{-1} \tilde{\mathbf{z}} \end{aligned} \quad (\text{B.65})$$

Recall that, by definition of $\tilde{G}(\tilde{\mathbf{x}})$:

$$\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} = \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \quad (\text{B.66})$$

which gives:

$$\begin{aligned} I - \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} &= I - \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} \right) \tilde{G}(\tilde{\mathbf{x}})^{-1} \\ &= I - I + \Lambda^{-1} \tilde{G}(\tilde{\mathbf{x}})^{-1} \\ &= \Lambda^{-1} \tilde{G}(\tilde{\mathbf{x}})^{-1} \end{aligned} \quad (\text{B.67})$$

Plugging this result into (B.65) and using $A^{-1} = I_{\tilde{N}} \otimes \tilde{R}^{-1}$ finally gives:

$$\hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \tilde{\mathbf{z}} \quad (\text{B.68})$$

For R_{SK} , reusing:

$$k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \quad (\text{B.69})$$

together with (B.67) yields:

$$\begin{aligned} R_{\text{SK}} &= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &\approx \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \times \Psi(\tilde{\mathbf{x}}) \Lambda \Psi(x)^\top \\ &= \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} \right) \Lambda \Psi(x)^\top \\ &= \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \left(\Lambda - \tilde{G}(\tilde{\mathbf{x}})^{-1} \right) \Psi(x)^\top \\ &= R + \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{aligned} \quad (\text{B.70})$$

Finally, when $\tilde{R} = \sigma^2 I_{d_z}$ with $\sigma > 0$, substituting $A^{-1} = \sigma^{-2} I_{\tilde{N}d_z}$ yields the announced simplifications. For such $\sigma > 0$, the matrix:

$$\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \succ 0 \quad (\text{B.71})$$

is SPD, as the sum of the SPSD matrix $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}})$ and the SPD matrix $\sigma^2 \Lambda^{-1}$. The map:

$$\sigma \in \mathbb{R}_+^* \mapsto \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \quad (\text{B.72})$$

is thus well-defined and continuous, and extends to $\sigma = 0$ when $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) \succ 0$, *i.e.* when $\Psi(\tilde{\mathbf{x}})$ has full column rank. As $\hat{z}_{\text{SK}}$ and R_{SK} depend on σ only through this inverse and an explicit factor σ^2 , both expressions remain valid at $\sigma = 0$ under this condition.

B.11 Proof of Proposition 3.16

TODO: à relire

The form is well defined. Let $\mathcal{H}_0$ denote the span of $\{k(x, \cdot), x \in \mathcal{U}\}$, and extend $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ to it by bilinearity. For $\psi = \sum_{i=1}^n a_i k(x_i, \cdot)$ and $\varphi = \sum_{j=1}^p b_j k(x'_j, \cdot)$ in $\mathcal{H}_0$, it comes that:

$$\langle \psi | \varphi \rangle_{\mathcal{H}_k} = \sum_{i=1}^n \sum_{j=1}^p a_i b_j k(x_i, x'_j) = \sum_{j=1}^p b_j \psi(x'_j) = \sum_{i=1}^n a_i \varphi(x_i) \quad (\text{B.73})$$

The last two expressions involve one of the two functions through its values alone, so neither representation matters and the form is well defined. It is symmetric and

bilinear by construction, and positive since k is SPSP. Taking $\varphi = k(x, \cdot)$ gives the reproducing property on $\mathcal{H}_0$:

$$\forall \psi \in \mathcal{H}_0, \forall x \in \mathcal{U}, \langle \psi | k(x, \cdot) \rangle_{\mathcal{H}_k} = \psi(x) \quad (\text{B.74})$$

It is definite. The Cauchy-Schwarz inequality holds for any positive semi-definite bilinear form, so (B.74) bounds every evaluation:

$$\forall \psi \in \mathcal{H}_0, \forall x \in \mathcal{U}, |\psi(x)|^2 \leq \langle \psi | \psi \rangle_{\mathcal{H}_k} k(x, x) \quad (\text{B.75})$$

A function of null norm thus vanishes everywhere, and $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ is an inner product on $\mathcal{H}_0$.

The completion is a space of functions. Equation (B.75) makes every Cauchy sequence of $\mathcal{H}_0$ pointwise Cauchy, hence pointwise convergent, which sends each class of the abstract completion to a function on $\mathcal{U}$. That map is injective. Indeed, let $(\psi_n)_n$ be a Cauchy sequence converging pointwise to 0, let $\varepsilon > 0$, and let N be such that $\|\psi_n - \psi_m\|_{\mathcal{H}_k} \leq \varepsilon$ for $n, m \geq N$. Fixing $n \geq N$ and writing $\psi_n = \sum_{i=1}^{n'} a_i k(x_i, \cdot)$, (B.73) gives:

$$\|\psi_n\|_{\mathcal{H}_k}^2 = \langle \psi_n | \psi_n - \psi_m \rangle_{\mathcal{H}_k} + \sum_{i=1}^{n'} a_i \psi_m(x_i) \quad (\text{B.76})$$

whose first term is at most $\varepsilon \sup_n \|\psi_n\|_{\mathcal{H}_k}$, finite since a Cauchy sequence is bounded, and whose second is a finite sum tending to 0 with m . Hence $\|\psi_n\|_{\mathcal{H}_k} \rightarrow 0$, and only the zero class is sent to the zero function. Passing (B.74) to the limit, $\mathcal{H}_k$ is an RKHS of reproducing kernel k .

Uniqueness. Let $\mathcal{H}$ be an RKHS of reproducing kernel k . It contains every $k(x, \cdot)$, hence $\mathcal{H}_0$, and its inner product agrees with $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ on the generators, hence on $\mathcal{H}_0$ by bilinearity. A function of $\mathcal{H}$ orthogonal to $\mathcal{H}_0$ verifies $\psi(x) = \langle \psi | k(x, \cdot) \rangle_{\mathcal{H}} = 0$ for every $x \in \mathcal{U}$, so $\mathcal{H}_0$ is dense in $\mathcal{H}$, which is therefore its completion $\mathcal{H}_k$.

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